A RESEARCH REPORT

UniverCell Holding GmbH and the battery industry

What the company is, how the industry around it arrived here, where it stands in 2026, and where it goes next. Written for the board of an Indian industrial company.

SUBJECT
UniverCell Holding GmbH
Flintbek, Germany

FIGURES CURRENT TO
20 October 2026
Every figure sourced

MONEY SHOWN IN
Indian rupees
Originals in brackets

SOURCES
249, numbered
Listed in the appendix

Contents

What is in this report

Fifteen chapters in eleven files, an annex of twenty-six tables and a source register. Word counts are exact, so a reader can budget.

History

- **01. How the industry got here, 2010 to 2026** (2,088 words). The 93 per cent real price collapse, LFP over NMC, China from assembler to chain owner.
- **02. The experience curve, and what it forecloses** (1,730 words). A 19 per cent learning rate, a 2030 pack floor of ₹6,667 to ₹8,095 per kWh (USD 70 to 85), and the industrial power bands behind it.

Where it stands

- **03. The industry as it stands in 2026** (2,173 words). 1.59 TWh of demand, over 4 TWh of nameplate, 40 per cent utilisation beside CATL's 96.9.
- **04. Concentration, trade and the export-control regime** (2,695 words). The chokepoint is processing, not mining, and the instruments carry dates.
- **05. Europe's answer, and what it cost** (2,401 words). Over 2 TWh announced, 1.2 to 1.7 TWh realisable, ₹1.67 lakh crore (over USD 17bn) consumed.
- **06. The company: UniverCell Holding GmbH** (3,261 words). Register, Series B, staff, one pending patent, products, absent certifications, 25 pros and cons.
- **07. The arithmetic nobody publishes** (1,736 words). Four published capacity numbers reconciled from the company's own inputs, step by step.
- **08. Dry coating: what it is, who has it, how long the lead lasts** (1,564 words). Eleven players ranked, and the date the lead expires.
- **09. The competitive landscape** (1,835 words). A collapsed European cohort, a pilot-scale dry-coating cohort, and Molicel.

Where it goes

- **10. Where this goes: scenarios to 2030 and 2035** (2,070 words). Three worlds, European demand and output, the price path on three learning rates.
- **11. What matures when: the technology S-curves** (1,544 words). Ten technologies ranked by what they do to a dry-coating specialist.
- **12. The rulebook, 2024 to 2031** (2,429 words). A disclosure regime, not a tariff: the dates that bite and the money with a 10 GWh wall.

The read

- **13. Strategic analysis** (3,098 words). Porter, PESTEL, value chain, BCG, TAM to SOM and VRIO, each inverting with the market boundary.
- **14. The read for India** (3,577 words). The import bill, the ACC PLI design against its delivery, announced against commissioned capacity, four constraints, six plays.
- **15. Conclusions, unknowns and the diligence list** (1,065 words). What the evidence supports and does not, five questions, the watch-list.

Reference

- **Annex. Reference tables** (14,160 words). The 26 tables the chapters point to, numbered A1 to A26 in first-reference order, each pointed at from a chapter.
- **Appendix. Source register** (3,251 words). All 249 sources, one line each, with a trust rating.

Summary

The argument in two pages

UniverCell Holding GmbH is a real, operating, revenue-earning electrode manufacturer with a cell pilot attached to it, and almost every number it publishes about itself describes the electrode business while almost every number its March 2026 investors bought describes the cell business. That gap is the finding, and it is reconstructible from the company's own documents.

0.86 per cent of the 1.5 GWh headline

WHAT THE CELL LINE WAS RUNNING AT IN JUNE 2026

A 116-fold gap, derived from the company's own 50,000 cells a month and its own 21.6 Wh per cell.

₹320.46 crore Series B, 4 March 2026

THE EQUITY BASE BEHIND THE RAMP

EUR 30m, roughly 0.2 per cent of what Northvolt consumed before failing.

2028 to 2029 when the lead expires

THE DRY-COATING ADVANTAGE, DATED

Unpatented, licensable from Fraunhofer IWS today, and industrialised by LG Energy Solution on its own 2028 guidance.

The thesis. This is an electrode-process business holding a real but unpatented three-year process lead into a policy-created demand window that opens with the battery passport on 18 February 2027 and closes as the Korean makers industrialise dry coating in 2028 and 2029. Its durable assets are the running merchant electrode line, over 1,000 tested electrode recipes, an in-house formation process and the engineers who hold all three. Its marketed asset, dry coating, is at pilot scale on its own documents and is the wasting half. For an Indian industrial group it is therefore a licensing, offtake and process-transfer counterparty, not an equity investment: no valuation could be established, **NOT FOUND**.

Finding one, the capacity arithmetic, published nowhere else. The company markets 1.5 GWh. Forty million cells a year, its fully funded target, at its own 21.6 Wh per cell is 0.864 GWh, which is 57.6 per cent of the headline, and a target cannot be smaller than an existing asset unless the two measure different things. So 1.5 GWh is an electrode-coating number. Cell assembly began in June 2026 at 50,000 cells a month: 0.0130 GWh a year, 0.86 per cent of the headline, a 116-fold gap. The 1.3 GWh claimed for 2027 needs roughly 100 times that run-rate in eighteen months, a sustained 29 per cent compound monthly growth rate, on one ₹320.46 crore (EUR 30m) round. The cross-check confirms the reading: 60 m per minute at 750 mm for 1,440 minutes is 64,800 m2, which is the advertised "64,000 m2", a daily coated area and not the floor area of roughly 6,100 m2 the founder himself describes. The 2030 target of 180 million cells, about 4 GWh, is the same electrode line at full utilisation, not a new factory.

Finding two, the peer set has collapsed. Of twelve Tier A European specialty cell names, seven are dead, shrunken, restructured or cancelled: Northvolt, CustomCells, Cellforce, Morrow, Leclanche's German units, ACC's two plants and Blue Solutions. Varta AG is an eighth by its second insolvency on 24 July 2026 and Freyr a ninth, having left the industry rather than lost in it. On the wider Western count, nine companies failed or

entered formal distress between January 2023 and October 2026, consuming a disclosed floor of ₹1.67 lakh crore (over USD 17bn), and every independent merchant-market European cell startup failed. Three are executing, each with a balance sheet behind it: Verkor on Renault, Lyten by acquisition, V4Smart on Porsche. The transferable lesson is not cost: the survivors have a committed buyer on the share register, and no casualty's stated cause was chemistry or process.

Finding three, the real comparator is Molicel, not CATL. Molicel, trading as Taiwan Cell Technology, runs approximately 1.5 GWh in Tainan against UniverCell's 1.3 GWh target, in the same high-drain 21700 and 18650 formats, into the same segments, with the design wins already taken at Archer Aviation, Vertical Aerospace and McMurtry. Against the Molicel P60C at an identical 6.0 Ah, UniverCell is 16.0 per cent lighter and 16.2 per cent better on watt-hours per kilogram, over 330 Wh/kg against 284, but **2.4 per cent worse on energy per cell**. The headline energy-density lead is about twelve grams of aluminium can replacing steel, which is exactly what the single pending patent covers, and any can supplier can copy it. What Molicel cannot copy is EU domicile with Battery Regulation compliance and content-rule eligibility.

The case for. The electrode business is installed and earning, FY2025 revenue a double-digit million euro figure. The venture is deliberately sized below where European cell ventures have died. End markets are qualification-locked rather than priced per kWh. Grant dependence is unusually low, approximately ₹25.50 crore (EUR 2.387m) of identified public money. The weight advantage reproduces independently at 16.0, 15.4, 11.9 and 12.5 per cent. The adverse record is clean after targeted search.

The case against. Dry coating, the premise of the round, is at pilot scale and unpatented, the intellectual property being one pending application on a housing lid. No customer has ever been named. No ISO 9001, ISO 13485, AS9100 or ECSS, UN 38.3 or IEC 62133 certification is publicly evidenced, so space and medical read as target markets rather than qualified supply. Hiring is roughly twelve months behind the company's own plan, with no CTO, CFO or VP Manufacturing. Real financials are not obtainable. The 1.5 GWh headline fails the Battery Booster Facility's 10 GWh floor twice over.

The India plays. License the process rather than the cell, ₹240 crore to ₹570 crore (USD 25m to 60m). Build India's first merchant electrode supplier, ₹380 crore to ₹860 crore (USD 40m to 90m), the highest-leverage move. Buy distressed European specialty capacity and the engineers with it, ₹140 crore to ₹760 crore (USD 15m to 80m). Hold an EU-domiciled vehicle meeting both blocs' content rules, ₹50 crore to ₹145 crore (USD 5m to 15m). Co-develop sodium-ion for Indian grid and two-wheeler storage, ₹95 crore to ₹285 crore (USD 10m to 30m). Own the specialty and defence-qualified niche, ₹190 crore to ₹475 crore (USD 20m to 50m), the best risk-adjusted.

The sentence to remember. India's constraint is process knowledge and the engineers who carry it, not money or demand, and a distressed European specialist sector holds more of what India needs than its market value suggests, but the duty-free equipment window closes on 31 March 2029 and the dry-coating lead expires around 2028.

Method

How this was built

Source hierarchy. 249 sources, each rated in the register. PRIMARY is the originating body's own publication, the register, the patent family, the CORDIS record, the Series B release, the company's pages and its NASA-hosted deck, and may be stated as fact. HIGH is an established institution reporting a primary fact it names. MEDIUM is trade press, law-firm or consultancy reporting, attributed and cross-checked. LOW is a vendor market model, data aggregator or self-administered rating, used for direction only. Where a LOW source contradicted a primary document the primary document won, and the contradiction is recorded rather than resolved silently.

What was accessible, and what was not. Accessible: the Kiel commercial register and its filing history, the patent family and four predecessor patents, the EIC grant record, the product pages, the twenty-page technical presentation, a third-party cell library, and ACEA, Eurostat, IEA, BNEF, Benchmark, Fraunhofer ISI and Transport and Environment publications. Not accessible: four years of filed accounts, behind a fee gate, so exact revenue, equity and net income are **NOT FOUND**; the shareholder names, paywalled; India's Tradestat series, **NOT FETCHED** behind a query interface; the methodology sections of nine vendor models. No management interview, no site visit, no independent measurement of any cell specification.

Confidence convention. Every material figure carries a tag. `HARD` is primary or corroborated. `SOFT` is single-sourced, secondary or date-uncertain. `EST` is this report's own derivation, inputs stated so a reader can re-run it. "HARD by absence" marks a finding established by exhaustive search returning nothing; **NOT FOUND** marks a gap that was looked for and is not filled by inference.

Currency. Fixed rates throughout, no trend applied: ₹95.24 to the US dollar, ₹106.82 to the euro, ₹123.00 to the pound. Other currencies convert at their rate of the same vintage. Rupee figures carry the original in brackets at first appearance in a chapter.

Where the vendor models disagree. Nine estimates of the 2025 global market span USD 47.53bn to USD 194.66bn, 4.1 times, none publishing its cell-versus-pack boundary. Three forecasts of India's 2030 cell demand span 115 to 272 GWh, 2.4 times. Two models of the 18650 and 21700 market disagree more than twofold on scope and roughly threefold on growth. None were averaged: each is labelled with its inferred scope and clustered, and the figures carrying the argument are rebuilt from physical inputs, gigawatt-hours deployed times a dated price per kilowatt-hour. Where no rebuild was possible, as with high-drain power tools, the number is **NOT FOUND** rather than borrowed.

Cut-off and limitations. All facts are as at **20 October 2026**. Three limits matter. The European capacity reference, Fraunhofer ISI's risk model, has a December 2024 cut-off, is roughly 22 months stale and is biased optimistic: everything since has removed capacity. No valuation is offered, the only figure in circulation being arithmetically impossible against the round. And the scenario work prices niches rather than predicting them: the bull and bear labels invert for a cell maker.

Chapter 01

How the industry got here, 2010 to 2026

Between 2010 and 2025 the volume weighted price of a lithium-ion pack fell about 93 per cent in real terms, and Chinese ownership of the processing chain, the European gigafactory collapse and the pivot from cars to grids follow from it.

The price collapse, year by year

The reference series is BNEF's annual survey: a volume weighted average across all end uses, restated in real terms each vintage.

YEAR	USD/KWH	₹/KWH	CHANGE
2010	1,200	114,288	USD 1,474 also circulates
2013	732	69,716	SOFT, secondary rendering
2014 to 2019	NOT FOUND	NOT FOUND	none located
2020	140	13,334	restated from 137, both primary
2021	132	12,572	minus 6%
2022	**151**	**14,381**	**plus 7%, the only rise**
2023	139	13,238	minus 14%
2024	115	10,953	minus 20%, largest since 2017
2025	108	10,286	minus 8%, record low, minus 93% on 2010
2026F	NOT FOUND	NOT FOUND	BNEF: prices "will decrease again"

Sources: BNEF vintages 2020 to 2025 HARD.[44,45,46,47,48] The 2014 to 2019 gap is genuine, not interpolated. The December 2025 survey breaks the ₹10,286 per kWh (USD 108) average open by segment, chemistry and region.[44,49]

MEASURE, 2025	USD/KWH	₹/KWH	NOTE
Pack, battery electric vehicle	99	9,429	second year running under 100
Pack, stationary storage	**70**	**6,667**	**minus 45% YoY, cheapest lithium-ion category for the first time**
Pack, two and three wheelers	133	12,667	most expensive segment
Cell, EV average	79	7,524	cell-series warning, Chapter 03
Pack, lowest observed LFP	50	4,762	one researcher sourced it, another could not confirm
China / North America / Europe	84 / approx 121 / approx **131**	8,000 / 11,524 / **12,476**	minus 13% / minus 4% / minus 8% YoY; **plus 44% and plus 56% over China, Europe widening**

The European Parliament's January 2025 series is a different vintage and scope: a European pack at ₹17,412 per kWh (EUR 163) against a Chinese pack at ₹12,391 (EUR 116), a gap of 40 per cent, with cathode active material about 50 per cent of cell manufacturing cost and anode active material about 10 per cent HARD. Use BNEF for the level and the Parliament for the gap, never blended.[118]

2022: the only reversal in the series

The rise to ₹14,381 per kWh (USD 151) shows what the experience curve does not control: learning continued, inputs moved faster the other way. Lithium carbonate peaked at CNY 575,000 a tonne in December 2022, roughly ₹77.1 lakh, and at about CNY 104,400 in October 2026 sits some 70 per cent below that peak yet 43.0 per cent up year on year, both rupee legs running through an approximate CNY 7.1 to the dollar and indicative only; cobalt touched near USD 82,000 a tonne (₹78.1 lakh) before a nine-year low near USD 10 a pound in February 2025; nickel spiked near USD 30,000 a tonne (₹28.6 lakh) in the March 2022 LME squeeze and now trades at USD 16,750 to 18,750; battery grade graphite ran at USD 8,000 to 12,000 a tonne (₹7.62 lakh to ₹11.43 lakh) against USD 3,000 to 6,000 delivered today {SOFT, EST on graphite}. [59,60,61,62] The pack consequence was plus 7 per cent real, China at ₹12,096 per kWh (USD 127), North America 24 per cent above China and Europe 33 per cent above. [47]

> **WHAT 2022 PROVES, AND IT GOVERNS EVERY FORECAST IN CHAPTER 02**
>
> The experience curve describes the cost of making things, not of buying inputs. In 2022 input inflation overwhelmed sixteen years of learning inside twelve months, so any price-floor projection needs a materials-bound slow case: Chapter 02's slow case exists because 2022 happened. BNEF's 2025 release is titled around prices falling "despite rising metal prices".

Chemistry: LFP displaces NMC, geographically too

LFP went from low teens per cent of global EV batteries deployed in 2020, precise figure NOT FOUND, to nearly 50 per cent in 2024, over 55 per cent in 2025 and **81.2 per cent** of Chinese EV installations, 625.3 of 769.7 GWh HARD. [32,43] At ₹7,714 per kWh (USD 81) against NMC's ₹12,191 (USD 128), a spread of ₹4,476 (USD 47), LFP packs run 37 per cent below NMC on BNEF's 2025 prices and more than 40 per cent below on the IEA's, unaveraged. [44] LFP carries no cobalt, policy-controlled now rather than geologically constrained: the Congolese export quota caps 2026 and 2027 at 96,600 tonnes a year against roughly 204,000 tonnes of 2024 output, a ceiling at under half of prior supply. [61] It advanced inside a Chinese chain holding roughly 95 per cent of global LFP cathode capacity, so chemistry and geography are one event. [113] Chapter 03 has the 2025 mix.

China: from assembler to chain owner

China made over 80 per cent of world cell output in 2025 and, upstream where substitution is slowest, **98 per cent of graphitization** HARD, a concentration that rose through the entire Western diversification effort and that the IEA projects into 2035. [32,35]

Three dated instruments made that leverage: the 15 July 2025 amendment to China's export-restriction catalogue on cathode and lithium processing technology, MOFCOM Announcement No. 58 of 9 October 2025 on high energy density cells, graphite anodes, advanced cathodes and key cell-manufacturing equipment, and Decision No. 70 of 7 November 2025 suspending No. 58 until 10 November 2026. [91] No. 58 targets the processing steps and the machines, not cells in general: most coating, calendering, winding and formation equipment is Chinese, so it sits on the critical path of every cell plant under construction outside China, India's included. Chapter 04 has the concentration table, the instruments in full and the live expiry question; Chapter 03 and Table A1 have the 2025 production and export totals.

Europe: the wave, and the 2023 to 2026 collapse

Europe's wave began with two battery IPCEIs, approved 9 December 2019 and 26 January 2021; at the peak, announced 2030 capacity exceeded 2 TWh a year against actual 2023 output of about 100 GWh. [119,118] Fraunhofer ISI's February 2025 risk model, on a December 2024 data cut, put realisation at 54 to 75 per cent of

announcements and counted over 700 GWh delayed or cancelled in Europe, roughly 240 GWh of it in Germany alone HARD; it is roughly 22 months stale here and biased optimistic, failures having kept coming after its cut-off. [138]

Then came the failures, company by company in Chapter 05, dated in Table A2: nine Western companies between January 2023 and October 2026, six of them European cell makers from Britishvolt to Leclanche's German units. [186, 187, 188, 189, 190, 148, 154, 157, 158, 165, 151] The disclosed floor across Northvolt, Li-Cycle, Morrow and Varta's bank write-off alone is over ₹1.67 lakh crore (USD 17bn); Britishvolt's and CustomCells' sunk figures are NOT FOUND, so the true total is higher.

Transport and Environment's 1 October 2026 assessment cuts both ways: European capacity could still reach 1.7 TWh by 2030, led by Germany, Hungary, Spain, France and the United Kingdom, but roughly 25 per cent of what was announced since 2022 is cancelled or on hold and Germany's 2035 pipeline has halved SOFT, while Benchmark counts 220 GWh of the 2030 European pipeline already cancelled or postponed SOFT. [113, 114, 39] Those three destruction measures must not be added: Fraunhofer counts absolute gigawatt-hours against a 2030 announcement set, Transport and Environment a share of announcements made since 2022, Benchmark only the 2030 pipeline, and the same projects recur across all three. After the Fraunhofer cut-off came ACC's Kaiserslautern and Termoli cancellations and the Morrow, Varta and Leclanche German insolvencies, February to September 2026.

Ownership tells the same story. Foreign control is measured six ways, five observations and one forecast, spanning 56 to 98 per cent because they measure different things, so they must not be averaged; the 56 is lowest because it is EU-only, excludes the United Kingdom, Norway and Switzerland, and was a forward expectation rather than an observation.

MEASURE OF FOREIGN CONTROL	VALUE	SCOPE AND VINTAGE	TAG
Asian manufacturers' control of European cell production capacity	98%	Europe, capacity, 2025	SOFT 55, 113
Non-European-owned share of active capacity	over 85%	Europe, active capacity, October 2026	SOFT 113
South Korean share of European production	72%	Europe, production, October 2026	SOFT 113
Korean share of European production capacity	approx three quarters	Europe, capacity, January 2025	HARD 118
Non-EU subsidiaries' share of EU production capacity	56%	EU only, expectation for 2025, ECA Special Report 15/2023	HARD 118, 120
Chinese-owned share of European cell production	approx 33%	Europe, forecast, 2030	SOFT 39

Chinese capacity is also arriving physically. CATL's wholly owned Debrecen plant began trial cell production on 22 September 2026, phase one at 34 GWh against a planned 100 GWh and roughly 9,000 employees; Stellantis and CATL have broken ground at Zaragoza on a fifty-fifty 50 GWh lithium iron phosphate plant costing ₹43,796 crore (EUR 4.1bn), targeted to start by end-2026; BYD targets mass production at Szeged in the fourth quarter of 2026, capacity NOT FOUND; and Gotion, on 80 per cent against InoBat Auto's 20, is building a 20 GWh first phase on 65 hectares at Surany in Slovakia, output primarily for export into the EU. [183, 184, 185, 164]

Every survivor has a captive carmaker, ACC with Stellantis, Verkor with Renault and its Dunkirk plant inaugurated at 16 GWh in December 2025, PowerCo with Volkswagen, and the consolidator is American: Lyten acquired Northvolt Sweden on 27 February 2026, signed an exclusive memorandum on the Northvolt Drei assets near Heide in Germany on 1 July 2026 and a letter of intent for Morrow in August 2026. Every independent merchant-market European cell startup of this period failed. [163, 161, 162]

2024 to 2026: slowdown, recovery, storage pivot

Germany switched off the Umweltbonus in December 2023, after a constitutional ruling on the Climate and Transformation Fund, and German BEV sales fell 27.4 per cent in 2024 to 380,609, roughly 13.5 per cent of its market; the EU managed 13.6 per cent that year, share rising in only 14 of 27 member states. Then recovery: 17.4 per cent on 1,880,370 BEVs in 2025 with Germany up 43.2 per cent, 20.7 per cent in the first half of 2026, and 21.7 per cent on 1,641,333 BEVs for January to August 2026 against 15.8 per cent a year earlier, all ACEA primary data HARD. No ACEA release for the nine months to September 2026 existed at the cut-off, so that share and volume are NOT FOUND. [76,72,73,74] The Commission's 16 December 2025 proposal to soften the 2035 car target, in Chapter 12, is the largest downside driver in any European cell-demand scenario SOFT. [75]

Composition moved too. Global EV sales passed 20 million in 2025, roughly a quarter of all new cars. [33] Deployed demand grew 29 per cent to 1.59 TWh, the EV share of it fell from 79 per cent in 2023 to 77 in 2024 and 75 in 2025 even as EV volumes grew 26 per cent, and shipments of roughly 1.897 TWh, up 31.7 per cent, leave a gap of roughly 300 GWh that is evidence of inventory build rather than a figure to average. [36,40] Storage grew 51 per cent in every region: global additions of 112 GW and 307 GWh in 2025 against a 2026 forecast of 158 GW and 459 GWh, China alone 61.1 GW and 173.1 GWh, the EU-27 a record 27.1 GWh on a cumulative 77.3 GWh against a roughly 750 GWh cumulative 2030 requirement. [52,53] Europe is the exception: its EV share of its own lithium-ion demand rose to 85 per cent from 77 while its storage share fell to eight per cent from 16, as China diversified to 19 per cent storage and North America to 26. [36] It is the world's most EV-dependent battery market and its storage boom has not arrived. [53] India is the mirror image, a compound 46.7 per cent a year storage requirement on the Central Electricity Authority basis, in Chapter 14. [206]

Chapter 02

The experience curve, and what it forecloses

Lithium-ion cost decline has behaved as an experience curve for three decades, so the 2030 price is less a forecast than an arithmetic consequence, landing below where a 2026 entrant with no cumulative volume can profitably sell a commodity cell.

The learning rate

Published rates differ on scope, not substance.[56, 57]

SOURCE	RATE	SCOPE	TAG
Ziegler and Trancik, *Energy and Environmental Science*, March 2021	24%	cylindrical cells, on price, 1992 to 2016	HARD
same	20%	all cell shapes	HARD
same	24 to 31%	cylindrical, energy density folded into service delivered	HARD
follow-on literature	16 plus or minus 3%	industry-wide	SOFT
recent secondary analyses	near 19%		SOFT
Nykvist and Nilsson, *Nature Climate Change*, 2015	6 to 9%	market leaders, secondary characterisation only, full text not read	SOFT

This report runs 19 per cent central, 24 fast, 13 slow. The 19 is not a mid-range pick: it independently reproduces BNEF's separately published 2030 forecast, the only non-circular test available. The 24 is the cylindrical rate, the scope for a 21700; the 13 is materials cost, not manufacturing learning, becoming binding, as in 2022.

The arithmetic, reproducible

The denominator is cumulative production, for which no published figure could be found, so it is constructed here and is an estimate: approximately 250 GWh cumulative pre-2016 (a roughly 20 GWh market in 2010 growing approximately 26 per cent a year EST); 70, 100, 150, 220, 330 and 440 GWh for 2016 to 2021 EST; 621 GWh for 2022 (950 divided by 1.53, from BNEF's "2023 demand plus 53 per cent" EST); **950 GWh for 2023** (BNEF, stated HARD); 1,233 GWh for 2024 (1,590 divided by 1.29, from Benchmark's "2025 plus 29 per cent" EST); **1,590 GWh for 2025** (Benchmark, stated HARD). The 2016 to 2025 sum is 5,704 GWh, which with the pre-2016 stock gives approximately **5,950 GWh, call it 6.0 TWh**, through end-2025 EST. This is not Benchmark's widely quoted "6 TWh pipeline", which is annual nameplate capacity.

The formula is `Price(t) = Price(2025) x (1 – LR) ^ log2( Cum(t) / Cum(2025) )`, with Price(2025) at ₹10,286 per kWh pack (USD 108), in real 2025 dollars converted at the fixed rate. On a base path of 2,000 GWh in 2026 rising to 3,800 GWh in 2030 and 6,300 GWh in 2035, additions of 14,500 GWh to 2030 give Cum(2030) of 20,500 GWh, a ratio of 3.417 and **1.774 doublings**; additions of 26,500 GWh to 2035 give 47,000 GWh, a ratio of 7.833 and **2.969 doublings**.

RATE	FACTOR 2030	2030 USD	2030 ₹	2035 USD	2035 ₹
13% slow, materials bound	0.781	84	8,000	71	6,762
19% central	0.688	**74**	**7,048**	**58**	**5,524**
24% cylindrical	0.615	66	6,286	48	4,571

All money in rupees, converted at a fixed rate stated in the method note.

₹7,048 per kWh in 2030

CENTRAL-CASE PACK PRICE, 19 PER CENT LEARNING RATE

USD 74. BNEF's independent 2030 forecast is ₹7,619 (USD 80): two unrelated methods, eight per cent apart.

BNEF forecast ₹7,619 per kWh (USD 80) for 2030 in 2023 dollars.[46] The extrapolation gives ₹7,048 (USD 74): a volume curve fit against a bottom-up model, eight per cent apart. A 2030 pack floor of **₹6,667 to ₹8,095 per kWh (USD 70 to 85)** is as robust as anything in this report, and the risk points cheaper, BNEF's 2023 forward view having undershot both years since. At 72 per cent cell-to-pack the 2030 central cell is about **₹5,048 (USD 53)**.

ITEM	USD/KWH	₹/KWH	NOTE
BNEF 2023 forward view for 2024	133	12,667	outcome 115, ₹10,953
BNEF 2023 forward view for 2025	113	10,762	outcome 108, ₹10,286; undershot two years running
2025 average cell	78 to 84	7,429 to 8,000	
Cell, lowest observed LFP, 2025	36	3,429	HARD; approx USD 25, ₹2,381 by 2030
2030 central cell	approx 53	approx 5,048	range USD 46 to 63, ₹4,381 to ₹6,000
2035 central cell	near 42	near 4,000	
Cell-to-pack ratio			82% published for 2021, 70% for 2023 BEV, 72% used here, the lower bound

Who captured the savings

A 93 per cent real decline is a transfer, and not to cell manufacturers. Buyers took the whole decline: across 39 reporting importers the implied unit price of lithium-ion imports fell 13.7 per cent, from ₹31.78 lakh a tonne (USD 33,364) in 2024 to ₹27.40 lakh (USD 28,785) in 2025, as volume rose 14.1 per cent from 2,803.29 kt to 3,197.54 kt and value slipped 1.6 per cent from ₹8.91 lakh crore (USD 93.53bn) to a preliminary ₹8.77 lakh crore (USD 92.04bn), on UN Comtrade via GTAIC SOFT with the unit prices computed here EST.[81] Processing-chain owners took the thick part of the value pool, the one scale leader volume and utilisation together, captive-customer cell makers survival, independent European merchant cell startups nothing. Europe as a jurisdiction took demand rather than supply, 98 per cent of its 2025 cell capacity controlled by Asian manufacturers, and paid twice: state aid in, private money destroyed, the batteries imported anyway.[55, 113] Table A3 has it party by party: value-pool shares, CATL's figures, the audited public money including both IPCEIs, the import values and Deloitte's loss estimates.[54, 42, 43, 119, 120, 191, 84]

THE INDIAN READ ON THE SAME CURVE

India captured the decline as a buyer: imports under HS 850760 reached ₹38,892 crore in calendar 2025 (USD 4,083.60m), volume up 75.12 per cent against value up 43.85 per cent, China supplying 93.26 per cent of the narrow HS 850760 line and 84 per cent of all lithium-ion SOFT. The unit price fell 17.86 per cent to ₹43.00 lakh a tonne (USD 45,150), and in the rolling year to April 2026 fell a further 17.68 per cent while volume rose 90.90 per cent and value reached roughly ₹44,000 crore to ₹47,000 crore (USD 4.63bn to 4.93bn) across three commercial estimates; Chapter 14 has the twelvefold growth since FY2018-19. [82, 206, 246] The one Indian firm that met the falling floor with a differentiated chemistry, Log9 Materials, is insolvent: roughly ₹150 crore invested in manufacturing, ₹571 crore (USD 60m) raised, admitted to the NCLT after defaulting on ₹6.7 crore, its lithium-titanate bet beaten by cheap Chinese LFP SOFT. [209] Chemistry bets in India have a body count.

What the curve forecloses for a 2026 entrant

The denominator is the world's cumulative production, not the entrant's: a new 1 GWh plant adds about 0.017 per cent to the 6.0 TWh stock in its first year, and to move one doubling on its own output it would have to make roughly 6 TWh, nearly four years of world production. The next two doublings come from incumbents: the entrant meets the decline as a price to match, not a cost reduction earned. Three 2030 numbers close the door, all EST at 72 per cent cell-to-pack: a commodity Asian cell at USD 50 to 60 per kWh (₹4,762 to ₹5,714); a European-made cell at about USD 74 (₹7,048), on a premium that has to narrow from 56 to 40 per cent to hold; the lowest-observed LFP benchmark at about USD 25 (₹2,381). German industrial power compounds it, at roughly 2.3 to 3.3 times China's and 2.6 to 3.9 times India's on the same bands in the second half of 2025 EST. [131]

JURISDICTION AND BAND	AS PUBLISHED	₹/KWH	TAG
Germany, Eurostat band IC, 500 to 2,000 MWh a year, H2 2025	22.64 EUR cents	**₹24.18**	HARD [131]
EU average, same band, H2 2025	18.37 EUR cents	₹19.62	HARD [131]
Germany, very large user, 70,000 to 149,999 MWh a year, H2 2025	14.40 EUR cents	₹15.38	SOFT [135]
Germany, eligible firm under the new industrial power price, 2026	approx 20.8 EUR cents blended	approx ₹22.22	EST derived
China, industry at 35 kV and above, Chengdu to Changsha, Sept 2026	CNY 0.57 to 0.81, approx 6.9 to 9.8 EUR cents	**₹7.4 to ₹10.5**	SOFT [147]
India, high-tension industrial, 2026	INR 6 to 9	**₹6 to ₹9**	SOFT state tariff orders

What is not foreclosed is everything that does not price per kWh. Qualification-locked niches, power tools, drones, medical devices, space, marine, critical infrastructure, buy on watt-hours per kilogram, cycle life on their duty profile, traceability and audited supply security, and pay multiples of the commodity price. Content rules do the same. The EU's SAFE instrument, live now, requires at least 65 per cent EU, EEA EFTA or Ukraine content by estimated cost of the end product across up to ₹16.02 lakh crore (EUR 150bn), [124] and the United States investment tax credit requires a storage project starting construction in 2026 to keep 55 per cent of direct costs with non-prohibited foreign entities, rising to 75 per cent after 31 December 2029, failure losing the entire 30 to 40 per cent credit. [129] These are eligibility gates, not price signals. Two openings follow from structure: coated electrode sits just upstream of cell assembly, sellable into everybody else's line, and unpatented process capability has to be bought as people and process, not licences.

THE NICHE IS A CLOCK, NOT A MOAT, AND THE LABELS INVERT

The experience curve is not an entrant's enemy; it is the timer on each premium niche. Every doubling of world cumulative output lowers the commodity price a niche buyer benchmarks against, so a 3x to 10x premium erodes from below. The bull and bear labels flip for a cell maker, fast cost decline being good for the world and bad for a specialist's pricing, so check which way a scenario table's labels point. The question is not whether a specialist beats the scale leader on cost, but how many years each niche has before the floor reaches it, and that differs for power tools and satellites.

Chapter 03

The industry as it stands in 2026

Lithium-ion ended 2025 as a 1.59 TWh industry worth roughly ₹16.19 lakh crore (USD 170bn) at pack level, up 29 per cent, with over 4 TWh of nameplate running at about 40 per cent while its largest producer ran at 96.9 per cent. That gap is the fact to price.

Demand in 2025, measured rather than modelled

Benchmark Mineral Intelligence measures demand deployed, used throughout; SNE Research measures shipments, including channel inventory. Totals for 2020 to 2022 could not be sourced and are left empty rather than back-cast; the 2024 total of 1,233 GWh is this report's arithmetic, 1,590 over Benchmark's 29 per cent, not published.

SEGMENT, 2025	GWH	SHARE	GROWTH
Electric vehicles	approx **1,192**	75%	plus 26%, IEA 1,200
Stationary storage, other industrial	approx **318**	20%	plus 51%, residual
Consumer and portable	approx **80**	5%	not stated
Total deployed	**1,590**	100%	plus 29%, Benchmark
Shipments, a different metric	approx 1,897		plus 31.7%, SNE

EV share was 79 per cent in 2023 on a 950 GWh total and 77 per cent in 2024, composition moving as volume compounds; the EV leg ran approximately 170 GWh in 2020, 340 in 2021 and 550 in 2022. Benchmark's EV figures of 750 GWh for 2023, 950 for 2024 and 1,192 for 2025 sit against the IEA's over 750, approximately 950 and 1,200 and SNE's 1,187 for 2025: agreement within one per cent in every year available, and the only validation obtainable. Rho Motion measured 2025 grid plus behind-the-meter storage at approximately 315 GWh, up roughly 50 per cent, about 240 GWh of it grid scale, filling the 318 GWh residual above, so any separately named industrial category is very small or already sits inside the storage and consumer buckets.[36, 32, 35, 40, 51]

The roughly 300 GWh gap between deployed and shipped is inventory, not a conflict to average away: first evidence the industry built faster than it sold. Growth has stepped down: EV battery usage was 608.5 GWh in the first half of 2026, up 20.0 per cent, and 844.2 GWh in the eight months to August, up 19.7 per cent, a 29 to 31 per cent industry now growing at about 20 per cent on its largest segment.[41, 42]

The market value question, and why nine vendors disagree by 4.1 times

Published 2025 estimates run USD 47.53bn to USD 194.66bn. All nine are vendor models `EST`; none publishes its cell versus pack boundary in the free abstract and the methodology sections are paywalled, so every scope below is inferred and labelled as inference.

VENDOR ESTIMATE, USD BN	INFERRED SCOPE	CLUSTER
Expert Market Research 47.53; IMARC 60.00; Future Market Insights 87.10	cells only or partial application sets	sub-segmented
Precedence 115.89; Fortune Business Insights 134.08; Polaris 136.39; Straits 137.40	cells, or a cell and pack blend	**cell level**
Grand View 151.00; MarketsandMarkets 194.66	packs, or packs plus systems	**pack level**

Scope resolves the 4.1 times spread: USD 47bn to 87bn is too small to be the global market at any sensible price, so sub-segmented, USD 116bn to 137bn approximately cell-level, USD 151bn to 195bn pack-level. Averaging objects of different scope describes nothing.

Rebuilt from two named, dated inputs: 1,590 GWh deployed at USD 79 per kWh gives approximately USD 126bn (₹12.00 lakh crore) at cell level, range USD 119bn to 161bn (₹11.33 to ₹15.33 lakh crore) on 1,590 to 1,897 GWh at USD 75 to 85; at BNEF's USD 108 per kWh volume weighted December 2025 pack price, USD 171.7bn, approximately USD 172bn (₹16.38 lakh crore), range USD 159bn to 209bn (₹15.14 to ₹19.91 lakh crore).[36, 44]

₹16.19 lakh crore global market value, 2025

LITHIUM-ION AT PACK LEVEL, EQUAL TO USD 170BN

Derived, not borrowed: 1.59 TWh deployed times USD 108 per kWh. Cell level roughly ₹12.00 lakh crore (USD 126bn). Nine vendor estimates for the year span 4.1 times.

Recommended headline, basis on its face: the pack and cell figures above, rounded, estimates from hard inputs on 1.59 TWh deployed. At 4.2 TWh and roughly USD 85 per kWh, 2030 pack level reaches approximately ₹33.81 lakh crore (USD 355bn): volume a vendor forecast, price an extrapolation here.[58]

Capacity by region: 2024, 2025, and what is actually committed for 2030

BASIS	GLOBAL	CHINA	EUROPE	UNITED STATES
2024 installed	**3.3 TWh**	approx **85%**	small, plus 10% YoY	approx 200 GWh, plus 50%
2025 installed	**over 4 TWh**, plus 30%	**over 80%**	**6 to 7%**	**6 to 7%**
2030, committed	approx **6.5 TWh**	approx **65 to 67%**		
2030, announced	**9 to over 10 TWh**	67 to 68% sited there, **77% Chinese-owned**	approx 1.2 TWh	approx 1.3 TWh
2030, regional shares		Asia-Pacific **70%**, from 78.4%	**13.7%**, from 7.8%	**14.7%**, from 6% in 2022

India and Indonesia opened first plants in 2024, 15 GWh combined; the EU and United States each grew installed capacity roughly 50 per cent in 2025 HARD. Announced rows and shares are SOFT.[35, 32, 37, 38] Committed and announced must not be blended: 6.5 TWh against 9 to 10 TWh is whether a second gigafactory wave happens. Over 240 gigafactories are operational today, over 400 projected by 2030 SOFT, but the IEA is blunter than that pipeline: American output under 350 GWh in 2030 and just over 400 GWh in 2035, though American committed capacity could make nearly twice projected 2035 American demand. India has announced approximately 290 GWh over 30-plus sites by 2030 against 1.4 GWh commissioned.[38, 34, 208, 194]

The utilisation paradox, explained properly

Over 4 TWh of nameplate against 1.59 TWh of deployed demand implies sector utilisation near 40 per cent, against roughly 30 per cent in 2024 (3.3 TWh, 1.0 to 1.2 TWh of demand); both derivations here EST. CATL ran 2025 at 96.9 per cent, up from 76.3 per cent.[54] Not a contradiction: output is concentrated, nameplate dispersed, so the residual sits idle. CATL and BYD took 54.6 per cent of global EV battery installations in the

eight months to August 2026, 39.4 and 15.1 per cent, and Chinese sector production rose from 1,096.8 GWh in 2024 to 1,755.6 GWh in 2025.[42,54] A tier one on 45 per cent of nameplate, 1,800 GWh, at 80 per cent makes about 1,440 GWh, leaving the other 2,200 GWh to make about 457 GWh, near 21 per cent: an assumed share, direction forced by the concentration data EST.

Three reasons a nameplate gigawatt-hour is not fungible. **Qualification, not steel, binds:** a line qualified for one customer's pack, chemistry and format needs re-qualification for another, years in aerospace, medical and defence. **Format is not interchangeable:** a prismatic LFP line cannot make a cylindrical 21700, and the cylindrical centre of gravity moves from 18650 to 21700 to the 46-series. **Ownership is not location:** Asian manufacturers held 98 per cent of Europe's 2025 cell capacity HARD, over 85 per cent of active capacity is non-European-owned, South Koreans alone make 72 per cent of European production, and Benchmark expects roughly 33 per cent of European-produced cells from Chinese-owned operations by 2030.[55,113,39]

> **WHAT 40 PER CENT AND 96.9 PER CENT MEAN TOGETHER**
>
> Idle capacity sits in tier two and tier three, overwhelmingly in China; qualified, high-specification, non-Chinese capacity is tight. Opposite conditions in one industry produce opposite prices: worsening pressure in commodity cells, scarcity pricing in qualified specialty cells. Any buyer needing qualified or non-Chinese supply cannot get cheap capacity on demand: the gap a specialist sells into, and where an Indian cell programme will stand.[32,36,54]

Price levels in 2026, by segment, chemistry and region

Annex Table A4 carries the full BNEF December 2025 price set by segment, chemistry and region, the volume weighted industry reference.[44,49] Stationary storage packs became the cheapest lithium-ion category for the first time, USD 70 per kWh (₹6,667), down 45 per cent. Europe prices about 56 per cent above China and widening. A European Parliament cross-check on January 2025 data puts the European pack at ₹17,412 per kWh (EUR 163) against China's ₹12,391 (EUR 116), Europe 40 per cent higher, with cathode active material approximately 50 per cent of cell manufacturing cost and anode active material approximately 10 per cent HARD; a different vintage and scope from BNEF, so take the level from BNEF and the gap from these.[118] That premium is the disadvantage every European cell maker carries, and why local-content policy exists. BNEF and the IEA frame the LFP discount differently, about 37 per cent cheaper than NMC at pack level against "more than 40 per cent cheaper on average" per kWh, not blended.[32]

> **DO NOT DRAW A CELL PRICE TREND LINE THROUGH 2024 AND 2025**
>
> The 2024 cell figure of USD 78 per kWh (₹7,429) is an all-segment cell price, published with a pack-only cost of USD 37 (₹3,524): the cell is 68 per cent of pack cost. The 2025 figure of USD 79 per kWh (₹7,524) is an EV cell average. Different series, so "cell prices rose in 2025" is not supportable; use the 2024 split only as the illustration that the cell is roughly 68 to 78 per cent of pack cost.[50,44]

At USD 79 per kWh a 21.6 Wh cylindrical 21700 cell is approximately ₹163 (USD 1.71) EST. Retail listings run far above that, but retail is not contract pricing and is low trust. Bulk contract pricing for premium high-drain 21700s from Molicel, Samsung or EVE is estimated at ₹286 to ₹571 a cell (USD 3 to 6), roughly ₹13,334 to ₹26,667 per kWh (USD 140 to 280), two to three and a half times the commodity floor EST: the band a specialty supplier must clear.

Chemistry as it stands, and the formats that bound a market

Annex Table A5 gives the 2025 chemistry and format positions row by row with confidence tags: an NCA share that is **NOT FOUND**, 8.0 GWh of semi-solid shipments, the 46-series ramp, the 37 to 75 per cent range of LFP 2030 forecasts.[32,43,36,67] LFP has won the volume market, over 55 per cent of global EV batteries deployed and 81.2 per cent of Chinese installations, while NMC grew absolutely and lost share. Format bounds the market as

tightly: the 21700 is right for power tools, drones, medical devices and light electric mobility, wrong for automotive volume, which cell-to-pack and cell-to-chassis architectures reinforce by favouring large prismatic and blade formats.

Sodium-ion: commercial, Chinese, and already carrying a body count

ACTOR	POSITION, 2026
CATL, Naxtra	mass production from December 2025 at GWh scale, to 175 Wh per kg; first sodium-ion cell to pass China's GB 38031-2025 safety standard; target 60 GWh a year by end-2026; 60 GWh HyperStrong supply agreement SOFT, HARD on the standard
Changan Nevo A06	first mass-market sodium-ion car, launched February 2026
Natron Energy, United States	**ceased all operations effective 3 September 2025**; ₹13,334 crore (USD 1.4bn) Rocky Mount plant cancelled; two sites closed; 95 staff terminated; assets to Sherwood Partners
Altris, Sweden, with Draslovka, Czechia	approximately ₹206 crore (EUR 19.3m) in-kind investment; production late Q3 or early Q4 2026
Share of the total battery market	single digits until 2030
Discarded figure	30% of stationary storage by 2030: circulates widely, appears alongside shares summing to 113%; not used

Chapter 11 carries the fuller field and the S-curve read.[63, 64, 66, 65, 182, 181]

> **THE SODIUM-ION POSITION IN ONE PARAGRAPH**
>
> The technology works and is certified; the business case is separate. Sodium-ion's cost argument assumed LFP would stay expensive, and it did not: LFP packs at ₹7,714 per kWh (USD 81), the lowest observed LFP cell at ₹3,429 (USD 36). Natron, the Western Prussian-blue champion, was liquidated having failed on funding before scale. Any proposition that does not name Natron and concede CATL's roughly 100 times scale lead is incomplete. The surviving case for a European or Indian entrant is specialty: cold climate, high cycle count, safety-critical, and where Chinese sourcing is barred.

What the 2026 snapshot establishes

Demand is large, compounding at roughly 20 per cent on its biggest segment, composition shifting from cars toward grids everywhere except Europe. Capacity is abundant in aggregate, scarce where qualified, priced at record lows. Chemistry is settled in the volume market, where LFP won inside a Chinese supply chain, and unsettled at the frontier, where sodium-ion is already Chinese and commercial. Nobody entering in 2026 makes money on cost per kWh: the defensible positions are those 40 per cent utilisation conceals, qualified supply, specified origin, formats the tier-one lines cannot produce.

Chapter 04

Concentration, trade and the export-control regime

China's grip sits at the refinery, the active-material plant and the machine shop, not the mine. It tightened through the Western diversification effort: refining concentration rose from roughly 82 per cent in 2020 to 86 per cent in 2024. The instruments that turn it into leverage are dated, and one expires within weeks.

The chokepoint is processing, not mining

Mineral endowment is distributed: Australia and Chile for lithium, Indonesia nickel, the DRC cobalt, graphite in Africa, Scandinavia and India. Dependence sits downstream, where ore becomes battery-grade material and material an electrode.

STEP IN THE CHAIN	CHINA'S SHARE	PERIOD	TAG
Graphitization	**98%**	2024	HARD
Battery-grade graphite refining	**94%**	2024	HARD
Anode active material	**over 90%**	2025	HARD
Cathode active material	approx **85%**, up to 90% by chemistry	2025	HARD
Cathode active material, LFP only	approx **95%**	2026	SOFT
Cobalt refining / lithium refining	**76% / 74%**	2024	HARD
Copper refining / nickel refining	47% / 31%	2024	HARD
Cell production, and nameplate capacity	**over 80%** each	2025	HARD

Source: IEA Global EV Outlook 2026, 2025 edition for capacity history, industry estimate for LFP cathode.[32,35,113] Where the West had a base, copper and nickel refining built for non-battery reasons, it holds half to two thirds; where the step is lithium-ion-specific, China holds 85 to 98 per cent. Graphite, among the cheapest cell materials, runs roughly ₹2.86 lakh to ₹5.71 lakh per tonne (USD 3,000 to 6,000) against ₹7.62 lakh to ₹11.43 lakh at the 2022 peak EST. The risk is not price. It is permission.

86% of global refining capacity

CHINA'S SHARE OF BATTERY-MATERIAL REFINING, 2024

Up from roughly 82 per cent in 2020, after four years of Western diversification policy.

The Inflation Reduction Act, the Critical Raw Materials Act, the Net-Zero Industry Act and Japanese and Korean supply-chain programmes all took effect between 2020 and 2024; refining concentration rose about four percentage points.[32] The IEA's 2035 projection, on announced policy, leaves China above 60 per cent of refined lithium and cobalt and roughly 80 per cent of battery-grade graphite, American cathode active material at half and anode a quarter of 2035 demand.[32,34] The dependence outlives a plant commissioned in 2027; what matters is where a programme sits relative to the chokepoint, and whether it is priced.

What the trade data can and cannot show

No fully reported 2025 ranking exists for HS 850760, the lithium-ion accumulator line: 2025 UN Comtrade filing is so incomplete that one aggregator's 2025 world total falls below its own 2024 total. Three labelled layers are used, not blended: 2023, the last fully reported year; China's 2025 customs aggregate via the China Passenger Car Association; and a rolling twelve months to mid-2026 from mirror data, 40 economies.

On the 2023 exporter ranking China held **55.97 per cent** of a ₹11.05 lakh crore (USD 116bn) world total across 116 reporters, then Poland 10.20 per cent, Hungary 8.90, Germany 5.35 and the Czech Republic 2.95; ranks six to ten are **NOT FOUND** in the accessible view.[79, 81] Ranks two, three and five are the assembly floor of Asian-owned European capacity: Europe's export strength is largely Korean and Chinese manufacturing inside the customs union. The full rankings, with the paywalled and NOT FOUND items, are Table A6.

China's own customs aggregate puts 2025 exports up **26 per cent** to ₹7.31 lakh crore (USD 76.8bn) from ₹5.80 lakh crore (USD 60.9bn), the EU taking **40 per cent** of that at ₹2.95 lakh crore (USD 31bn) on a share up 4 points, the United States down **22 per cent** by value to ₹1.13 lakh crore (USD 11.9bn), the average unit price down 18 per cent to ₹14.29 lakh per tonne (USD 15,000), and December steeper than the annual figure SOFT.[83] Mirror data to mid-2026 lifts China's share of world trade to **66.10 per cent by value** from 61.05 and **78.18 per cent by tonnage** from 75.42 SOFT.[80] The quarterly detail, rolling-year levels and destinations sit in Table A1.

> **A LABEL ROUTINELY MIS-READ**
>
> The 66.10 and 78.18 per cent figures are China's share of world lithium-ion accumulator **exports**, through partner import statistics, not of world imports; the aggregator's phrasing invites that misreading. China is the largest exporter of cells, Germany the largest importer. Reversing the two turns the supplier of the world's battery output into its absorber.[80]

On the **world import ranking** Germany led the United States in 2023 and still does, at roughly 22 to 25 per cent: no recent overtaking.[79, 80, 85] The dated overtaking is on **China's destination list**, where Germany became China's largest lithium-ion export market from **May 2025**, displacing the United States, on ₹97,764 crore (USD 10.265bn) in the first nine months of 2025, up 29.87 per cent and 18.5 per cent of China's exports SOFT.[83] Only the second carries a date.

One tension: China's exports to Germany, ₹2.70 lakh crore (USD 28,319.15m), almost equal Germany's reported imports from all origins, ₹2.69 lakh crore (USD 28,210.16m), on a different period cut. Read literally that makes every German import under this code Chinese, not credible when Poland and Hungary alone exported USD 22.1bn in 2023. So the Chinese customs aggregate is used for **rank and direction**, the 2023 Comtrade ranking for **level** EST. EU imports of Chinese lithium-ion batteries reached ₹1.75 lakh crore (EUR 16.4bn) in the first half of 2026, up **39 per cent**, against a 2024 baseline of roughly ₹2.99 lakh crore (EUR 28bn) of battery imports in total, ₹2.35 lakh crore (EUR 22bn) of it Chinese; the EU Trade Commissioner's line was that the status quo "is not an option".[84, 191]

The American substitution

The United States pushed Chinese cells out, and it made almost no difference to China. Its share of American accumulator imports fell **19.01 percentage points, 65.11 to 46.10 per cent**; exports to the United States fell ₹78,525 crore (USD 8,244.92m) while exports elsewhere rose ₹1.91 lakh crore (USD 20,024.62m), a net ₹1.12 lakh crore (USD 11,779.70m) gain and 1,278,051.98 more tonnes outside America SOFT.[80] The redirected volume was roughly **2.4 times the volume lost** EST; Germany, Australia, India, Vietnam and the Netherlands absorbed it, India at **plus 73.26 per cent** in twelve months.

Tariffs plus content rules work at national level: a 19 percentage point share shift in one year is a large effect. But it does not reduce global dependence; it relocates it onto markets without content rules, and India receives that by arithmetic, not choice.

Across the 39 reporting importers, 2025 preliminary value fell 1.6 per cent, ₹8.91 lakh crore to ₹8.77 lakh crore (USD 93.53bn to 92.04bn), volume rose 14.1 per cent, 2,803.29 kt to 3,197.54 kt, and the implied unit price fell 13.7 per cent, ₹31.78 lakh to ₹27.40 lakh per tonne (USD 33,364 to 28,785) SOFT, computed here EST. [81] Two honesty notes: the source reports volume growth as plus 12.79 per cent against the plus 14.1 per cent its own data points give, unexplained, and the 2025 value decline is partly a reporting artefact of incomplete filings. Neither touches the finding, that volume growth runs fourteen to sixteen points ahead of value growth.

₹27.40 lakh per tonne

IMPLIED IMPORT UNIT PRICE FOR LITHIUM-ION ACCUMULATORS, 2025

USD 28,785, down about 14 per cent in a year: far more battery for slightly less money.

India is the extreme case: calendar 2025 imports under HS 850760 reached ₹38,892 crore (USD 4,083.60m), up **43.85 per cent by value but 75.12 per cent by volume**, unit price down 17.86 per cent to ₹43.00 lakh per tonne (USD 45,150), China at **93.26 per cent** of the narrow code and **84 per cent** all-lithium-ion, consistent because the narrower code carries the higher share. On the rolling year to April 2026, three commercial estimates give roughly ₹44,000 crore to ₹47,000 crore (USD 4.63bn to 4.93bn), volume up 90.90 per cent, unit price down a further 17.68 per cent SOFT; Table A7 has both lines in full. [82, 206, 246] India absorbs the price collapse, it does not escape the dependence.

The policy layer: landed cost

The United States duty stack on HTS 8507.60:

MEASURE, HTS 8507.60	RATE	EFFECTIVE	TAG
MFN base rate	**3.4%**	standing	SOFT
Section 301, lithium-ion EV batteries	**25%**, from 7.5%	**27 Sep 2024**	HARD
Section 301, lithium-ion non-EV batteries	**25%**, from 7.5%	**1 Jan 2026**	HARD
Section 301 forced-labour action (HTS 9903.05.31)	12.5% ad valorem, products of China	**24 Jul 2026**	SOFT
Chinese cell, defensible stacked rate	**approx 28.4%**	2026	EST
EU-origin cell, all-inclusive ceiling	**15%**, no stacking on MFN	**1 Jul 2026**	SOFT

Sources: the USTR Section 301 modifications determination for the 2024 and 2026 staging, secondary tariff references for the MFN base and the July 2026 action. [86, 87, 88] Subheadings the trusted source cites as exempt from the forced-labour duty: HTS 8507.60.00, lithium carbonate, lithium oxides and hydroxides, nickel and cobalt ores, oxides and sulphates. [88] The sources conflict on the forced-labour duty and this report does not resolve it; it trusts the exemption of HTS 8507.60.00. Vendor claims of a stacked 40.9 per cent and of 82 per cent are low trust, unused: the defensible figure is 3.4 plus 25, approximately 28.4 per cent, to check against the Federal Register.

A German cell lands in the United States at roughly **13 percentage points less duty** than a Chinese cell, 15 per cent against approximately 28.4 per cent, before the prohibited-foreign-entity rules EST, widening the preference well beyond the tariff gap: from 2026 at least 55 per cent of a storage project's direct costs must be

non-prohibited, rising to 75 per cent after 2029, the entire 30 to 40 per cent investment tax credit the penalty for failure SOFT. [129, 130] The 15 per cent ceiling excludes aluminium, still at 50 per cent under Section 232, biting if cell housings ship separately.

No EU anti-dumping or anti-subsidy case on lithium-ion rechargeable cells from China is open at 20 October 2026, HARD **by absence**. The measures around it:

EU TRADE DEFENCE MEASURE	STATUS AT 20 OCTOBER 2026	TAG
Lithium-ion rechargeable cells from China, AD or AS case	**NONE OPEN**	HARD by absence
Battery electric vehicles from China, countervailing duties	**7.8% to 35.3%** company-specific, imposed 29 Oct 2024, five-year term	HARD
Price-undertaking route for Chinese BEVs	Commission guidance issued 12 Jan 2026	HARD
Primary alkaline manganese dioxide cells from China, anti-dumping	Opened **2 Jul 2026** on a complaint by VARTA Consumer Batteries. **Not lithium-ion**	SOFT
Other 2026 actions on China	Tyres 4.3% to 45.3%; polyamide yarns 60% to 67.6%; silicon-steel safeguard; PHEVs monitored	SOFT
Foreign Subsidies Regulation	In force since July 2023; remedies include repayment, capacity limits and mandatory R&D sharing	SOFT

Sources: Commission materials on the BEV duties and the price-undertaking guidance, a law-firm note on the primary-cell case, trade reporting for the rest. [89, 90] Citing the primary alkaline case as lithium-ion would be wrong. Brussels is tightening on everything around the cell but not the rechargeable cell itself, even with Chinese imports up 39 per cent in the first half of 2026: the open European policy question for 2027, and the largest potential upside for a European cell maker EST.

CBAM's definitive regime began 1 January 2026: importers must be authorised declarants accounting for embedded emissions, with a 50-tonne annual de minimis exempting roughly 90 per cent of importers while retaining about 99 per cent of emissions. Covered sectors: cement, iron and steel, aluminium, fertilisers, electricity, hydrogen. [121] **Batteries are not in scope. Neither finished cells nor packs are CBAM goods in 2026.** The battery constraint is instead Regulation (EU) 2023/1542, through carbon-footprint declaration, performance classes and eventually maximum thresholds: a dry-coating energy saving is a Battery Regulation argument, not a CBAM argument. The 2028 expansion was proposed by the Commission on **17 December 2025**, roughly 180 downstream products from **1 January 2028**, 94 per cent of them intermediate industrial goods with a steel and aluminium share averaging 79 per cent; the Council's general approach of **12 June 2026** covers about 200 goods, and Parliament voted **464 to 50** in 2026 to expand beyond 450 items with anti-circumvention measures. Whether CN 8507 appears is **NOT FOUND**, not confirmed either way. [122, 123] Batteries are unnamed there, but enclosures, racks, module frames and BESS containers are exactly the assemblies targeted: a cost-of-imported-systems effect, not a cost-of-imported-cells effect EST.

China's export-control regime, every date exact

DATE	INSTRUMENT	EFFECT	TAG
15 Jul 2025	MOFCOM and MOST amendment to the *Catalogue of Technologies Subject to Export Prohibitions and Restrictions*	**Licensing on battery cathode technology exports**: fourth-generation LFP and LMFP preparation, plus lithium processing	SOFT

DATE	INSTRUMENT	EFFECT	TAG
9 Oct 2025	**MOFCOM and GAC Announcement No. 58 of 2025**, effective 8 Nov 2025	Dual-use licensing: cells and packs at or above **300 Wh/kg** under HS 85076000; **artificial graphite anode materials** with equipment and technology; advanced cathode materials; **key equipment**, winding, stacking, filling, pressing, formation; third-generation LFP added	SOFT; thresholds **low confidence**
7 Nov 2025	**MOFCOM and GAC Decision No. 70 of 2025**	**Suspends** Nos. 55, 56, 57 and 58, plus MOFCOM Nos. 61 and 62, effective immediately, "starting from Friday (November 7, 2025) **to November 10, 2026**"	HARD
28 Sep 2026	MOFCOM account of China to United States talks on the Kuala Lumpur arrangement	Suspension **reported** extended from 10 Nov 2026 to 10 Jan 2027	SOFT, **unresolved**

Sources: law-firm and trade notes for the July 2025 amendment and No. 58's scope, thresholds from a low-confidence vendor source; Global Times, Pillsbury and CIRS on Decision No. 70.[91,94,92,93,95]

In force today, 20 October 2026. The July 2025 catalogue licensing on cathode, LFP and lithium-processing **technology** is live and was never suspended: a Chinese company cannot transfer its highest-generation LFP cathode know-how abroad without approval, while older generations remain transferable. The **product-level** controls of No. 58, on cells, graphite anode and equipment, are **suspended, not repealed**. The hard expiry date, carried by a numbered MOFCOM instrument, is **10 November 2026**.[92,93,95] The extension to **10 January 2027** is reported, corroborated across two independent sources in one research lane and carried only by low-trust aggregators in two others, and **no numbered MOFCOM announcement amending Decision No. 70 could be located**. So: 10 November 2026 statutory, the extension reported and not confirmed, a live verification item. No other new Chinese battery export-control instrument in 2026 could be found: **NOT FOUND**.

If No. 58 returns, for a European builder. The clause that bites is equipment, not cells. Coating, calendering, winding and formation lines are overwhelmingly Chinese, on the critical path of every cell plant built outside China. In-house European coating, anode, cathode and formation hedges it, as does a dry process dropping the solvent recovery train.

For an Indian builder. The mirror image: a programme built on imported Chinese lines, Chinese artificial graphite anode and Chinese LFP cathode carries all three exposures at once, the licensing decision sitting with a government that is also the shareholder of the competition. July 2025 already bars licensing the highest-generation LFP cathode know-how into India without Beijing's approval.

> **THE ONE-LINE VERSION FOR THE BOARD**
>
> The scarce thing in 2026 is not lithium. It is cobalt tonnes, Chinese permission to use Chinese graphite, cathode and anode processing, and Chinese equipment. All are policy-controlled, not geological: they change on a published date, not over a mining cycle. The most consequential date here is **10 November 2026**, and the instrument that would extend it has not been found {EST on HARD inputs}.

Chapter 05

Europe's answer, and what it cost

Europe announced more than 2 TWh of cell capacity for 2030, will realise 1.2 to 1.7 TWh of nameplate and 0.8 to 1.1 TWh of output, and has consumed at least ₹1.67 lakh crore (over USD 17bn) of disclosed capital doing it. What works is 98 per cent Asian-controlled: the question is ownership, not tonnage.

What the Fraunhofer model does, and when its data stops

The reference estimate for the announcement-to-reality gap is Fraunhofer ISI's capacity risk model of February 2025, blog 28 April 2025.[138] Each announced project is scored against eleven risk factors, converted into a realisation probability and aggregated into a band. Whether that aggregation is Monte Carlo is NOT FOUND; the shape fits probabilistic aggregation rather than a flat deterministic haircut, but the technique is not asserted. It prices plant-level execution risk, not demand. Its cut-off is December 2024, roughly 22 months stale here, and the staleness runs one way: everything since has removed capacity.

The published figures: announced European capacity for 2030 above 2,000 GWh per year; realisation probability 54 to 75 per cent; implied nameplate approximately 1,200 to 1,700 GWh; realistic output after delays and scrap 800 to 1,100 GWh HARD; implied output-to-nameplate ratio approximately 65 to 67 per cent {EST, derived from the two ranges}; European cell demand 800 to 1,300 GWh per year; Europe at approximately 10 per cent of global production capacity against approximately 25 per cent of global demand HARD.[138]

The Net-Zero Industry Act's indicative battery objective is 90 per cent of EU annual demand produced domestically in 2030, which the European Parliament's research service puts at 550 GWh of capacity.[117, 118] The IEA's 2025 measurement gives Europe 6 to 7 per cent of global nameplate cell capacity against Fraunhofer's approximately 10 per cent of production capacity: different metrics, different vintages, unreconciled.[32]

> **THE MODEL IS 22 MONTHS STALE AND BIASED OPTIMISTIC**
>
> Date-stamp the Fraunhofer numbers. Four events post-date the December 2024 cut-off, none in the model: ACC's cancellation of Kaiserslautern and Termoli (February 2026), Morrow's bankruptcy (6 May 2026), Varta's insolvency filing (24 July 2026), Leclanche's two German subsidiaries at the Amtsgericht Offenburg (23 September 2026). The 54 to 75 per cent band describes a world that no longer exists: a ceiling, not a central case EST.

The gigawatt-hours that were announced and never built

Four independent measures of project-level destruction exist. They do not sum and must not be added: scopes and vintages differ, projects recur. Fraunhofer ISI, on December 2024 data, counts over 700 GWh delayed or cancelled in Europe against its 2030 announcement set, approximately 240 GWh in Germany HARD.[138] Transport and Environment's *Ready to Scale*, 1 October 2026, puts roughly 25 per cent of European cell capacity announced since 2022 as cancelled or on hold, and has Germany's 2035 pipeline halved SOFT.[113, 114] Benchmark Mineral Intelligence, restated in later coverage, counts 220 GWh of the 2030 pipeline cancelled or postponed SOFT.[39] The fourth is the post-cut-off record, February to September 2026: ACC's Kaiserslautern and Termoli cancellations, and Morrow, Varta and Leclanche's German units insolvent HARD / SOFT.

They agree on direction: between a fifth and a third of what Europe announced will not exist, Germany worst hit. The same T&E work records 25 per cent attrition yet has capacity reaching 1.7 TWh by 2030, led by Germany, Hungary, Spain, France and the United Kingdom.[113] The tonnage gap is closing; the ownership gap is not.

Who owns the capacity that works

Six measures of foreign control span 56 to 98 per cent because they measure different things, five observations and one 2030 forecast, never to be averaged:

- **98 per cent**, Asian control of European cell production capacity, 2025 SOFT [55, 113]
- **over 85 per cent**, non-European-owned share of **active** European capacity, October 2026 SOFT [113]
- **72 per cent**, South Korean share of European production, October 2026 SOFT [113]
- **approx three quarters**, Korean share of European production capacity, January 2025 HARD via EPRS [118]
- **56 per cent**, non-EU subsidiaries' share of EU production capacity, expectation for 2025, ECA Special Report 15/2023 HARD via EPRS [118, 120]
- **approx 33 per cent**, Chinese-owned share of European cell production, forecast for 2030 SOFT [39]

The Court of Auditors' 56 per cent is lowest because it is EU-only, excludes the United Kingdom, Norway and Switzerland, and was a 2023 forward expectation, not an observation. This report uses T&E's over 85 per cent for active capacity: most recent, most clearly scoped, the only one separating capacity that runs from capacity announced.[113] The 98 per cent is a 2025 measure of Asian control specifically SOFT, recorded as the fact base has it, not promoted. Those three form a trajectory on widening scope.

On the split cited by Deloitte, materials extraction and processing is 50 to 60 per cent of value added, cell production 15 to 30 per cent.[55] The materials half is Chinese wherever the plant stands: roughly 85 per cent of cathode active material capacity, over 90 per cent of anode, 94 per cent of graphite refining, 98 per cent of graphitization.[32] European owners hold under 15 per cent of active capacity.

2 to 5 per cent of the battery value pool

WHAT EUROPEAN-OWNED CELL MANUFACTURING CAPTURES

Under 15 per cent European ownership of active capacity times cell production at 15 to 30 per cent of value added. An estimate, not a measurement.

Europe got employment, tax base and grid offtake; know-how, procurement, yield curves and profit report to Seoul, Ningde or Shenzhen. Deloitte estimates European battery companies forgo ₹1.12 lakh crore (EUR 10.5bn) of profit over four years, with lost value added possibly reaching ₹10.68 lakh crore to ₹16.02 lakh crore (EUR 100bn to 150bn) by 2030 {EST, vendor}.[55]

The arriving capacity is Chinese. CATL's wholly owned Debrecen plant in Hungary began trial cell production on 22 September 2026, phase one 34 GWh against a planned 100 GWh, roughly 9,000 employees.[183] Stellantis and CATL, fifty-fifty, broke ground at Zaragoza, Spain on 50 GWh of lithium iron phosphate for ₹43,796 crore (EUR 4.1bn), targeted to start by end-2026.[184] BYD, wholly owned, targets mass production at Szeged, Hungary in the fourth quarter of 2026, capacity NOT FOUND.[185] At Surany, Slovakia, Gotion (80 per cent) and InoBat Auto (20 per cent) build a 20 GWh first phase on 65 hectares, output primarily exported into the EU.[164] InoBat, Europe's champion of the low-volume, high-specification proposition European policy says it wants, ended as the 20 per cent minority partner.

The casualty list

Annex Table A2 has dates, stated causes and capital for all twelve. The five the argument rests on:

COMPANY	EVENT AND DATE	CAPITAL CONSUMED OR DESTROYED
Northvolt, Sweden	Chapter 11, 21 November 2024; Swedish bankruptcy 12 March 2025	approx **₹1.43 lakh crore (USD 15bn)** raised[187]
Li-Cycle, Canada	Filed May 2025; Glencore took all assets for ₹415 crore (USD 43.6m)	**₹16,191 crore (USD 1.7bn)** raised, approx 97 per cent destroyed[186]
Morrow Batteries ASA, Norway	Bankruptcy 6 May 2026, weeks after production began	over **₹4,593 crore** (NOK 5.1bn, approx EUR 430m EST), with a ₹562 crore (NOK 550m) Innovation Norway loan[158,159,160]
Varta AG, Germany	StaRUG July 2024; self-administration, Amtsgericht Stuttgart, **24 July 2026**; approx 3,260 employees	**₹3,044 crore (EUR 285m)** written off, 2024 equity wiped out; FY2024 revenue ₹8,471 crore (EUR 793m), net loss ₹689 crore (EUR 64.5m)[154,155,153]
CustomCells, Germany	Insolvent 30 April and 5 May 2025; rescued by ABACON family offices with SALVIA, 1 July 2025	**NOT FOUND**[148,149,150]

Nine Western battery or value-chain companies failed or entered formal distress between January 2023 and October 2026, six of them European cell makers: Britishvolt, Northvolt, CustomCells, Morrow, Varta and Leclanche's German arm. Morrow was the third Nordic failure after Northvolt and Freyr. In the margin, Blue Solutions, Bollore's solid-state subsidiary, planned a ₹23,500 crore (EUR 2.2bn) Mulhouse plant and on 8 September 2026 announced a fourth-generation refocus and a Bluebus shutdown affecting up to 41 positions: contraction, not failure.[168]

₹1.67 lakh crore raised or written off

FOUR WESTERN BATTERY FAILURES, JANUARY 2023 TO OCTOBER 2026

Northvolt, Li-Cycle, Morrow and Varta's bank write-off alone. A conservative floor, not a total.

The total is a floor. The ₹1.67 lakh crore sums the four disclosed figures above. **Britishvolt's and CustomCells' actual sunk capital is NOT FOUND**, as is Cellforce's, Leclanche's and Ascend Elements'; Britishvolt's GBP 4bn was a plan, not a spend, and is excluded. True destruction is higher by an amount this report cannot size, the aggregate is an estimate from hard and soft inputs, and the difference between ₹1.67 lakh crore and USD 17bn at ₹95.24 is the euro legs of the sum, not a data conflict. EST

The stated causes

Eight diagnoses are on the record, as the companies, their owners and the trade press gave them. All are about capital and customers; not one about chemistry or process.[186]

Morrow's 1 GWh Arendal plant was, per trade press, **"far too small to supply a major car manufacturer"**, with no investors for expansion; its chief executive cited early-industrialisation capital needs, higher capital costs, industrialisation delays and a restrained investment market, LFP prices down nearly 50 per cent between 2022 and 2025 SOFT.[158,186] Northvolt was ramp failure, not demand failure, the shortfall triggering BMW's cancellation of an approximately ₹21,364 crore (EUR 2bn) contract in June 2024 SOFT.[187] CustomCells was single-customer concentration: its largest customer, the eVTOL developer Lilium, collapsed owing tens of millions HARD.[148] Varta cited a structural 2027 financing gap, a worsened market, weaker demand, currency effects and the loss of a major button-cell customer, reported as Apple cancelling an AirPods contract in May 2026 SOFT.[154] Porsche said Cellforce no longer had a sufficiently viable long-term perspective, having already withdrawn from the EU's ₹6,836 crore (EUR 640m) five-project funding round in November 2025 SOFT.[151,152] Leclanche called the filing for its German units precautionary, further funding not secured SOFT.[156,157] For the

sector, Emma Nehrenheim of the European Battery Alliance describes a **"valley of death"** between project financing and profitability, needing bridging capital that never arrived in time, and calls European investors insufficiently bold SOFT . [186]

ACC judged the conditions for restarting its paused projects unlikely to be met, and that one repays rereading: Europe's flagship carmaker-owned joint venture cancelled two gigafactories, shifted to lower-cost LFP with CATL in Spain, and in February 2026 publicly called building new battery factories totally irresponsible SOFT . [165, 166] Its own shareholders are evidence that European carmakers will not pay a premium for European cells.

What the survivors have in common

TRAIT	WHAT THE RECORD SHOWS
A carmaker on the share register	ACC (Stellantis), Verkor (Renault) and PowerCo (Volkswagen) all have one. Every independent merchant startup failed [186]
Scale matched to committed offtake	Capex ahead of contract is the universal failure mode: Northvolt built for 16 GWh and shipped 1. Verkor's 16 GWh at Dunkirk is sized to Alpine volume [163]
Price-insulated end markets	All who competed on dollars per kWh against Chinese LFP lost: packs at ₹7,714 per kWh (USD 81) in 2025 against a China average of ₹8,000 (USD 84), prices down roughly 50 per cent between 2022 and 2025. No casualty had this
Process capability that ships, not a chemistry promise	Tesla needed roughly seven years to industrialise dry electrodes; LG Energy Solution targets 2028
A capital structure that survives the valley of death	No casualty crossed the gap between project finance and profitability [186]
A usable regulatory instrument	The EU's ₹16,023 crore (EUR 1.5bn) Battery Booster Facility requires 10 GWh of planned annual capacity, an EV focus, and the applicant's first full commercial-scale EV cell production globally. Likely qualifiers: ACC, Verkor, PowerCo [105, 109]

Europe's answer to a decade of ramp failures was an interest-free loan, up to 60 per cent of eligible costs and at most ₹5,341 crore (EUR 500m) per project, designed around the failure mode seen at Northvolt and Morrow and open only to those who had already solved it: Asian incumbents fail the first-globally test, sub-10 GWh specialists the capacity test.

> THE BASE RATE, AND THE INDIAN READ-ACROSS
>
> Of the independent, non-carmaker-anchored European merchant cell startups that attempted industrial-scale production between roughly 2015 and 2026, this report identifies **none that reached sustained profitability and at least four that failed outright**: Britishvolt, Northvolt, CustomCells and Morrow, with Varta's automotive ambition a fifth by another route and Leclanche's German cell business a sixth. The success rate at gigafactory scale is approximately zero EST .
>
> The transferable finding for an Indian board is not Europe's cost base: the binding constraint is a committed buyer on the share register, not technology, subsidy or announced capacity. India's stack is supply-side. The ACC production-linked incentive has delivered roughly 2.8 per cent of its 50 GWh target, its FY2027 allocation was cut 44 per cent to ₹86 crore while Auto PLI was doubled, and as of a Lok Sabha reply reported 21 July 2026 **no company had claimed a single rupee**. [194, 195, 198] A scheme that pays for capacity reproduces Europe's outcome; one that binds an offtaker does not.

The question this leaves for the company chapters

Two readings exist, and the case for any European specialist turns on which is right. The **geographic** reading: European power costs two to three times Asian power, labour more, permitting slower, the materials chain Chinese at every upstream step, a European pack carrying a 56 per cent premium at the same specification; the failures were inevitable.

The **scale and market** reading: Northvolt built sixteen gigawatt-hours for a commodity market before it could make one, Morrow's single gigawatt-hour was both too large to finance privately and too small to interest a carmaker, CustomCells had one customer, Varta had a 2027 debt maturity in a shrinking consumer market. None of that is about latitude: a one-gigawatt-hour plant selling qualification-locked cells where the criterion is watt-hours per kilogram rather than rupees per kilowatt-hour is a different business, not a smaller Northvolt, and a premium fatal with Volkswagen may suit a defence integrator or a satellite programme.

That reading is coherent, arguably correct and untested at industrial scale. The company that tested the nearest version, CustomCells, was killed by one customer's non-payment, and its founders now run the subject of this report.

Chapter 06

The company: UniverCell Holding GmbH

UniverCell is a real, operating, revenue-earning electrode manufacturer with a cell business attached, and almost every number it publishes describes the electrode business while almost every number its investors bought describes the cell business.

The corporate record

Registered 20 to 22 May 2019 at the Amtsgericht Pinneberg under HRB 14346 PI, seat in **Elmshorn**; moved to Flintbek, near Kiel, on 3 November 2020 with a capital increase and fully revised articles. Electrode foils have been made "since November 2020": production began on relocation. Status aktiv: no insolvency, Schutzschirm or StaRUG filing found; the trademark dates from 6 February 2026; ten active shareholders, names paywalled: NOT FOUND. Full record, and a purpose wider than the marketing: **Table A8**. "Founded in 2019 in Kiel" is unsupported, and managing directors Dr Stefan Alexander Permien and Marius Strack each represent **jointly with another**: neither binds the company alone.[16, 17, 22]

Share capital, and what the Series B actually did

Capital rose in four steps: ₹26.71 lakh (EUR 25,000) in May 2019, ₹2.14 crore (EUR 200,000) on 3 November 2020, ₹2.99 crore (EUR 280,000) on 27 October 2021 by a resolution of 9 September 2021, and ₹4.31 crore (EUR 403,528) on 2 February 2026. Shareholder lists are filed for 22 May 2019, 17 November 2020, 27 October 2021 and then **two separate 2026 dates**, 13 February and 8 April, bracketing the March close. The ₹320.46 crore (EUR 30m) round announced 4 March 2026 lifted nominal capital by only ₹1.32 crore (EUR 123,528), so approximately ₹319.18 crore (EUR 29.88m) of it is share premium, or Agio. EST Nominal capital says nothing about valuation, and Tracxn's "USD 8.8m post-money" is arithmetically impossible against that round: **discard** it. **No UniverCell valuation could be established. NOT FOUND.**[16, 17, 27]

₹320.46 crore Series B, 4 March 2026

THE EQUITY BASE BEHIND THE RAMP

EUR 30m, co-led by DTCF, IKA and WIKA with EIC Fund support, roughly 0.2 per cent of what Northvolt consumed.

Funding and public money

Nine instruments: ₹26.71 lakh of founders' capital, May 2019; undisclosed angel money in 2020 and 2021 from Robert Dahl (Karls Erdbeerhof) and Martin Stolz (Kaufhaus Stolz); a February 2024 seed of approximately ₹44.76 crore (USD 4.7m), investors not named SOFT; the EIC Accelerator grant C2C-CSA, GA 101161461, 1 July 2024 to 31 October 2026, ₹25.50 crore (EUR 2,387,000) of Horizon Europe money on a ₹36.43 crore (EUR 3,410,285) project, **sole beneficiary, no consortium**; the Series B, ₹320.46 crore, 4 March 2026, "with strong support from the EIC Fund"; SIB:DE, March 2026 to February 2029, ₹154.89 crore (EUR 14.5m) of BMFTR money across 25 partners under EDAG Production Solutions, UniverCell's share NOT FOUND SOFT; and PLUS-LIBER, announced 9 October 2026, amount and funder NOT FOUND. **IPCEI and EuBatIn: NOT FOUND. Direct state aid, Schleswig-Holstein or federal: NOT FOUND.**

Identified public money is approximately ₹25.50 crore, plus unquantified shares of the ₹154.89 crore SIB:DE budget and of PLUS-LIBER, against roughly ₹332.39 crore (USD 34.9m) of equity: **unusually low grant dependence for a European battery scale-up**. EST The EIC grant **expired on 31 October 2026** and was single-beneficiary, so it is no evidence of industrial partnership, though it put the EIC Fund on the cap table.[10, 16, 19, 20, 21, 27]

Financials: a balance sheet and a paywall

Four accounts are on file, FY2021 to FY2024, the last filed 9 July 2026; none of the figures that matter is public. Obtainable: an FY2023 Bilanzsumme, or balance-sheet total, of ₹131.97 crore (EUR 12,354,495) SOFT, and FY2025 revenue as a **"zweistelliger Millionenbetrag"**, a double-digit million euro figure, so ₹106.82 crore to ₹1,057.52 crore (EUR 10m to 99m) SOFT, implying roughly ₹1.18 crore to ₹1.39 crore (EUR 110,000 to 130,000) per employee at 83 to 92 staff EST. Plausible for a contract coater, but **electrode-foil revenue**: cells reached series production only in June 2026. **Exact revenue, equity, net income and loss, every year: NOT FOUND.** The documents are fee-gated post-DiRUG at unternehmensregister.de, earlier years in Bundesanzeiger; a paid pull of the FY2024 Jahresabschluss is the highest-value open data item.[17,18,23]

The founders, the staff, and a missed hiring plan

Both founders are ex-CustomCells. Permien took a chemistry PhD at the Christian-Albrechts-Universitaet zu Kiel on lithium-ion battery materials and reaction mechanisms, the 2017 thesis honoured as the best in its faculty, then joined Custom Cells Itzehoe GmbH SOFT; **his dissertation title is NOT FOUND**. Strack, the chief operating officer, lives in Hamburg and is an ex-CustomCells named inventor; **his prior titles and education are NOT FOUND**. The state agency credits them jointly with "10-plus years in battery design, electrode coating, cell production and battery development".[12,21,26] Nine staff are named, two of them quality managers and **no CTO, CFO or VP Manufacturing** EST; the absent tenth is **Thomas Yu, PhD, Product Development Lead**, author of the only substantial technical disclosure.[2,11]

Published headcounts: 60 with "100 planned by year-end", WTSH, 2024 to 2025 HARD; "approx. 100 employees by end of 2025", slide 2 of the deck, January 2026 HARD; "more than 82", Series B release, 4 March 2026 HARD; 83 in dpa syndication that month SOFT; **92** on univercell.group/company in October 2026 HARD; and Tracxn's modelled 55 at 30 September 2026, contradicted by the company's own site that day, so **discarded**. Use 92: **there is no headcount decline**. But 100 were promised by end-2025 to a NASA-hosted audience in January 2026, and "more than 82" six weeks later: **hiring is roughly twelve months behind plan**.[2,10,11,21,27] The nine roles advertised in October 2026 are **all operations and back office, no R&D**: a factory ramp and ERP build-out.[3]

The CustomCells lineage, resolved

No asset purchase or corporate link was found. The link is human and intellectual-property: **both founders are named inventors on four patents assigned to Custom Cells Itzehoe GmbH**, which filed for insolvency in spring 2025. Permien is named on DE102018006416A1, published 20 February 2020, EP3948980A1, 9 February 2022, and DE102017011769A1, 19 June 2019; Strack on EP3611775A1, 19 February 2020.[15] UniverCell is the successor team in that vacated market. But CustomCells ran the same strategy with over 200 staff and thirteen years, and failed when one anchor customer, Lilium, stopped paying. **That precedent is not an analogy. It is personal.** EST

Sites and footprint

Five locations: Flintbek, Konrad-Zuse-Ring 1, headquarters and main electrode production HARD. **Moorsee**, near Kiel, is an acquired "newly factory expansion" "under 10 minutes away" where "pre-existing infrastructure allows rapid conversion and achieving SOP": size, capex and start of production NOT FOUND. **Russee** appears only in maintenance advertisements for "Flintbek and Russee" HARD. Offices in South Korea and the USA HARD. **Actual floor area is approximately 6,100 m2**, Permien's own production and storage figures HARD; the repeated 64,000 m2 is a **daily electrode output**, resolved in Chapter 07. Five locations on 92 people is thinly spread EST; South Korea is where the dry-coating equipment and materials chain sits.[11,21]

Intellectual property: one pending application

UniverCell Holding GmbH has exactly one patent family, EP4712256A1. It is pending, not granted, and it covers the cell housing lid. "Energy storage cell and manufacturing method therefor" names Leif Flessau, Thomas Yu, Stefan Permien and Marius Strack, carries priority of 12 September 2024 and published 18 March 2026, two weeks after the Series B close; an A1 is an application with a search report, not a grant. The family runs to WO2026057533A1 (PCT) and TW202620949A. Claim 1 covers the housing and a lid whose opening is a degassing port, plug-sealed afterwards, whose insert forms part of the first terminal: gas removal during formation, since high-silicon anodes and new electrolytes gas more.

That family covers the marketed housing and lid, which the deck labels "patented design for superior performance". **Industrial dry coating has no UniverCell patent at all, zero after exhaustive assignee and full-text search.** Nor has the tabless jellyroll, broadly pre-empted by Tesla, nor in-house formation; the 1,000-plus recipe library is trade secret by nature EST . [14] **Everything UniverCell markets as its moat is unpatented process know-how**: a deal buys people and process, not a portfolio. EST

> **A CITATION THAT MUST NOT BE REPEATED**
>
> A search-engine summary attributes patent JP5705408B2 to UniverCell Holding. It is false: JP5705408B2 is a 2015 Japanese filing predating the 2019 incorporation. One family, EP4712256A1, no granted patent. [14]

The technology, explained without jargon

Wet coating disperses electrode powder in a solvent, usually N-methyl-2-pyrrolidone, then drives it off in a long dryer, an electrode plant's largest energy consumer and floor area. **Dry coating removes the solvent and therefore the dryer.** The foil page specifies "Solvent type: water or NMP" and an 80-metre, twelve-zone dryer; the deck offers only "pilot production of dry-process electrode" and dry coating **compatibility**. [4, 11]

> **WHAT THE PUBLISHED SPECIFICATIONS ACTUALLY DESCRIBE**
>
> A dry line needs no solvent and no 80-metre dryer. UniverCell's installed, revenue-earning, 1.5 GWh-rated capacity is therefore **conventional wet and water-based coating**, and dry coating, the premise of the Series B, is at pilot scale. The published 750 mm and 60 m/min describe the **wet** line; the dry line's width and throughput are **NOT FOUND**, so benchmarking them against Fraunhofer's DRYtraec pilot at 240 mm and up to 10 m/min compares a wet line with a dry line. Verification request number one. [4, 11, 145]

Use **27 per cent at cell level** for the energy saving: the CORDIS text for GA 101161461, the deck and Permien's April 2026 abstract all say it. HARD The 30 to 40 per cent in press coverage is the **electrode process step** alone, in no company document, never to be averaged with the first. [12, 19, 24]

Other claimed attributes: a footprint up to 60 per cent smaller SOFT ; all anodes and cathodes made in house HARD ; over 1,000 tested recipes, the 1,000th customer electrode order 26 January 2026 HARD ; chemistries from high-nickel NMC to LFP, LCO, LMFP, LMNO, LTO and XNO HARD ; five-plus years in SiC and SiOx, recipes validated at more than six suppliers, "100s km of Si anode produced" HARD ; validated suppliers for all raw materials, "100% material sourcing from EU and North America is viable" HARD . [1, 4, 7, 11]

Products and published specifications

Electrode foil is the actual revenue product. The page adds 0.1 m/min at the slow end, continuous and intermittent patterns, substrate to 6 um, anodes in graphite, LTO and Si/C, cathodes in NCA, NMC 811, NMC 9XX and LFP. Unstated: areal loading in mg/cm2, tolerances, single versus double-sided coating, m2 per day. The milestone is over 30 km of electrode a day, in 2022. [4, 11]

The **PouchCell** is 80 mm x 390 mm x 25 mm, custom dimensions available, in NCA, NMC 622, NMC 811 and "Nickel-Rich LFP". Automated production of 300 mm x 80 mm pouch cells ran from 2021, the format since grown to 390 mm. **No capacity in Ah or Wh, no energy density, no C-rate, no cycle life and no temperature range is stated anywhere.** [5, 11]

CylindricalCell 21700, the flagship. The website publishes no numbers, only adjectives and "Prototypes are available now"; figures exist only in the deck and the About:Energy and Voltt library. [6] The full UC21700-HE specification, with 21700-HP sampling targeted Q3 2026 and operating temperature range **NOT FOUND**, is **Table A9**. [11, 13] Four caveats come before 330 Wh/kg is quoted.

1. **Pilot-batch numbers, not qualified production.** The deck's cycle-life chart reads "result from initial 21700-HE pilot batch", the site offers prototypes, and one slide gives "targeted capacity 6.0 Ah" against "typical capacity 6.0 Ah" in another. HARD
2. **The test condition shifts inside one slide.** Over 750 cycles is claimed at 1C/1C; the pilot result behind it reads "0.5 CC Chg/DChg, 100% DoD". Halving the rate makes 750 cycles much easier. HARD
3. **329 Wh/kg, 851 Wh/L and 6C continuous are not credible together.** 851 Wh/L exceeds anything in volume production in a 21700, and cells above 320 Wh/kg typically rate 1C to 2C: a best-case composite. EST
4. **Over 750 cycles is thin for the stated markets.** Satellites, critical-care medical devices and critical infrastructure need longer life or conservative depth of discharge; a high-silicon anode is the likely reason. EST

Where the 330 Wh/kg advantage comes from

Its own benchmark, sourced to About:Energy and manufacturer data, puts the tabless 21700 HE at 6.0 Ah, approximately 63 g, over 330 Wh/kg, 20.79 Wh per cell, against the multi-tab Molicel P60C (6.0 Ah, 75.0 g, 284 Wh/kg, 21.30 Wh), the multi-tab M65A (6.5 Ah, 74.5 g, 314 Wh/kg, 23.39 Wh), the tabless Reliance RS60 (6.0 Ah, 71.5 g, 302 Wh/kg, 21.59 Wh) and the tabless EVE 50PL (5.0 Ah, 72.0 g, 250 Wh/kg, 18.00 Wh). Against the P60C, the only cell at an identical 6.0 Ah, it is 16.0 per cent lighter and 16.2 per cent better per kilogram, but **energy per cell is 2.4 per cent lower**, and re-housed in the P60C's 75 g can at constant energy it rates 277 Wh/kg, **below** 284. EST Against the M65A: plus 5.1 per cent per kilogram, 11.1 per cent behind per cell. The 15 to 20 per cent lighter claim checks out exactly: 16.0, 15.4, 11.9 and 12.5 per cent against the P60C, M65A, RS60 and 50PL. EST [11, 13, 22]

12 grams of housing, per cell

WHERE THE 330 WH/KG ADVANTAGE LIVES

Effectively all of the headline energy-density lead is mass reduction, not electrochemistry, and exactly what the single pending patent covers.

The moat is **aluminium packaging**, not dry coating and not chemistry: genuine where watt-hours per kilogram is the buying criterion and cost per kilowatt-hour is not, in power tools, drones, aviation and satellites, but no lead over Asia, nothing on volumetric energy, copyable by any can supplier. EST The thermal benchmark, a lower surface temperature than the P60C, Amprius SA17 and EVE 50PL from 0.5C to 6C, is credible: aluminium conducts heat four times better than steel. EST [11]

Projects, partners and consortia

Four carry the weight. **IKA** runs the dry-coating programme through to series production, on CONTERNA continuous kneading, and is **also a Series B co-lead investor**, with WIKA, DTCF and the EIC Fund. **Varta sits beside UniverCell in the SIB:DE sodium-ion consortium**, Germany's largest, with EAS Batteries, E-Lyte Innovations and ZSW: partners, not comparables. SOFT **Fraunhofer FFB** in Muenster hosted Permien's April 2026 talk in its own production session: UniverCell is **inside** that ecosystem. The Alliance for Cyber Security (BSI) tie is **a membership, not a product certification**. Also named: Litona, PEM Motion, Nanoramic (2021, Neocarbonix), Akku Vision (August 2021, cells in e-bike production since end-2022), and a PLUS-LIBER consortium with IBU-tec and Fraunhofer IFAM. SOFT **Indian or Asian partner: NOT FOUND.** [8,9,10,12,20,29,30]

> THE NASA CONNECTION, STATED PRECISELY
>
> Thomas Yu presented "Innovations in 21700: Designed for High Performance Special Applications" on 20 January 2026, hosted on nasa.gov. **This is a conference presentation at a NASA-hosted battery workshop, not a NASA contract, qualification or endorsement.** It is also the best technical disclosure the company has made, and evidence of real United States aerospace business development. [11]

Customers and end markets

No customer is named anywhere: not the website, the Series B release, the deck or any press coverage, a finding by absence. [1,2,10,11] What is stated: "first multi-million-euro commercial contracts" HARD; "customers in the United States, which has recently entered into a multi-million-euro commercial agreement", the singular and plural slip making it **one** US customer HARD; those contracts at "in the tens of millions of euros" SOFT; the 1,000th customer electrode order, 26 January 2026 HARD; more than six validated silicon suppliers HARD; **deliberate avoidance of automotive** HARD; and of Chinese partnerships SOFT. On size the release says "multi-million" and dpa "tens of millions": **trust the release**, the controlled and more conservative, and flag the discrepancy. [23]

Four company documents give four different end-market lists. The authoritative one, the Series B release, names space systems, medical technology and critical infrastructure, singling out satellites and critical-care medical devices; slide 3 of the deck, the CORDIS record and WTSH variously add power tools, e-mobility, energy storage, aerospace, e-bikes, marine, stationary storage, e-aviation and robotics HARD. The breadth is a datapoint: it is still discovering which niche pays. **Defence and Bundeswehr: NOT FOUND**, no tender, no programme, no MIL-STD reference. **Tender wins: NOT FOUND.** [19,21]

Certifications: searched, not found

Nine standards searched, **every one NOT FOUND as of October 2026**: ISO 9001; IATF 16949, automotive, which the company avoids, so likely deliberate; ISO 13485, **required for critical-care medical devices**; AS9100 and EN 9100, **required for aerospace supply**; ECSS, **required for European satellite programmes**; UN 38.3, **mandatory to ship any lithium cell by air, sea, road or rail**; IEC 62133; ISO 14001; ISO 45001. One item is **HELD**: Alliance for Cyber Security (BSI), a membership, not a product certification HARD. [1,2,3,4,5,6,10,11]

Charitably, they are held but unpublished, normal for a contract manufacturer audited by customers; UN 38.3 almost certainly is.

Sceptically, **zero** published certifications reads as **pre-qualification**: prototypes, not flight hardware. Space and medical are target markets, not qualified supply.

The adverse record, searched and clean

Litigation, Rechtsstreit, Klage: **NOT FOUND**. Insolvency, Schutzschirm, StaRUG: **NOT FOUND**, status aktiv. Executive departures: **NOT FOUND**, both founders managing directors since 2019, reconfirmed at the register entry of 18 October 2024. Recalls, safety incidents, thermal events: **NOT FOUND**. German-language search on Kritik, Verzoegerung, Probleme, Insolvenz, Klage: nothing negative. Kununu and Glassdoor: **NOT FOUND**, unusual for a 92-person German employer but an absence of data, not negative data. ProvenExpert: seven reviews at 5.00 out of 5, a self-administered widget, evidence of nothing.[16, 17, 31] It calls itself "a successful gigafactory" and "Europe's leading Gigafactory for electrodes and batterycells": unverifiable superlatives, not to be repeated, when on nameplate it is roughly 5.3 times smaller than CATL Arnstadt at 8 GWh and 27 times smaller than VW PowerCo Salzgitter at 40 GWh. EST German Wikipedia's 1.5 GWh entry repeats the company's figure.[3, 28]

PROS AND CONS

Evidence for each line: **Table A10**.

THE PROS

1. **A real, installed, earning electrode business**, with **price-insulated end markets**: automotive avoided, the rest qualification-locked.
2. **Sized below where European cell ventures died**: 0.2 per cent of Northvolt's approximately ₹1.43 lakh crore (USD 15bn).
3. **Low grant dependence**: approximately ₹25.50 crore (EUR 2.387m) against ₹332.39 crore of equity.
4. **The weight advantage is real, and reproduced here**: 16.0, 15.4, 11.9, 12.5 per cent.
5. **The one patent covers the one differentiator**, the housing and lid.
6. **A clean adverse record**, and **headcount 60 to 92**, Germany's largest independent custom-cell business.
7. **Inside the German research establishment**; IKA equipment partner and Series B co-lead.
8. **Dry coating is a prerequisite on most all-solid-state roadmaps.**

THE CONS

1. **Cell output is 0.86 per cent of the headline, a 116-fold gap**, and two own headline numbers differ 1.7 times in one release.
2. **Dry coating, the Series B premise, is pilot-scale and unpatented**, and the whole IP position is one pending application on a packaging feature.
3. **Zero named customers, ever**; the one anchor a single United States customer.
4. **No certification evidenced for its stated markets**, none even for shipping cells.
5. **Hiring roughly twelve months behind its own plan.**
6. **Thin bench, slow structure**: no CTO, CFO or VP Manufacturing; five sites; joint signature.
7. **The energy-density advantage fails a controlled comparison**, and the cell specification is internally inconsistent, a best-case composite.
8. **No real financials, so a board cannot model the burn.**
9. **Excluded from the ₹16,023 crore (EUR 1.5bn) Battery Booster Facility**: it requires 10 GWh planned.
10. **The dry-coating lead expires around 2028 to 2029: three years, not ten**, and Lyten, well capitalised, reaches Heide from 2028, 60 to 70 km from Flintbek.
11. **Format-bounded too**: 21700 is giving way to the 46-series.

Chapter 07

The arithmetic nobody publishes

UniverCell publishes four different capacity numbers that cannot all be true at once, and reconciling them shows that the 1.5 GWh gigafactory is an electrode plant with a cell pilot attached, running at 0.86 per cent of its advertised nameplate.

The four numbers in conflict

Seven of the company's own claims collide: **1.5 GWh**, as "ready to scale beyond 1.5 GWh", from the Series B release and the site, repeated by Batterieforum and Wikipedia HARD as a quote; **50,000 21700 cells a month** in series production from June 2026 HARD; "more than **40 million** battery cells per year", the ramp the Series B funds HARD; **approximately 1.3 GWh by 2027**, Permien in press coverage SOFT; **180 million cells, approximately 4 GWh, by 2030**, slide 19 HARD; **64,000 m2 of electrode per day**, homepage HARD; and **over 30 km of electrode per day**, 2022, slide 7 HARD. [1, 10, 11, 22, 23, 25, 28]

Step one: establish watt-hours per cell, and validate the basis

The assumption used throughout: **21.6 Wh per 21700 cell, 6.0 Ah multiplied by 3.6 V nominal**, both the company's own, 6.0 Ah in the deck and the Voltt library, 3.6 V on slide 15. [11, 13] The library's 3.45 V would give 20.7 Wh; 3.6 V is deliberate, the company's figure and the **conservative** choice, since a higher watt-hour figure makes every gap below look smaller. It validates against slide 19's 2030 ramp, "+180 millions cells (approx 4 GWh) per year".

```
4,000,000,000 Wh / 180,000,000 cells = 22.2 Wh per cell
```

UniverCell's own cells-to-gigawatt-hours conversion is approximately 22 Wh per cell, so 21.6 Wh is right and marginally conservative. EST from HARD inputs. [11]

Step two: what 50,000 cells a month actually is

```
50,000 cells/month x 12 months = 600,000 cells/year 600,000 cells x 21.6 Wh = 12,960,000 Wh = 12.96 MWh/yr = 0.0130 GWh/yr
```

The honest run-rate: **0.0130 GWh a year, 12.96 MWh, of cell assembly at the June 2026 production start.**

0.86 per cent utilisation of the 1.5 GWh nameplate

WHAT THE CELL LINE WAS RUNNING AT IN JUNE 2026

An estimate from the company's own 50,000 cells a month and its own 21.6 Wh per cell. A 116-fold gap.

Step three: the gap table

CLAIM	GWH/YR AT 21.6 WH PER CELL	PER CENT OF THE MARKETED 1.5 GWH	MULTIPLE NEEDED TO REACH 1.5 GWH
Actual: 50,000 cells a month, June 2026	**0.0130**	**0.86%**	**116x**
Series B target: "more than 40 million cells a year"	0.864	57.6%	1.7x

All money in rupees, converted at a fixed rate stated in the method note.

CLAIM	GWH/YR AT 21.6 WH PER CELL	PER CENT OF THE MARKETED 1.5 GWH	MULTIPLE NEEDED TO REACH 1.5 GWH
Permien: "approx 1.3 GWh by 2027"	1.300	86.7%	1.2x
Marketed nameplate	1.500	100%	1.0x
2030 target: 180 million cells a year	3.888, rounded by the company to 4	259%	not applicable

All derived figures are estimates from the company's own inputs. EST Sensitivity: at 20.7 Wh the run-rate falls to 0.01242 GWh and the gap widens to **121 times**, so 116 is conservative.

Step four: the three findings that follow

(a) The cell line runs at 0.86 per cent of the headline capacity. To fill 1.5 GWh of **cells** requires:

```
1,500,000,000 Wh / 21.6 Wh = 69,444,444 cells/year = approx 5,787,000 cells/month
```

They are making 50,000. EST

(b) "40 million cells a year" and "1.5 GWh" are mutually inconsistent by 1.7 times, in the same press release.

```
40,000,000 cells x 21.6 Wh = 864,000,000 Wh = 0.864 GWh
```

At the company's own watt-hours per cell the **fully funded** target reaches about 58 per cent of what is advertised as already available, and a target cannot be smaller than an existing asset unless the two measure different things. **Therefore 1.5 GWh is not a cell-assembly number. It is an electrode-coating number.** EST from HARD inputs. [10]

(c) The 1.3 GWh-by-2027 claim requires roughly 100 times the June 2026 run-rate in about eighteen months.

```
1,300,000,000 Wh / 21.6 Wh = 60,185,185 cells/yr = approx 5,015,000 cells/month 5,015,000 / 50,000 = 100.3x Compound monthly growth = 100.3^(1/18) − 1 = approx 29% per month, sustained
```

A 29 per cent compound monthly growth rate in output, for eighteen consecutive months, on ₹320.46 crore (EUR 30m). EST The most demanding number in the plan.

29 per cent compound monthly output growth

WHAT THE 2027 TARGET REQUIRES, EVERY MONTH FOR 18 MONTHS

An estimate from the company's own 50,000 cells a month and its own 1.3 GWh by 2027. Sustained, on one Series B.

Step five: the cross-check that 1.5 GWh is the electrode line

Three checks, all on the company's electrode line at 750 mm and 60 m/min maximum. [4]

Check one, the advertised daily area reconciles.

```
60 m/min x 0.75 m x 1,440 min/day = 64,800 m2/day
```

Which is the advertised "64,000 m2 per day": not a floor area but the coater's maximum, to within one per cent. EST

Check two, the 2022 milestone reconciles.

```
30,000 m/day x 0.75 m = 22,500 m2/day = approx 35% of theoretical maximum
```

About 35 per cent of maximum is normal utilisation, so the two company numbers are consistent. EST

Check three, the cell-equivalence of the electrode line. Assumption stated plainly: a 6.0 Ah 21700 jellyroll contains roughly **0.107 to 0.117 m2 of coated foil**, approximately 0.8 to 0.95 m of wound length at 62 to 65 mm electrode height, cathode and anode together. That range is this report's estimate; the conclusion holds across it. EST

BASIS	M2/DAY	M2/YR AT 330 OPERATING DAYS	CELLS/YR AT 0.107 M2	GWH/YR AT 21.6 WH	GWH/YR AT 0.117 M2
Demonstrated 2022 output, over 30 km/day	22,500	7,425,000	69.4m	**1.50**	**1.37**
Theoretical maximum, 60 m/min x 750 mm x 24h	64,800	21,384,000	199.9m	**4.32**	**3.95**

- **Demonstrated** 2022 throughput converts to approximately **1.4 to 1.5 GWh a year of cell-equivalent output: that is where "1.5 GWh" comes from.**
- The **theoretical maximum** converts to approximately **4.0 to 4.3 GWh, matching the company's own 2030 target of 180 million cells and approximately 4 GWh.**

So **the 2030 target is not a new factory. It is the existing electrode line at full utilisation**, which is why Moorsee is framed as "rapid conversion" rather than greenfield: lower capex than the 2030 number implies, cell assembly untouched. EST, assumptions stated; inputs HARD. [1, 4, 11]

The footprint collision, resolved

The same 64,000 m2 number is used two incompatible ways, once in a NASA-workshop document. The four readings: "64,000 m2 of **electrodes per day**", homepage HARD; "Total facility size 64,000 m2", **slide 2 of the deck** HARD; "1.5 GWh per year **on an area of 64,000 m2**", Batterieforum Deutschland SOFT; and, from the state agency **quoting Permien**, "approximately 3,300 m2 for production and 2,800 m2 storage", about 6,100 m2 HARD.

> THE RESOLUTION, AND IT IS NOT CLOSE
>
> The **per-day** reading is correct; "facility size" is an error the company propagated into its own deck and from there into third-party profiles. The proof is the founder's own words: approximately 3,300 m2 of production plus 2,800 m2 of storage is roughly **6,100 m2, about one tenth of 64,000 m2**. Even adding Moorsee and Russee, 64,000 m2 of building is implausible for 92 employees, and check one confirms the per-day reading, because 60 m/min at 750 mm for 1,440 minutes is 64,800 m2. **Never cite 64,000 m2 as a floor area. Say "approximately 64,000 m2 of electrode coated per day", or say nothing.** [1, 11, 21, 25]

Scale context, against named comparators

Ten comparators, against the 1.3 GWh 2027 target and marketed 1.5 GWh: Chinese sector production of 1,755.6 GWh in 2025, CATL alone 333.57 GWh of China installations, **over 250 times the planned 2027 output** SOFT; **EVE Energy's Malaysia Phase 1**, 21700 for power tools and two-wheelers, **680 million cells a year**, 17 times the 40-million-cell target, 1,133 times the June 2026 run-rate (HARD on EVE, EST on the ratios); VW PowerCo Salzgitter, 40 GWh from 17 December 2025, 31 and 27 times; CATL Arnstadt, 8 GWh from December 2022, 6 and 5.3 times; Porsche Cellforce, 20 GWh, cancelled, closed May 2026; Northvolt Drei at Heide, also in Schleswig-Holstein, 60 GWh, paused, assets to Lyten HARD; Leclanche, 0.2 GWh, German units insolvent 23 September 2026; and **Molicel (TCC) Tainan**, approximately 1.5 GWh now, a 1.8 GWh Kaohsiung project, a 3.3 GWh target: the same order of magnitude, **the real peer** SOFT.

Two do the work. **Morrow Batteries**, Arendal, 1 GWh, insolvent 6 May 2026, was "far too small to supply a major car manufacturer" per its administrator: same size, different segment, the warning. The answer is that UniverCell's segments are ones where 1 GWh is a large share of demand, **which holds only out of automotive and volume power tools.** And **Fraunhofer FFB**'s open pilot at Muenster, electrode capacity approximately 6.8 GWh a year, completion end-2027, is five times its size. SOFT
28, 43, 54, 143, 157, 158, 162, 175, 178, 180

The honest formulation for a board

> WHAT TO SAY, AND WHAT NOT TO SAY
>
> UniverCell has approximately 1.5 GWh-equivalent of **electrode coating** capacity: real, installed, already earning ₹106.82 crore to ₹1,057.52 crore (a double-digit million euro figure) selling coated foil to other people. Its own **cell assembly** started in June 2026 at approximately 13 MWh a year, 0.86 per cent of that headline. **The gigafactory is an electrode plant with a cell pilot attached. The Series B is the money to turn the pilot into a line.** That concedes the asset, refuses the superlative, and locates the risk: an eighteen-month ramp of one hundred times, on one round of funding.

The questions this arithmetic generates

Six questions follow, each with a verifiable answer: whether the 1.5 GWh is electrode or cell capacity, since 40 million cells a year is 0.864 GWh and the two cannot both describe cells; the **dry** line's width and throughput in mm and m/min; the run-rate in cells a month now and in every month since June 2026; Moorsee's capex and start of production; the merchant-electrode share of 2025 revenue and the largest customer's share; and which certifications are held, with numbers and dates. An answer of "electrode" resolves the contradiction and costs nothing but a superlative; "cells" would contradict the press release. **That is why question one goes first.**

Chapter 08

Dry coating: what it is, who has it, how long the lead lasts

Dry electrode coating is a real advance and UniverCell is credibly in the global top three on operating industrial activity. But its published line specification describes a wet line, its dry route is at pilot scale on its own documents, Fraunhofer licenses the concept to anyone, and the lead lasts three years, not ten.

What the process actually is

A conventional electrode is made wet: powder, carbon and binder slurried in N-Methyl-2-pyrrolidone for most cathodes or water for some anodes, coated onto foil and dried in a multi-zone oven, the plant's largest energy and floor-area consumer. The published specification is that route: 0.1 to 60 m per minute, width to 750 mm, substrates to 6 um, continuous and intermittent patterns, an 80 metre twelve-zone dryer, "solvent type: water or NMP".[4]

The dry route deletes solvent and oven: powders blended with a mechanically fibrillised binder are laminated or roller-pressed onto foil. VW PowerCo presses "powdery basic materials directly onto the foil" with Koenig und Bauer;[174] Fraunhofer IWS offers DRYtraec, a roll-to-roll dry transfer coating platform, under licence;[145] LiCAP sells its Activated Dry Electrode process;[172] Sakuu dry-prints both electrodes on its Kavian platform.[177] UniverCell publishes no binder chemistry; its partners include fluoropolymer producer Daikin Chemical Europe, consistent with a fibrillised-binder route. EST [242]

Only the dryer vanishes, and the energy and floor-area saving sits there. Failure modes move rather than disappear: slurry stability and crack control on high-silicon anodes become powder dosing uniformity, fibrillisation control and areal-loading tolerance at width and speed.

The savings claims and their scopes

The 27 per cent lower energy per cell used here comes from the CORDIS record for the EIC Accelerator grant, slide 11 of the technical deck and the chief executive's April 2026 abstract, "energy savings of up to 27% at cell level". HARD [19, 11, 12] The 30 to 40 per cent energy figure is scoped to the electrode step, the up to 60 per cent footprint figure to the factory, both from Series B press coverage, not a company document. SOFT [24] Different denominators: not in conflict, never averaged. HARD [19] SOFT [24]

27% lower energy per cell

UNIVERCELL'S DRY-COATING SAVING, AT CELL LEVEL

The 30 to 40 per cent figure in press coverage is the electrode process step alone; two of the company's own primary documents say 27 per cent at cell level.

Eight vendor claims sit behind the debate, UniverCell's two above and six more, none audited or comparable:

- Tesla, approx 70% lower emissions, cell production versus wet coating SOFT [77]
- VW PowerCo, approx 30% energy, 15% less floor space, cell production, both electrodes HARD [174]
- LG Energy Solution, 17% to as much as 30% cost, electrode manufacturing; approx 30% less land, factory HARD [169]
- Duerr and LiCAP, up to 65% less space, 70% lower energy, electrode line SOFT [173]
- Sakuu, 55% lower CO2, 60% smaller footprint, 20% lower capex, 30% utility opex, electrode and plant SOFT [177]

- AM Batteries, approx 50% lower plant energy, 40% equipment capital saving, electrode manufacturing SOFT [176]

Removing the dryer saves tens of per cent of electrode-step energy and some floor area; nobody claims it halves cell cost.

The honest global ranking

PLAYER	OPERATING SCALE OCTOBER 2026, STATED MASS PRODUCTION, AND WHY
1 Tesla	Mass production, 4680, Giga Texas, both electrodes fully dry. Benchmark, not rival: no merchant cells or electrodes SOFT [77, 78]
2 VW PowerCo	Gigafactory built, both electrodes, dry start guided 2026 or 2027, the guidance itself HARD ; series dry use unconfirmed, NOT FOUND [174, 175]
3 UniverCell	Both electrodes in house, sub-GWh, dry coating at pilot scale, 21700 series production June 2026. Roughly 1,000 times smaller than ranks 1 and 2, yet shipping custom electrodes on a 1,000th order, January 2026 [4, 11, 7]
4 Sakuu	Both electrodes dry-printed. "Hundreds of metres" delivered; NCM cell at 83% retention after 4,000 cycles, fully dry cathode; no volume; SPAC announced October 2026 SOFT [177]
5 AM Batteries	Merchant electrode; engineering pilot and customer centre. First rolls to a major cell maker October 2024; DOE award 31 August 2026, approx 2 GWh electrode, 1.5 GWh cell [176]
6 LG Energy Solution	Full electrode; Ochang pilot complete. Mass production 2028 per LG's June 2026 publication, 2029 for dry LFP per April 2026 reporting: different products, not a contradiction. HARD company, SOFT press [169, 170]
7 Samsung SDI	46-series. Two pilots, the 6,500 m2 S-Line at Suwon and DryEV at Cheonan, processes to transfer to the coming Ulsan plant. No mass-production date, NOT FOUND SOFT [171]
8 Duerr and LiCAP	Equipment plus process IP. Chassieu proof of concept; LiCAP runs a 300 MWh dry line in California; no pilot host [172, 173]
9 Fraunhofer IWS	Licensable platform, four prototype systems. The licensor, not a producer [145]
10 SK On, Panasonic, CATL, BYD	SK On "coordinating" timing across pilots in Korea, Hungary and the US; Panasonic 4680 at Wakayama; CATL reviewing large cells; BYD weighing solid-state. Pilot and evaluation, no dates, NOT FOUND; over ₹4,000 crore (USD 420m) of dry-line orders, 2025 and 2026 {SOFT, single aggregator} [170, 171]
11 Anaphite	Cathode dry coating, Series A. The one genuine dry-coating startup peer [27]

Two names do not belong: 24M's SemiSolid is a thick-electrode slurry route, Chasm Advanced Materials a carbon-nanotube business. EST

> **THE SPECIFICATION QUESTION THAT DECIDES WHERE RANK 3 BELONGS**
>
> The only published specification, quoted above, carries an 80 metre twelve-zone dryer and "water or NMP" solvent. A solvent-free line needs no dryer, so those numbers describe the wet line. The deck lists only "pilot production of dry-process electrode" and calls the 21700 "dry coating compatible". The dry line's width and throughput are NOT FOUND, so benchmarking against Fraunhofer's 240 mm at 10 m per minute compares a wet line with a dry pilot. {EST from HARD inputs} [4, 11] Verification request one: which line do those figures describe, and what are the dry line's width, speed and areal-loading tolerance?

Flintbek's revenue-earning line is not dry but conventional wet, water-based coating, on double-digit-million-euro revenue from merchant coated foil HARD [11, 23] and a 2022 milestone of over 30 km of electrode a day. HARD [23] The 64,000 m2 on the homepage is that daily output, not floor area; HARD [1] the site is approximately 6,100 m2, 3,300 m2 production and 2,800 m2 storage, on the chief executive's account. HARD [21]

The Tesla counterweight

It must be stated or the case reads as cheerleading: Tesla needed roughly seven years and very large sums, its chief executive called dry electrode a mistake and "way harder" than expected at the 2025 shareholder meeting, and 4680 performance was reported underperforming on key metrics in May 2026. SOFT [77, 78] Validated as possible, not easy: the strongest argument for a real moat, and the strongest reason to demand proof before paying.

Duerr and LiCAP: no host

Europe's best dry-coating equipment path lost its host when Porsche's Cellforce closed. Duerr has partnered LiCAP Technologies of California since 2023 on the Activated Dry Electrode process, claiming roughly one tonne of carbon saved per 10 kWh of electrode capacity, with Cellforce at Kirchentellinsfurt the planned host, due in operation 2026. HARD [172] The Chassieu platform was presented as X.Cellify DC in October 2025, Duerr seeking gigawatt-scale pilot partners. SOFT [173] Cellforce closed completely in May 2026.[151]

> **A PRE-ASSEMBLED CONSORTIUM THAT IS SHORT ONE BALANCE SHEET**
>
> Duerr supplies equipment, LiCAP the process IP; neither has what an operator with a running electrode line, 1,000-plus tested recipes and qualified customers brings. It has a validated process, no anchor pilot partner and a public search for one. For an industrial group needing dry-electrode capability and unable to license from Tesla, this is the cheapest entry point; the window closes as the Koreans industrialise in 2028 and 2029. Inference from public positions, not an announced transaction. EST

The moat, and the date it closes

Three primary-sourced facts cap it.

No patent. Exactly one family, EP4712256A1 with PCT and Taiwanese counterparts, pending not granted, claiming a cell housing whose lid opening serves as a degassing port during manufacturing. It claims neither dry coating nor the tabless jellyroll, and was filed in Taiwan, not China, not India, not Korea. Exhaustive assignee and full-text searches found zero dry-coating patents: the marketed moat is unpatented know-how. HARD [14]

Licensable by anybody. Fraunhofer IWS licenses DRYtraec commercially, supports installation, works with plant makers on larger lines, and launched a ₹39.52 crore (EUR 3.7m) Dresden DRYplatform in November 2025. HARD [145] UniverCell sits inside that ecosystem: its April 2026 talk was hosted by Fraunhofer FFB in Muenster. HARD [12, 146]

Equipment already ordered. The dry-line orders at rank 10 are CATL, BYD, Panasonic and LG; PowerCo guides 2026 or 2027. SOFT [170, 174]

Defensible instead: more than 1,000 tested electrode recipes, the in-house formation process, the customer relationships behind that 1,000th order, and that one mechanical feature, filed in Taiwan where Molicel sits.[7, 14] EST Sulphide solid electrolytes suit neither NMP nor water slurry, and high-silicon anodes are hard to wet-coat, so dry coating is a capability every all-solid-state and high-silicon entrant must buy. EST [68, 69]

Top three on operating industrial dry-electrode activity, roughly twenty-fifth to nowhere on scale. The lead expires around 2028 to 2029: a three-year advantage to monetise by 2029, not a decade-long moat. EST

Chapter 09

The competitive landscape

UniverCell's real peer set is neither CATL nor the silicon-anode materials firms data vendors list against it: a European specialty cohort that has largely collapsed, a dry-coating cohort still at pilot scale, and one Taiwanese company of almost exactly its size already selling the same product into the same segments.

Tier A: European specialty cell makers

The cohort whose order book UniverCell contests, mostly gone. Chemistry, format and segment focus is in **Table A11**, ownership, funding and dated status changes in **Table A12**. Capacity and status:

PLAYER	CAPACITY AND STATUS, OCTOBER 2026
UniverCell, Flintbek DE	approx 1.3 GWh 2027 target, actual June 2026 run-rate 0.013 GWh; scaling
CustomCells, Itzehoe DE	pilot, no GWh published; restructured after insolvency
V4Smart, Ellwangen DE	low-volume premium; alive and investing
Leclanche, Willstaett DE	approx 2 GWh planned; German units insolvent
Morrow, Arendal NO	1 GWh built; insolvent, Lyten LOI
Northvolt into Lyten, SE, NO, DE	16 GWh acquired plus 1 GWh; expanding by acquisition
Verkor, Dunkirk FR	16 GWh, 50 GWh by 2030; ramping to plan
InoBat and GIB, Surany SK	phase 1 20 GWh; under construction
ACC, FR, DE, IT	Douvrin only; two plants cancelled
Cellforce, Kirchentellinsfurt DE	pilot; closed
Saft and EnerSys, FR, US, DE	not disclosed at cell level; incumbent and investing
Blue Solutions, Mulhouse FR	plan only; refocusing

Three contest it directly: CustomCells, the structural analogue 50 km away, now approximately 80 staff against UniverCell's 92; V4Smart, captive to Porsche; Saft with EnerSys, holding the space, defence and submarine accounts.

The count. Seven of twelve are dead, shrunken, restructured or cancelled: Northvolt, CustomCells, Cellforce, Morrow, Leclanche's German units, ACC's two plants, Blue Solutions. Varta AG is an eighth: second insolvency filing dated 24 July 2026 in the fact base, a trade-press account naming the court, the four entities and the administrator on 28 July; this report uses 24 July and flags the discrepancy. HARD [154, 155] Freyr is a ninth, Europe's second most funded cell startup after Northvolt, rebranded as T1 Energy, moved to Austin, pivoted to solar: it did not lose, it left. SOFT [167] Three are executing: Verkor on an anchor OEM, Lyten by acquisition, V4Smart on Porsche's balance sheet.

THE MORROW DIAGNOSIS, PUT AGAINST UNIVERCELL'S OWN NUMBER

Morrow's administrator: the plant was "far too small to supply a major car manufacturer" at 1 GWh, no investors for expansion, LFP prices down nearly 50 per cent between 2022 and 2025. SOFT [158] UniverCell's 2027 target is 1.3 GWh: the same size. The answer is segment, not scale. In power tools, medical, marine, space and e-aviation, 1 GWh is a large segment share, not a rounding error on a car programme, and the customer pays several times commodity price. It holds only while it stays out of automotive and commodity power tools; chasing volume makes the diagnosis apply. EST

TWO ADJACENCIES THAT ARE OFTEN MIS-FILED

Pack integrators are customers, not competitors. BMZ Group at Karlstein runs an indicative 5 GWh of pack capacity and serves medical technology, one of UniverCell's named segments; Webasto is expanding pack assembly at Schierling and Dangjin. German integrators are the natural first buyers of European-made, Battery-Regulation-compliant cells, the most actionable read in Tier A. That 5 GWh is a low-trust trade listing, indicative. SOFT **The sodium-ion claim is not unique.** Altris of Sweden with Czech Draslovka works the same Prussian-white cathode family as UniverCell's partner Litona, roughly two years ahead, on a ₹206 crore (EUR 19.3m) in-kind industrial investment, production slated late Q3 or early Q4 2026. HARD [182] Natron Energy, the Western Prussian-blue champion, ceased operations 3 September 2025. HARD [65] Faradion, same field, is Reliance-owned: the Indian benchmark already bought a European cell IP platform, in the UK, in 2021.[181]

Tier B: the dry-coating technology competitors

Operating scale is ranked in chapter 08; money and business model are in **Table A13**. AM Batteries is the closest business-model competitor, a transatlantic mirror image selling merchant dry electrode, not cells, EU public money against US.[176] Sakuu's pre-money SPAC value is roughly 17 times UniverCell's raise, on validation not product. EST [177] Duerr with LiCAP is the one entry point, time-bound by LG's industrialisation.[172, 169]

Tier C: the Asian price floor

Global EV battery usage in H1 2026 was 608.5 GWh, up 20.0 per cent year on year: CATL 242.7 GWh for 39.9 per cent, 39.4 per cent January to August 2026; BYD 87.7 GWh, 14.4 per cent; LG Energy Solution 52.6 GWh, 8.6 per cent; EVE Energy 3.4 per cent. CATL and BYD are 54.3 per cent of the world; seven Chinese firms in the top ten hold 72.4 per cent. SOFT [42]

The decisive fact is one Malaysian plant. EVE Energy's Kulim site in Kedah makes 21700 cells explicitly for power tools and electric two-wheelers, the segments UniverCell names: phase one rated 680 million cells a year, first cell 16 February 2025, investment announced up to ₹4,022 crore (USD 422.3m), HARD [178] phase two completed 26 February 2026, adding a planned 10 to 15 GWh of storage. HARD [179] ASEAN rather than China, tariff-efficient for these segments. That is approximately 17 times UniverCell's funded target of over 40 million cells a year and 1,133 times its June 2026 run-rate of 50,000 cells a month. {EST from HARD inputs} Any Indian buyer of high-drain 21700 cells is probably buying there already.

₹163 (USD 1.71) per cell

THE COMMODITY FLOOR FOR A 21.6 WH 21700

At BNEF's global average EV cell price of ₹7,524 per kWh (USD 79). A premium high-drain cell plausibly contracts at ₹286 to ₹571 (USD 3 to 6), and UniverCell must land above that band.

Four price bases bound the problem: that commodity floor, an EST on a HARD price; [44] premium high-drain retail listings, low trust, e-commerce only, ₹476 to ₹1,143 (USD 5 to 12) a cell or ₹20,953 to ₹24,762 (USD 220 to 260) per kWh; a bulk OEM band, this report's estimate not a sourced price, ₹13,334 to ₹26,667 (USD 140 to 280) per kWh EST ; and the European pack premium over China, plus 56 per cent. HARD [44]

UniverCell cannot compete at ₹163 a cell and must sell to demand paying a European-sovereignty and custom-engineering premium over an already premium Taiwanese or Chinese cell. Power tools and e-bikes are on its own website; against EVE's volume those are its weakest claims. Two vendor models of the 18650 and 21700 market disagree more than twofold on scope and roughly threefold on growth, so neither is used. EST

What the data vendors get wrong

Tracxn's profile lists nine competitors, six powder or additive suppliers rather than cell or electrode makers: Sila Nanotechnologies, silicon anode, ₹15,334 crore (USD 1.61bn); Nexeon, silicon anode, ₹2,905 crore (USD 305m); NanoOne, cathode material, public; Cenate, silicon anode, ₹37.9 crore (USD 3.98m); Advano, silicon anode, ₹260 crore (USD 27.3m); Ten-Nine Technologies, electrode additive, funding not stated. Only Anaphite, dry coating, ₹215 crore (USD 22.6m), is genuine; Nyobolt, fast-charge cells and systems, ₹1,533 crore (USD 161m), is defensible; EAMPL, unfunded and not established: discard.

Read from the page on 30 September 2026, the set fails on five counts. HARD [27] It mixes value-chain layers: Sila against UniverCell's ₹332.39 crore (USD 34.9m) is a 46-fold gap meaning nothing. It omits every genuine competitor, no CustomCells 50 km from Flintbek, no Molicel in Taiwan. It mislabels the business, "developer of custom electrodes" dropping the cells the Series B funds. Its record is unreliable: one founder where the company names two, 55 employees at 30 September against the company's own 92 that day. And the alternatives are paywalled: PitchBook, Crunchbase and CB Insights sets NOT FOUND, so no list should be attributed to them.

The error carries a signal: the classifier read electrode language, 1,000-plus tested recipes, an Electrode Foil page, electrode as a service, and clustered it with silicon-anode materials firms. The marketing describes a materials and process business while the Series B funds cell manufacturing, and investors and procurement read it the same way. EST

The comparator that matters: Molicel, not CATL

The real comparator is Molicel, trading as Taiwan Cell Technology: approximately 1.5 GWh in Tainan against UniverCell's 1.3 GWh target, a 1.8 GWh Kaohsiung project funded at roughly ₹3,333 crore (USD 350m) towards 3.3 GWh, the same high-drain 21700 and 18650, the same segments of eVTOL, drones, motorsport, power tools and robotics, the design wins taken: Archer Aviation's production Midnight, Vertical Aerospace's VX4, a McMurtry Automotive partnership. Its newest cell, the INR-21700-P70X of July 2026, is cited at 340 Wh/kg and 30C. {HARD on the wins, SOFT on the P70X} [180]

Its own benchmark, from About:Energy and manufacturer datasheets: UniverCell 21700 HE, tabless, 6.0 Ah, approx 63 g, over 330 Wh/kg, 20.79 Wh; Molicel P60C, multi-tab, 6.0 Ah, 75.0 g, 284 Wh/kg, 21.30 Wh; Molicel M65A, multi-tab, 6.5 Ah, 74.5 g, 314 Wh/kg, 23.39 Wh; Reliance RS60, tabless, 6.0 Ah, 71.5 g, 302 Wh/kg, 21.59 Wh; EVE 50PL, tabless, 5.0 Ah, 72.0 g, 250 Wh/kg, 18.00 Wh.

At 6.0 Ah against the P60C it is 16.0 per cent lighter, 16.2 per cent better on watt-hours per kilogram, but 2.4 per cent **worse** on energy per cell; re-housed in the P60C's 75 g can at constant energy it reads 277 Wh/kg, below 284. Against the best-in-class M65A it is 15.4 per cent lighter, 5.1 per cent better per kilogram, a lead narrowing against 314 Wh/kg, and 11.1 per cent worse per cell. The advantage is mass, roughly 12 g of aluminium can against steel, not electrochemistry. {EST from HARD inputs} [11, 13] The press claim of power-tool cells 15 to 20 per cent lighter checks out. EST [22]

Three things UniverCell has that Molicel does not, one hard to copy: unpatented industrial dry coating, copyable; merchant custom electrode supply, revenue-earning but copyable, AM Batteries better funded; EU domicile with Battery Regulation compliance and local-content eligibility, structural and not copyable by a Taiwanese or Chinese producer. The aluminium tabless housing, 12 g lighter, any can supplier can copy, on one pending application.

Molicel is cheaper, flight-qualified, roughly fifteen years more experienced, the same size. EST [180] UniverCell publishes no ISO 9001, ISO 13485, AS9100 or ECSS, UN 38.3 or IEC 62133 certification anywhere as of October 2026, so the space and medical positioning reads as target markets, not qualified supply: the most important question for management. {HARD by exhaustive absence} [11] Stated cycle life of over 750 cycles, the pilot result annotated at half the claimed rate, is thin for satellites and critical-care devices. [11]

There is no evidence of any asset purchase or corporate link between UniverCell and CustomCells, and none should be asserted; both founders are named inventors on four patents assigned to Custom Cells Itzehoe GmbH. [15] The team now running Germany's largest independent custom-cell business, on headcount, built the last one; a single customer's non-payment killed it.

Chapter 10

Where this goes: scenarios to 2030 and 2035

Europe arrives at 2030 roughly a tenth short of cell self-sufficiency in every scenario this report can construct. The binding question is not whether Europe has enough cells but who owns them: that answer decides whether a 1.5 GWh specialist is a strategic asset or a rounding error.

The three worlds and their drivers

Three worlds frame what follows, a bear "Fortress China", a base "Managed Dependence" and a bull "European Rearmament", on six drivers, bear/base/bull. The 90 per cent CO2 cut proposed on 16 December 2025 is adopted in all three: with heavy plug-in hybrid and e-fuel use and slippage at the 2028 review; with plug-in hybrids near 15 per cent of 2030 sales and the car target holding; or with local-content conditionality. BEV share of about 15 million new cars in 2030: 40/55/65 per cent. Announcement No. 58 is reinstated at or after the lapse, licensing cells at or above 300 Wh/kg, cathode and anode material and cell-making equipment; rolled forward; or reinstated against enforced EU local content, making the cost shock a moat. Attrition worsens from 25 per cent of announcements to about 45, holds at the 25 to 30 per cent of 2022 to 2026, or halts. Storage build-out reaches about 90/138/220 GWh a year by 2030, the base SolarPower Europe's medium case, the bull pulled toward roughly 750 GWh cumulative. Learning rate per doubling: 13/19/24 per cent.

The newest driver: the Commission's proposal swaps the 100 per cent zero-emission 2035 target for a 90 per cent CO2 cut, admits plug-in hybrids, range extenders, mild hybrids and combustion past 2035, offsets the residual 10 per cent with EU low-carbon steel or sustainable fuels, and cuts the 2030 van target from 50 to 40 per cent. SOFT [75] The most EV-dependent market feels it most. HARD [36]

European cell demand, GWh a year

Fraunhofer ISI publishes 800 to 1,300 GWh for Europe in 2030. HARD [138] The build-up here, on about 15 million new cars a year across the EU, EFTA and UK, bear/base/bull: BEV share 40/55/65 per cent, so 6.0/8.25/9.75m cars at 62/65/68 kWh, giving 372/536/663 GWh of cells; plug-in hybrids at 18 kWh add 2.7/2.25/1.5m units, the base 15 per cent of 15m, for 49/40/27 GWh; vans, trucks and buses 80/120/200; storage 110/160/280, base the EU's 138 plus the UK and Nordics; two and three-wheelers, tools, consumer, industrial and niche 70/80/110 GWh. Totals 681/936/1,280 GWh, rounded to 700/1,000/1,300, then 1,100 at 9 per cent, 1,750 at 12 and 2,500 at 14 to 2035. EST Bear sits below Fraunhofer's floor, which pre-dates the December 2025 proposal.

European capacity and output

Announced 2030 capacity exceeds 2,000 GWh. Fraunhofer's eleven-factor risk model puts realisation at 54 to 75 per cent: 1,200 to 1,700 GWh nameplate, 800 to 1,100 GWh of output after delays and scrap, a 65 to 67 per cent ratio; T&E agrees at the top, October 2026, 1,700 GWh. HARD [138] SOFT [113] Nameplate of 1,000/1,350/1,700 GWh at 0.60/0.65/0.70 utilisation and yield gives 600/878/1,190 against demand of 700/1,000/1,300, short by 100/122/110 GWh or 14/12/8 per cent; in 2035, 1,300/1,900/2,600 gives 845/1,290/1,900 against 1,100/1,750/2,500, short by 255/460/600 GWh or 23/26/24 per cent. EST

The 2030 gap is narrow, and T&E judges medium-probability projects enough for the Industrial Accelerator Act. SOFT [113] It does not last, the 2035 shortfall widening as demand compounds at 9 to 14 per cent, the comfort a one-off from a 2020 to 2024 wave now 25 per cent cancelled, its German 2035 component halved. EST SOFT [113]

"ENOUGH CELLS IN EUROPE" IS NOT "ENOUGH EUROPEAN CELLS"

Over 85 per cent of active European capacity is non-European-owned, South Korean makers holding 72 per cent. SOFT [113] Benchmark expects about 33 per cent of Europe-made cells from Chinese-owned plants by 2030. SOFT [39] CATL began trial production at Debrecen on 22 September 2026, phase one 34 GWh against 100 GWh planned; Stellantis and CATL target 50 GWh at Zaragoza by end-2026, BYD Szeged in Q4 2026. SOFT [183, 184] These raise the share Chinese companies hold while cutting the share made in China: re-hosting, not de-risking. The ownership gap matters, not the tonnage. EST

THE CAPACITY DATA IS 22 MONTHS STALE AND BIASED OPTIMISTIC

Fraunhofer ISI's model has a December 2024 cut-off, and every European failure since falls in 2026: ACC's cancellations, Morrow's bankruptcy weeks after commercial production began, Varta AG's insolvency and Leclanche's two German filings. SOFT [158, 154, 157] Every 2030 capacity figure inherits that optimism: date-stamp it or do not use it.

Cell and pack price, rupees and dollars per kWh

Price follows the learning curve: Price(t) = Price(2025) x (1 minus LR) ^ log2(Cum(t)/Cum(2025)), Price(2025) ₹10,286 per kWh (USD 108) for the average pack on cumulative output through 2025 of about 6.0 TWh. HARD [44] Additions of 14,500 GWh to 2030 give 1.774 doublings, 26,500 GWh more from 2031 to 2035 give 2.969, so on the three rates: EST [44, 56]

	2030	2035
Pack at 13/19/24%, from ₹10,286 (USD 108)	₹8,000/₹7,048/₹6,286 (USD 84/74/66)	₹6,762/₹5,524/₹4,571 (USD 71/58/48)
Cell, from ₹7,429 to ₹8,000 (USD 78 to 84)	₹6,000/₹5,048/₹4,381 (USD 63/53/46)	₹5,048/₹4,000/₹3,238 (USD 53/42/34)
European cell, from ₹11,619 to ₹12,476 (USD 122 to 131)	₹7,048, premium 40%	₹4,762 (USD 50), premium 20%
Lowest LFP cell, from ₹3,429 (USD 36)	about ₹2,381 (USD 25)	about ₹1,810 (USD 19)

BNEF's forecast of ₹7,619 (USD 80) corroborates the 2030 base, but BNEF has undershot twice running: do not adopt it unamended. HARD [44, 56] EST

₹7,048 per kWh

BASE-CASE GLOBAL PACK PRICE IN 2030 (USD 74)

Two independent methods, BNEF's published forecast and a 19 per cent learning curve, land within 8 per cent of each other.

Caveat: the bear price varies only the rate, holding volume constant; a true bear has both. On a bear volume path about 12,150 GWh of additions to 2030 give 1.597 doublings, and at 13 per cent the pack lands near ₹8,286 per kWh (USD 87), not ₹8,000 (USD 84). EST Weak demand means weak price decline: protective of margin, bad for volume.

THE BULL AND BEAR LABELS INVERT FOR A CELL MAKER

Fast cost decline is bullish for electrification, bearish for anyone selling cells. In the bull case the commodity cell reaches ₹4,381 per kWh (USD 46) by 2030 and ₹3,238 (USD 34) by 2035, the lowest LFP cell about ₹1,810 (USD 19). A specialist's price falls fastest where demand is highest, so a column labelled "bull" without saying bull for whom reads backwards. EST

Per cell, the commodity floor for a 21.6 Wh 21700 is ₹163 (USD 1.71), at BNEF's average cell price of ₹7,524 per kWh (USD 79), falling to about ₹109 (USD 1.14) by 2030 on the base path; a bulk premium high-drain contract is estimated at ₹286 to ₹571 (USD 3 to 6), two to three and a half times it: the band a specialist must clear. EST

China's share, and global demand

Two metrics are conflated. Cells made in China were over 80 per cent of 2025 output, running to 78/70/60 per cent in 2030; the share of Chinese-headquartered companies including overseas plants was about 73 per cent of January to August 2026 EV installs, CATL alone 39.4 per cent, running to 80/75/65 and CATL to 42/38/32. HARD [32, 42] EST The base cross-checks: 70 per cent of 3,800 GWh is 2,660 GWh in China against nameplate of roughly 4,225 to 4,355 GWh, so 61 to 63 per cent utilisation, plausible where the sector runs near 40 per cent and CATL at 96.9. EST [35] SOFT [54] Tier-one qualified capacity is full, idle capacity tier two and three: brutal commodity pricing, continued tightness in qualified supply. EST

Global demand, 1,590 GWh in 2025, reaches 3,000/3,800/4,600 GWh in 2030, compounding at 13.5/19.0/23.7 per cent, then 4,600/6,300/8,000 GWh in 2035 at 8.9/10.6/11.7. HARD [36] EST The brackets hold the anchors: McKinsey's 4.2 TWh in 2030 and 6.8 TWh in 2035, just above base EST [58]; the IEA's EV-only about 3 TWh in 2030, about 4 TWh on current policies to almost 5 TWh on stated policies in 2035, 9 TWh net-zero HARD [32]; a pack-level market of roughly ₹33.81 lakh crore in 2030 (USD 355bn) against ₹16.19 lakh crore in 2025 (USD 170bn); utilisation from about 40 to 58 per cent on the committed 6,500 GWh path. EST [44, 35] Bear reads the IEA's EV-only number as the whole market; base is the learning-curve denominator.

Leading indicators

Table A14 lists fourteen indicators with October 2026 readings, triggers and dates. The largest discrete events: Announcement No. 58's suspension lapses on 10 November 2026 SOFT [92]; and an EU anti-dumping initiation on HS 8507.60, where no case was open as of 20 October 2026, HARD by absence. [84]

What each scenario means, for a specialist and for an Indian entrant

The subject is small: about 1.5 GWh marketed, an electrode-coating not a cell-assembly number, a June 2026 cell line at roughly 13 MWh a year, ₹320.46 crore (EUR 30m) raised in March 2026. EST [11] HARD [10] Against 700 to 1,300 GWh of demand it is no supply-side factor: the scenarios set its niche price and the value of its domicile.

India buys in all three: calendar 2025 imports under HS 850760 were ₹38,892 crore (USD 4,083.60m), value up 43.85 per cent and volume 75.12, so it absorbs the price collapse rather than escaping it. SOFT [82] Without domestic cells the FY2030 bill is ₹1.30 lakh crore to ₹2.20 lakh crore a year (USD 14 to 23bn), on the 272 GWh anchor forecast at ₹4,762 or ₹8,095 per kWh (USD 50 or 85); on the 115 GWh floor forecast it starts at ₹0.55 lakh crore. EST [202] Do not present the upper bound as a point estimate.

	BEAR	BASE	BULL
Clearing cell price, demand	₹6,000, Europe about ₹8,381 (USD 88); 700 GWh	₹5,048, Europe ₹7,048; 1,000 GWh	₹4,381, Europe fastest; 1,300 GWh

	BEAR	BASE	BULL
The specialist, what kills it	niches commoditise before qualification completes; killed by customer concentration, the CustomCells and Lilium mode, or by losing engineers to Lyten	0.3 to 0.6 GWh of real output by 2030, two or three profitable niches, a ₹1,068 to ₹2,136 crore (EUR 100 to 200m) business; killed by a second round on worse terms	the reference European specialty supplier, process licensing attached; risk of acquisition
India, the move	cheap cells, but No. 58 licence-gates equipment and graphite anode, so access binds, not price: buy opportunistically, build nothing	imports cheap and Chinese, no cell content mandate: license the process, not a chemistry, or build the first merchant electrode business	EU-origin cells carry a scarcity rent, one EU subsidiary meets both blocs' rules: move early
Hard deadline	10 Nov 2026, China's suspension lapses	31 Mar 2029, India's duty-free window for cell-making capital goods closes SOFT [200]	1 Jan 2029, the Industrial Accelerator Act applies

The ramp arithmetic is scenario-independent: 1.3 GWh by 2027, from 50,000 cells a month, needs roughly 100 times the June 2026 run-rate in about 18 months, a sustained 29 per cent compound monthly growth rate. EST [10]

Content rules written for security, not climate, move the order book. A United States storage project starting construction in 2026 needs 55 per cent of direct costs non-prohibited-foreign-entity, 75 after 31 December 2029, or loses the entire 30 to 40 per cent credit. SOFT [129] SAFE, live now on up to ₹16.02 lakh crore (EUR 150bn), sets a floor of at least 65 per cent EU, EEA EFTA or Ukraine content by estimated cost of the end product. HARD [124] Europe's carbon rules are the slower second wave, deferred by a delegated act that does not exist. SOFT [103]

India in October 2026: roughly 60 GWh of pack assembly, about 23 GWh a year of storage-system assembly, roughly 1 GWh of nameplate cell capacity whose only operating plant made approximately 1.9 MWh in Q3 FY2026, one plant ever localised at electrode level, TDSG at Hansalpur in August 2025, past 80 per cent domestic value creation, against a Central Electricity Authority requirement of 34.72 GWh in FY2026-27 rising to 236.22 GWh by FY2031-32, 46.7 per cent a year. SOFT [214, 243, 209, 206] The unoccupied square is the electrode: the step from 25 to 60 per cent domestic value addition without solving cathode or anode chemistry. EST

Chapter 11

What matures when: the technology S-curves

Of the ten technologies that will define cell manufacturing to 2035, three strengthen the case for a dry-coating specialist, four erode it, two are near neutral, and the one that is its premise is both asset and open question, with a lead of about three years rather than ten.

The ten, ranked by how much they matter to this company

RANK, TECHNOLOGY	CURVE POSITION
1 Dry electrode, the asset	early commercial, volume 2028 to 2030
2 46-series cylindrical, threat	growth, 2027 to 2030 automotive
3 Silicon anode, help	early commercial, 2028 to 2032 in EVs
4 LFP, threat	late growth, maturing
5 All-solid-state, long-dated	pre-commercial, 2030 to 2035 volume
6 Lithium-sulphur, via Lyten	pre-commercial, 2029 onward
7 High-nickel NMC, banked	mature, premium
8 Sodium-ion, optionality	early growth China-led, 2028 to 2030
9 Cell-to-pack and cell-to-chassis	mainstream, China
10 Semi-solid, neutral	shipping, China

The dated board of 21 programmes, owner, guidance and status: **Table A15**.

The three that help

Silicon anode is the best credible adjacency. Slurry stability, binder migration and volume change on cycling defeat solvent routes, so dry routes suit high-silicon anodes, and today's suppliers sell into the drone, eVTOL and defence segments this specialist names: Group14 with Sionic at or above 330 Wh/kg and 842 Wh/L with about 1,200 cycles, plus Amprius and Sila. SOFT [68] Its own claims, more than five years in silicon-carbon and silicon oxide, recipes validated with more than six suppliers, hundreds of kilometres produced, are company statements, not measurements. HARD [11]

All-solid-state is the strongest long-dated hedge, mechanically. Sulphide electrolytes are incompatible with NMP and water slurries, so dry processing is a prerequisite for most roadmaps, yet no all-solid cells were in cars in 2026, guidance runs from Samsung SDI's H2 2027 to Nissan's FY2028, volume 2030 to 2035. SOFT [69] What ships is semi-solid, up 51.9 per cent to 8.0 GWh in 2025, WeLion and Qingtao nearly half: a Chinese-led step this specialist is not on. SOFT [67]

High-nickel NMC is the right chemistry here. Nickel chemistries grew absolutely in 2025 while losing share, and remain the basis for high-power small-format cylindrical cells. HARD [36] Europe retains NMC cathode capability, and Korean makers holding 72 per cent of European production are a supply base. SOFT [113]

The four that threaten, and the one that cuts both ways

LFP sets a floor that keeps falling. LFP grew 48 per cent in 2025, the fastest of any chemistry, and LFP packs at ₹7,714 per kWh (USD 81) against NMC at ₹12,191 (USD 128), a 58 per cent nickel premium, set the comparison price in every customer conversation. HARD [36, 44] China holds roughly 95 per cent of LFP cathode

capacity, and fourth-generation LFP and LMFP know-how is licence-gated since July 2025, never suspended. SOFT [113, 91]

The 46-series caps the addressable market by format, not capacity. The cylindrical centre of gravity migrates from 18650 to 21700 to 46xx: Samsung SDI in volume since March 2025 for BMW, Rivian and Audi, LG's backlog above 440 GWh, Panasonic's 4680 delayed past end-March 2026. SOFT [171, 170] A tabless 21700 suits power tools, drones, medical devices and light e-mobility, not automotive.

Sodium-ion at scale is Chinese, certified and three to four years ahead. CATL's Naxtra is in certified mass production at up to 175 Wh/kg, the first sodium-ion cell to pass GB 38031-2025, with 60 GWh targeted by end-2026 and a 60 GWh HyperStrong agreement, roughly 100 times European scale. Natron Energy, the Western Prussian-blue champion, ceased operations on 3 September 2025 when its board failed to raise funding, cancelling a ₹13,334 crore plant (USD 1.4bn) and 95 jobs. SOFT [63, 64, 66] HARD [65] Altris with Draslovka is two years ahead of the German consortium: a sovereignty and option play, not leadership. HARD [182]

Lithium-sulphur arrives wearing a rival's name. The chemistry is pre-commercial, 2029 the earliest credible date, so the threat is its owner: Lyten holds 16 GWh and Northvolt Labs, targets Heide, 60 to 70 km from Flintbek, for 2028, is hiring more than 600 staff over twelve months and took a Morrow letter of intent in August 2026. HARD [161, 162] From 2028 an American-owned rival in defence, aerospace, drones and data centres bids for the same engineers, grid and customers. EST

Cell-to-pack cuts both ways. CATL's Qilin fills up to 72 per cent of pack volume with cells, favouring large prismatic and blade formats and designing small cylindrical cells out. SOFT In power tools, drones, medical devices, marine and space, cell-level replaceability is required.

Dry electrode: the asset and the question

> **THE PUBLISHED SPECIFICATION DESCRIBES THE WET LINE**
>
> The published electrode-foil specification is a wet, water-based line: 750 mm maximum width, 0.1 to 60 m/min, an 80 metre twelve-zone dryer, "solvent type: water or NMP". HARD [4] A dry line needs neither solvent nor dryer, and the technical deck lists only "pilot production of dry-process electrode". HARD [11] So the installed, revenue-earning, 1.5 GWh-rated capacity is wet coating, and dry coating, the entire premise of the March 2026 round, is at pilot scale. Its width and throughput are NOT FOUND, so benchmarking wet-line numbers against Fraunhofer's 240 mm and 10 m/min dry pilot compares two processes. EST

Use the 27 per cent energy saving at cell level, in two company documents and the founder's April 2026 abstract; the repeated 30 to 40 per cent figure is the electrode step, not the cell, and press-only. Do not average them. HARD [19, 11, 12] SOFT [24]

Dry coating attacks Germany's power-cost disadvantage rather than subsidising around it, but is hard: Tesla needed roughly seven years and billions of dollars for the only confirmed full-dry cathode at mass scale, the hard half. SOFT [77, 78] Series dry coating would be a two to four year lead over LG Energy Solution's 2028 target, but Fraunhofer licenses DRYtraec openly, PowerCo guides 2026 or 2027, and the platform where LiCAP holds a 300 MWh dry line in California and Duerr the equipment wants a gigawatt-scale host now: treat the lead as three years, not ten. SOFT [145, 174, 172] Its intended host closed in May 2026, so an operating electrode line, recipe library, qualified customers and balance sheet would complete it before Korean industrialisation closes the window in 2028 to 2029. EST

From EV-led to storage-led

Benchmark's 2025 demand of 1.59 TWh, up 29 per cent, has EVs at 75 per cent against 77 in 2024 and 79 in 2023, storage up 51 per cent in every region, portable and consumer down to five from six per cent. HARD [36] Storage packs became the cheapest category for the first time in 2025, ₹6,667 per kWh (USD 70), down 45 per cent. HARD [44] SOFT [52]

By region: Europe the most EV-dependent, 85 per cent EV against 8 storage, up from 77 and down from 16; China 77 and 19, storage up from 16; North America's EV share declining, storage 26, up from 16. HARD [36] The EU's record 27.1 GWh sits against the roughly 750 GWh cumulative 2030 requirement in chapter 10. HARD [53] Europe's storage boom has not happened: the open door is 2027 to 2030, where exposure to a softened 2035 target is concentrated.

The genuinely new 2026 demand pools

Table A16 sizes seven pools, data-centre backup, drones, defence, space, eVTOL, robotics and medical, with vendor spreads, mechanisms and relevance to 1.5 GWh. Defence and NATO drones are closed to Chinese cells by rule, not price: they suit a small European producer, not CATL. EST

0.3 GWh per year

UKRAINE'S FPV DRONE CELL DEMAND AT 2025 VOLUMES

Small in energy, closed to Chinese supply by politics, and equal to one fifth of a 1.5 GWh plant's headline capacity.

SANITY-CHECK THE ROBOT NUMBER BEFORE IT REACHES A BOARD

Global humanoid shipments exceeded 50,000 units in 2026 at typically under 2 kWh a pack, about 0.1 GWh. Reaching 74.2 GWh in 2035 needs roughly 37 million robots a year at 2 kWh each, about 15 million at 5 kWh, against global passenger car production of around 75 million. The 74 GWh figure is a heroic upper bound, not a forecast: take the pool seriously for the cell specification, not the vendor model. EST [70]

What helps, what threatens, and the Indian read

What helps. All-solid-state, where dry coating is a prerequisite. Silicon anode. High-nickel NMC, the working chemistry. European sovereignty policy, which turns domicile into a procurement requirement. The non-automotive niches. EST

What threatens. The LFP cost floor. The 46-series. CATL's certified sodium-ion. Lyten's 16 GWh in identical niches. Dry-coating know-how commoditising: open Fraunhofer licensing, PowerCo's guidance, the unhosted Duerr and LiCAP platform, LG's 2028 target. EST

For an Indian reader the ranking inverts into a shopping list: process capability and format are what an entrant cannot buy from China. Cathode know-how is licence-gated, equipment and graphite anode controls lapse back into force on 10 November 2026 unless extended, and Chinese commissioning engineers cannot enter India, roughly 2,000 visas in 2024 against about 200,000 in 2019. SOFT [91, 92, 211] A solvent-free electrode process is manufacturing capability, not chemistry intellectual property: outside the Chinese control regime, aimed at the one constraint India cannot buy around.

Chapter 12

The rulebook, 2024 to 2031

Europe's battery rules are a disclosure and verification regime, neither tariff nor carbon tax. They fall on whoever places a cell on the EU market: a cost to an Asian importer, an argument for a European one. What moves orders is content floors written for security, and the instrument paying cash to European cell makers sets a test UniverCell fails twice.

The instrument, amended three times

Regulation (EU) 2023/1542, adopted 10 July 2023, applies generally from 18 February 2024, deferring Art. 11 removability, Art. 17 conformity assessment, Ch. VI economic-operator duties and Ch. VIII waste management.[96] Its consolidated text of 13 August 2026 carries three amendments: 2024/1781 on ecodesign, 2025/1561 delaying due diligence, and 2026/1738, amending Annex I for vehicle circularity and leaving the 18 February 2027 passport date. SOFT [97]

Carbon footprint (Art. 7), the passport (Art. 77, Annex XIII), recycled content (Art. 8) and performance and durability attach only to rechargeable industrial batteries above 2 kWh[99]: a 21700 cell at roughly 0.021 kWh is nowhere near it. A cell sold for end use falls under the most similar category's requirements, the strictest governing[96]: the merchant maker's duty is to be its customers' data source.

The duties split three ways. Cell maker, pack integrator above 2 kWh and importer of Asian cells all carry CE conformity, safety, substance restrictions, labelling and the QR code; Art. 11 removability binds none, reaching portable and light-means-of-transport batteries only. What a 0.021 kWh cell escapes, integrator and importer carry directly: performance and durability documentation, the carbon-footprint declaration, class and threshold, the passport they create, host and register against a unique identifier, and recycled content. The cell maker is left with performance data where the cell is itself placed on the market for end use, evidence of recycled cobalt, lithium, nickel and lead in its own active materials, producer registration in each Member State, and due diligence above ₹427 crore (EUR 40m) turnover. No tariff, no local-content rule: for a Chinese cell the European importer is both responsible economic operator and producer, standing behind data from tiers that owe it nothing on grid intensity, cross-tier verification and recycled-content proof. EST

The six dates that bite

The full calendar, 2023 to 2050, by class and article, is Table A17. Six dates bite: printed labelling and a nominal carbon-footprint class on 18 August 2026 SOFT [99]; the QR code, battery passport and removability on 18 February 2027 HARD [96, 99]; due diligence above the turnover threshold on 18 August 2027 HARD [100, 101]; the maximum carbon-footprint threshold on 18 February 2029 SOFT [99]; recycled content phase 1 on 18 August 2031, cobalt 16, lead 85, lithium 6 and nickel 6 per cent; and phase 2 on 18 August 2036, cobalt 26, lead 85, lithium 12 and nickel 15 per cent. HARD [98]

> **THE CARBON-FOOTPRINT CLOCK HAS NOT STARTED, AND MOST COMPLIANCE VENDORS SAY OTHERWISE**
>
> Article 7's dates are conditional: the obligation applies on the later of the calendar date and 12 months (declaration) or 18 months (class and threshold) after the delegated or implementing act takes force. That act is not adopted: an EV draft went for feedback in April 2024 and had not reached the Official Journal by late September 2026, reportedly over grid-electricity modelling. So 18 February 2025 (EV) and 18 February 2026 (industrial above 2 kWh) have not triggered {SOFT, strong}[103], and vendors presenting 18 February 2026 as live have a commercial interest in urgency. The Commission allows fields whose obligation has not started to be left empty on 18 February 2027. Expect the bite between 2027 and 2029; treat any compliance-driven order book dated from February 2026 as overstated. EST

Due diligence: delayed and gated low

Due diligence starts 18 August 2027, moved from 18 August 2025 by Regulation (EU) 2025/1561, adopted 18 July 2025 and in the Official Journal on 30 July 2025; only the start and guidelines dates moved, scope unchanged, guidelines now due 26 July 2026, not 18 February 2025. Annex X covers cobalt, natural graphite, lithium and nickel and their compounds, verified by a notified body.[100] The exemption in force is net turnover below ₹427 crore (EUR 40m) in the year before the last, tested group-wide; the circulating ₹1,602 crore (EUR 150m) figure is not law but the un-enacted Omnibus IV proposal, which would exempt mid-caps under 750 employees at or below it, or a balance sheet at or below ₹1,378 crore (EUR 129m), still not in force in mid-September 2026. HARD [101, 102] UniverCell, ₹320.46 crore raised (EUR 30m) in March 2026 and about 92 people, carries it as a customer-side data burden. EST

The money, and its wall

The Battery Booster, announced December 2025 in the Commission's automotive package, is ₹19,228 crore (EUR 1.8bn): ₹16,023 crore (EUR 1.5bn) of interest-free Innovation Fund loans for cell producers in ramp-up, ₹3,205 crore (EUR 300m) for critical raw materials. Alongside it the Commission claimed **₹38.99 lakh crore (EUR 365bn) of additional EU GDP and 170,000 jobs by 2030**, a political magnitude rather than a forecast. {HARD as a Commission claim}[104, 108] Established 9 June 2026 from the Innovation Fund, that is emissions-trading auction revenue SOFT [107], the Facility ran one call, 28 July to 30 September 2026, lending up to 60 per cent of eligible costs, at most ₹5,341 crore (EUR 500m) per project. HARD [105, 106] Eligibility needs European Economic Area manufacturing, cells for electric-vehicle use and ramp-up phase at the opening; two conditions then decide access, the applicant's first full commercial-scale EV cell project globally and **planned annual capacity of at least 10 GWh**. HARD [105]

Most cell capacity in Europe cannot apply: LG Energy Solution, Samsung SDI, SK On, CATL and Envision AESC already run commercial-scale EV cell plants somewhere and fail the "first globally" test, leaving ACC, Verkor and PowerCo the reported qualifiers. SOFT [109] Europe's home-grown makers kept dying between pre-series and commercial, Northvolt in March 2025, Morrow on 6 May 2026. UniverCell fails twice: a 1.5 GWh marketed headline against the 10 GWh floor, itself an electrode-coating number not cell assembly, and no EV lead. Selected projects, applicant count and allocation are **NOT FOUND** at 20 October 2026, awards and first payments targeted before end-2026; the criteria for the ₹3,205 crore raw-materials strand are **NOT FOUND** too.[105] Adjacent: BATT4EU ₹9,881 crore (EUR 925m, 2021 to 2027), InvestEU ₹21,364 crore (EUR 2bn, 2026 to 2027). SOFT [108]

10 GWh minimum planned annual capacity

THE BATTERY BOOSTER FACILITY'S FLOOR, AGAINST UNIVERCELL'S 1.5 GWH HEADLINE

Also its first commercial-scale EV cell plant anywhere. Two tests, two failures.

Benchmarks, state aid, local content

Table A18 sets out the Critical Raw Materials Act, RESourceEU, the Net-Zero Industry Act, the Clean Industrial Deal, both IPCEIs and the Industrial Accelerator Act (IAA). The CRMA's 65 per cent single-country cap is already breached on the three materials that matter: China refines 74 per cent of lithium, 76 per cent of cobalt and 94 per cent of battery-grade graphite, and the cap is unenforceable[32]; CRMA and NZIA numbers are benchmarks without penalty. EST UniverCell is in no IPCEI participant list, **NOT FOUND**, against 59 companies in 12 Member States and up to ₹65,160 crore (EUR 6.1bn) of state aid.[118, 119]

The binding local-content rule sits not in the Battery Booster, whose call text has none, but in the IAA, proposed 4 March 2026: Union-origin phasing for storage escalating after three years to the cells, non-price auction criteria, procurement from 1 January 2029. HARD [110] Conflating the December 2025 package with the July 2026 call makes a wrong slide. The IAA creates a scarcity rent on EU-origin cells, priced into 2027 and 2028 tenders though the binding date is 2029. EST

CBAM DOES NOT TOUCH A BATTERY CELL, AND SAYING OTHERWISE IS THE EASIEST ERROR TO MAKE

The definitive Carbon Border Adjustment Mechanism began 1 January 2026 in cement, iron and steel, aluminium, fertilisers, electricity and hydrogen, importers as authorised declarants for embedded emissions. Neither cells nor packs are CBAM goods in 2026. HARD **by absence from the sector list** [121] A 50-tonne de minimis exempts roughly 90 per cent of importers while retaining about 99 per cent of embedded emissions. The 2028 expansion was proposed 17 December 2025 for roughly 180 downstream products from 1 January 2028, 94 per cent of them intermediate industrial goods and the steel and aluminium share averaging 79 per cent; the Council's approach of 12 June 2026 covered about 200 goods, and Parliament voted 464 to 50 to go beyond 450 items, with anti-circumvention measures. Whether CN 8507 appears is **NOT FOUND**, not confirmed either way. [122, 123] Enclosures and containers are plausibly 2028 CBAM goods; the cell inside is not. The constraint on batteries is Regulation 2023/1542. EST

Germany: research restarted, power unfixed

Table A19 carries the federal research line from the 2024 crash to the 2026 restart, the Fraunhofer FFB pilot fab, the power subsidy with its NACE test, and Schleswig-Holstein: a grid **91.2 per cent renewable** in 2025, generating roughly three times what the state consumes HARD [136, 137], with **NOT FOUND** for any state battery strategy or funding line; its levers are power, permitting and land.

Power is the unfixed problem: **₹24.18** per kWh (22.64 EUR cents) in Germany's Eurostat band IC, 500 to 2,000 MWh a year, in H2 2025, about 23 per cent above the EU average and 2.3 to 3.3 times Chinese and 2.6 to 3.9 times Indian industrial power. HARD [131, 147] At or below 1.5 GWh the company pays in that band, not the 14.40 cent (₹15.38) large-user rate, and the 2026 subsidy blends to ₹2.00 (1.87 cents) per kWh, an 8 per cent cut on a ₹24.18 base: margin repair only. EST

In its favour: that 91 per cent feed-in is Germany's best carbon-footprint input, a number for an Art. 7 declaration, and dry coating removes electrode manufacture's most electricity-intensive step, 27 per cent at cell level. {HARD, the company's own figure}

Defence and space: rules that bite now

SAFE, up to ₹16.02 lakh crore (EUR 150bn) within the ₹85.46 lakh crore (EUR 800bn) ReArm Europe and Readiness 2030 frame, is live: EU, EEA EFTA or Ukraine content at least 65 per cent of end-product cost, third countries capped at 35 per cent, category-two projects keeping design authority. HARD [124, 125] Applied to battery-electric military systems, drones to microgrids, it creates demand Chinese imports cannot meet.

65 per cent of end-product cost

MINIMUM EU, EEA EFTA OR UKRAINE CONTENT IN A SAFE-FUNDED DEFENCE PURCHASE

Live now. A floor on cost, not a disclosure duty. The market behind it is NOT FOUND, not to be invented.

The One Big Beautiful Bill Act, enacted 4 July 2025, bars firms with specified China-linked ties from the US clean-energy credits, Sections 48E, 45Y and 45X, with Treasury and IRS guidance in Notice 2026-15 of February 2026. For storage the Material Assistance Cost Ratio, non-prohibited over total direct costs, must reach 55 per cent for construction starting in 2026, then 60, 65, 70 and 75 per cent for 2027, 2028, 2029 and after 31 December 2029; failure voids the entire 30 to 40 per cent investment tax credit. SOFT [129, 130] Cells are the largest cost line in a storage system, so a non-Chinese, non-US-tariffed European cell is one of few qualifying options: the likeliest explanation for UniverCell's reported multi-million-euro US agreement. EST

On space, ESA's European Components Initiative has run since 2004 toward an ITAR-free posture for sensitive satellite components, and geographical return favours a qualified European cell, Saft of France near a single point of European supply: a strategic gap, not a crowded market, though qualification runs for years on tiny volumes, so margin and credibility, not volume. EST [126, 127] Two things are **NOT FOUND** and may not be invented: any quantified EU defence or space battery market, and the export-control treatment of a German-made cell sold to a US space customer, most such satellites having moved from the US Munitions List to the Export Administration Regulations under ECCNs 9A515 and 9E515 in 2013 to 2014. No ITAR conclusion without specialist review. [128]

India: supply-side only

Chapter 14 carries India. The ACC PLI programme, approved May 2021, pays for output, sets no content requirement on cell buyers, and sets a minimum facility 10 to 50 times a specialty plant. Against a 50 GWh target and ₹18,100 crore: 1.4 GWh commissioned, **2.8 per cent of target**, **zero rupees claimed**, the line cut 44 per cent in Budget 2026-27 to ₹86 crore. SOFT [193, 194, 195, 198] Starved by its own milestones, not cancelled: little is commissioned, so no incentive is payable. EST

The duty stack is inverted-duty protection: 20 per cent basic customs duty on finished cells (HSN 85076000) against zero duty on 85 types of cell-making capital goods to 31 March 2029. SOFT, conflicting [199, 200] So exporting cells into India is tariff-disadvantaged, licensing technology and selling equipment or coated electrode tariff-advantaged, and electrode foil may classify outside heading 8507. EST

India asks only 20 per cent local content on viability-gap-funded storage, live now, and 25 per cent domestic value addition in two years rising to 40 per cent in five in the ACC PLI's third tranche [228, 196]; no approved-list regime for cells exists, **NOT FOUND** as of October 2026. [243] The solar file is the sequencing lesson: ALMM List-II's domestic-cell requirement slipped from 1 June 2026 to 1 January 2027, approved cell capacity being only about 16 per cent of module capacity [247], and once integrated makers took their own output barely 1.9 per cent of standalone module makers' needs reached the open market. [248] India's ratio is worse, roughly 60 GWh of pack assembly on about 1 GWh of cells. EST

How hard each rule bites

Eleven instruments, ranked by enforceability and date certainty, not political prominence, this report's judgement EST: the US prohibited-foreign-entity and MACR cost-ratio test first, then SAFE's content floor, both live; the Industrial Accelerator Act, reaching storage above 1 MWh; the Battery Regulation's labelling, QR and passport, then its due diligence, then, well behind, its carbon footprint; CBAM, which does not reach batteries; India's viability-gap content floor, reachable without cells; the CRMA and NZIA benchmarks, which carry no penalty; the Battery Booster Facility, open to almost nobody; and last the ACC PLI, which has paid nothing.

Europe's carbon rules are the slower second wave, held up by a missing delegated act; what moves orders now is American and European content rules, turning "not Chinese" into a procurement requirement. India's stack is supply-side only, no content mandate on cells, on a cell base of roughly 1 GWh: the gap a European process-technology holder sells into, with the duty-free equipment window closing 31 March 2029. EST

Chapter 13

Strategic analysis

Every framework inverts with the market boundary: inside the specialty arena UniverCell faces a structurally mediocre industry it can survive, inside the commodity cell arena a hostile one it cannot. The one force worse in the niche is supplier power, exactly what its in-house electrode capability was built to answer.

CONDITIONS IMPOSED ON THE FRAMEWORKS KEPT

Research assessed SWOT, PESTEL, BCG, the value chain and the strategy canvas as unusable or decorative. SWOT is not reproduced, Chapter 06 owning the pros and cons, nor the canvas, whose factors and scores would have been invented. On the rest: every PESTEL entry carries a date or a number, BCG states relative share as a band against niches this report defines, the value chain a value-added estimate per stage, and an uncertain cell says NOT FOUND.

Segment definition, before anything is applied

The arena: custom and semi-custom cells in lots of 10,000 to 10,000,000 a year, high mix, sold where qualification not price gates the buy, plus merchant coated electrode foil sold as a product to third-party cell makers, not cells alone.[4] Inside sit the 18650 and 21700 cylindrical formats and custom pouch, high-nickel NMC and NCA to 92 per cent nickel, graphite and graphite-silicon to 80 per cent silicon, LTO, LMFP and specialty variants, sold into power tools, drones, eVTOL, medical, satellites, marine, robotics, critical infrastructure and defence across the EU and EEA plus the mandated non-Chinese slice of US demand.[11] Realised price inside is ₹13,334 to ₹28,572 per kWh (USD 140 to 300) EST against a commodity floor of ₹3,429 to ₹8,000 per kWh (USD 36 to 84).[44] Outside sit single-SKU programmes above roughly 50 million cells a year, the 46-series, prismatic, blade and cell-to-chassis, commodity LFP for volume EV and grid storage, passenger-car traction, utility-scale storage, consumer electronics, and price-led demand everywhere.[10, 11, 19, 21]

The funded target, forty million cells a year at 21.6 Wh, is 0.864 GWh:[10] three times Ukraine's estimated annual FPV drone pack demand, three million drones in 2025 at about 0.1 kWh a pack, and six per cent of EVE Energy's Malaysian plant at 680 million 21700 cells a year. EST from HARD [178]

0.86 GWh fully funded annual cell target

WHAT THE SERIES B BUYS

Three times Ukraine's estimated FPV drone pack pool, six per cent of one Malaysian plant: arena choice makes it strategic or trivial.

The commodity arena fails on five counts. **Scale:** CATL alone installed 242.7 GWh in the first half of 2026, 281 times the funded target;[41] PowerCo's Salzgitter is 40 GWh, 46 times it, producing since December 2025;[175] Morrow Batteries went bankrupt on 6 May 2026 at 1 GWh, judged too small for a carmaker.[158] **Price:** European packs run about 56 per cent above Chinese, ₹12,476 against ₹8,000 per kWh (USD 131 against USD 84), and the learning curve puts the 2030 global average cell near ₹5,048 per kWh (USD 53) against a European cell at ₹7,048 (USD 74). EST [44] **Format:** the cylindrical centre of gravity is migrating to the 46-series, Samsung SDI in 46-phi mass production since March 2025, an LG backlog above 440 GWh, cell-to-pack designing small cylinders out of the vehicle. SOFT [171, 170] **Customer structure:** automotive is gated by IATF 16949, which the company neither holds nor needs, having explicitly avoided automotive;[21, 28] every European cell survivor has a carmaker shareholder, ACC with Stellantis, Verkor with Renault, PowerCo with Volkswagen, UniverCell none.[186]

The base rate: of the independent, non-OEM-anchored European merchant cell startups attempting industrial-scale production between roughly 2015 and 2026, none is identifiably profitable and at least four failed outright. EST

Porter's five forces, scored for two arenas at once

Scoring: 1 benign for an incumbent in that arena, 5 punishing; a higher total is worse. EST throughout.

FORCE	SPECIALTY NICHE	COMMODITY EV CELL	GAP	WHAT THE GAP IS MADE OF
Competitive rivalry	**3**	**5**	minus 2	An 8% fall into fourfold overcapacity
Buyer power	**3**	**5**	minus 2	Cancelled billion-euro contracts
Supplier power	**5**	**4**	**plus 1**	**Leverage is bought with volume**
Threat of entry	**3**	**4**	minus 1	Qualification, not capital; know-how licensable
Threat of substitutes	**3**	**4**	minus 1	Format and chemistry migration
Total, lower is better	**17 of 25**	**22 of 25**	**minus 5**	

The published basis for all ten scores is **Table A20**.

> THE VERDICT, AND THE PART USUALLY MISSED
>
> The niche is the better arena, 17 of 25 against 22 of 25, four of the six points of improvement from rivalry and buyer power alone. Not a good industry; a survivable one. The decisive detail moves the wrong way: supplier power is worse in the niche, leverage over an upstream 85 to 98 per cent Chinese being bought with volume. The load-bearing asset is therefore in-house anode and cathode production, not dry coating, and it is what an Indian entrant cannot buy from China at any price: cathode technology is licence-gated and commissioning engineers cannot get a visa.[91, 211]

PESTEL, every entry dated or quantified

All twenty-four entries, dated or quantified and tagged for a European specialty cell maker, are **Table A21**. The six that carry the argument:

- **P, tailwind.** OBBBA prohibited-foreign-entity rules, enacted 4 July 2025: 55 per cent non-prohibited direct costs in 2026, 75 per cent after 2029; guidance Notice 2026-15, February 2026. SOFT [129]
- **P, ambiguous.** July 2025 Chinese cathode technology licensing never suspended; Announcement No. 58 suspended only to 10 November 2026, an extension to 10 January 2027 reported, not confirmed.[91, 92]
- **Ec, headwind.** Pack ₹10,286 per kWh (USD 108) in 2025, minus eight per cent; 2030 pack floor roughly ₹6,667 to ₹8,095 per kWh (USD 70 to 85) on two methods. EST [44]
- **S, tailwind for a European holder of engineers.** 1,000 to 1,500 trained process engineers per GWh; India would need roughly 36,000 for its announced gigafactories, and issued roughly 2,000 visas to Chinese nationals in 2024 against 200,000 in 2019.[209, 211]
- **T, headwind.** Dry electrode industrialisation by others: PowerCo 2026 or 2027; LG 2028, 2029 for dry LFP; Samsung SDI two pilots, no date, NOT FOUND.[174, 169, 171]
- **L, tailwind.** Battery passport mandatory 18 February 2027 for LMT, industrial above 2 kWh and EV batteries; printed labelling 18 August 2026.[96, 99]

Eleven tailwinds, seven headwinds, five ambiguous, one neutral. Legal tailwinds can be repealed or deferred, as the carbon-footprint regime already has for want of a delegated act; economic headwinds are continuous, and a gap of two to three times on power and 56 per cent on pack price does not pause.

The value chain, and where the profit actually sits

Value added splits Deloitte's structure: materials extraction and processing 50 to 60 per cent, cell production 15 to 30 per cent, the cell-band split an estimate.[55] By stage, with UniverCell's participation EST: mining 10 to 15%, no; refining 15 to 20%, no; active materials 20 to 25%, no, bought in on a validated multi-supplier base with an all-EU and North America option stated;[11] **electrode manufacture 5 to 10%, yes, and the revenue business**;[4,7] cell assembly 8 to 15%, yes from June 2026 at 0.0130 GWh a year;[10] module, pack and BMS 10 to 15%, no, its customers, German pack integrators included SOFT; system integration 10 to 20%, no; recycling 2 to 5%, no. Who controls each stage, with the published share, is **Table A22**.

The bands are judgements and do not sum to 100 per cent at either extreme, 80 per cent low and 125 per cent high; bands, because the source splits the chain in two, not eight.

5 to 10% of battery value added

THE ELECTRODE STAGE, WHERE UNIVERCELL PLAYS

Refining and active materials carry 35 to 45 per cent on this split and are 74 to 98 per cent Chinese-controlled. The company is not in the profit pool.

UniverCell buys from the stage 2 and 3 pool, at 74 per cent for lithium refining to 98 per cent for graphitization. A value pool is not a margin: stage 4 has no dominant controller and almost no merchant competition, so pricing power there can beat a 20 to 25 per cent stage facing a 90 per cent incumbent in fourfold overcapacity. For 92 people with ₹320.46 crore (EUR 30m) the 5 to 10 per cent stage is the only rational choice; an Indian group with capital should take both, stage 3 with Epsilon and Altmin, stage 4 localised in India once, by TDSG at Hansalpur in August 2025, figures in Chapter 14.[215,216,214]

BCG growth-share matrix: an assessment, not a measurement

No segment accounts, revenue or capacity are published, so relative share is a band against niches this report defines, definitions that move the answer; vendor growth models disagree. EST throughout.

BUSINESS LINE	SEGMENT GROWTH	RELATIVE SHARE	QUADRANT AND CASH
Merchant electrode foil	NOT FOUND; proxy 10 to 30% a year, nameplate capacity up 30% in 2025 to over 4 TWh and committed 2030 capacity near 6.5 TWh[32,35]	**HIGH** in European merchant custom electrode; only named global peer AM Batteries, pre-volume[176,7]	**Star.** Sole cash engine, funding the rest
Cylindrical 21700 cells	9 to 27% a year, vendors disagreeing two-fold on scope: 9.3% for 18650 and 21700, 25.8% for 21700 alone, 26.5% on a demand model	**VERY LOW** in premium high-drain 21700; EVE's one plant is seventeen times the funded target and 1,133 times the June 2026 run-rate[178,180]	**Question mark.** Absorbs the ₹320.46 crore
Custom pouch cells	NOT FOUND; supply contracting by exit[148,157]	**NOT SCOREABLE.** No capacity, Ah, Wh, energy density, C-rate, cycle life or temperature range published[5]	**Probable dog**, unresolved. Small
Sodium-ion, Litona under SIB:DE	High from a tiny base: CATL in mass production from December 2025 targeting 60 GWh by end-2026, single digits of the market until 2030[63,64,181]	**VERY LOW** in European sodium-ion; neither UniverCell nor Litona in the European field survey, where Altris with Draslovka has a ₹206 crore (EUR 19.3m) partner, cathode chain and production late Q3 or early Q4 2026[182,181]	**Question mark with a casualty:** Natron Energy ceased operations 3 September 2025, a ₹13,334 crore (USD 1.4bn) factory cancelled[65]

The cash finding. Three lines consume cash; one generates it, marketing subordinating it to the cell story. A leading data vendor labelled UniverCell a "developer of custom electrodes" with a competitor set of six silicon-anode and cathode materials companies that are in fact potential suppliers.[27] The only cash cow is the only high-share business, and ₹320.46 crore funds two question marks and one unscoreable line against roughly 29 per cent compound monthly growth for eighteen months.

TAM, SAM and SOM, with the arithmetic on the page

Global lithium-ion at cell level in 2025 was approximately ₹12.00 lakh crore (USD 126bn): 1,590 GWh at roughly ₹7,524 per kWh (USD 79), range ₹11.33 lakh crore to ₹15.33 lakh crore (USD 119bn to 161bn) across 1,590 to 1,897 GWh at ₹7,143 to ₹8,095 per kWh (USD 75 to 85). EST from HARD [36, 44] Useless as a denominator: the company's capacity is smaller than the forecast error.

Six published vendor segments, 2026, at pack and system level {EST, vendor}: space ₹40,763 crore (USD 4.28bn) at 7.3 per cent CAGR; portable medical ₹30,001 crore (USD 3.15bn) at 11.6 per cent; data-centre backup ₹33,620 crore to ₹47,239 crore (USD 3.53bn to 4.96bn), the 40 per cent spread a scope difference, backup units against backup plus storage; military ₹16,762 crore to ₹24,477 crore (USD 1.76bn, 2.52bn and 2.57bn from three vendors), a 46 per cent spread; drone and UAV ₹15,715 crore (USD 1.65bn) at 8.2 per cent; and eVTOL on a cells-only scope ₹1,214 crore (USD 127.5m), its USD 0.9bn alternative whole-aircraft and not additive. Mid-point USD 15.6bn, extremes USD 13bn to 18bn, or ₹1.24 lakh crore to ₹1.71 lakh crore; high-drain power tools excluded for want of a credible cell-demand figure, NOT FOUND.[178] Cells were 68 per cent of pack cost on BNEF's 2024 split, roughly 73 per cent on the 2025 implication, but specialty assemblies carry more non-cell content, so the conversion below is 40 to 60 per cent. EST [50, 44]

LAYER	VALUE	ARITHMETIC
TAM, cell level	USD 5.2bn to 10.8bn, ₹49,525 crore to ₹1.03 lakh crore	USD 13bn to 18bn x 0.40 to 0.60
SAM	USD 0.62bn to 2.59bn, ₹5,905 crore to ₹24,667 crore	x 0.30 to 0.40 geography x 0.40 to 0.60 format and qualification
SOM, funded target	₹1,152 crore to ₹2,467 crore (USD 121m to 259m)	40,000,000 x 21.6 Wh = 0.864 GWh = 864,000 kWh at USD 140 to 300/kWh; 5 to 42% of SAM, midpoint near 12%
SOM, June 2026 run-rate	₹17 crore to ₹37 crore (USD 1.8m to 3.9m)	50,000 x 12 x 21.6 Wh = 0.0130 GWh = 12,960 kWh, same prices; 0.07 to 0.63% of SAM
Electrode revenue today	₹106.82 crore to ₹1,057.52 crore (EUR 10m to 99m), FY2025 SOFT [23]	Three to sixty times the cell line

Geography keeps 30 to 40 per cent: Europe is almost 15 per cent of EV battery deployment and 6 to 7 per cent of global cell capacity but a higher share of defence, space and medical procurement, plus US demand where the material assistance cost ratio forces non-Chinese sourcing.[32, 129] Format and qualification keeps 40 to 60 per cent: arena formats only, no 46-series or prismatic, and no near-term flight-qualified space or implantable medical, no certification being evidenced and Saft incumbent on decades of flight heritage. EST [11, 126]

Cross-check: at ₹13,334 to ₹28,572 per kWh the TAM implies **17 to 77 GWh a year** of specialty demand, EVE's plant at 14.7 GWh just below. EST

₹5,905 to ₹24,667 crore serviceable addressable market

USD 0.6BN TO 2.6BN OF SPECIALTY CELL DEMAND REACHABLE

Four times wide, because the vendor models underneath it are; every filter is stated, so a reader can re-run it.

The reframe. The numerator is a plant and the denominator's error bar is wider, so share of market is not the live variable; realised rupees per kWh by niche is, and which 0.864 GWh gets sold decides the return. Against the commodity cell, price is highest in space and critical-care medical, on tiny volumes, monopsony buyers, ECSS and ISO 13485 lock-in, qualification costing years with none evidenced today;[126, 11] high in defence and drones, where SAFE's 65 per cent content floor excludes Chinese cells by law and two to five year qualification locks buyers in, live now;[124] high in eVTOL and robotics, mass-sensitive to twelve grams of aluminium, qualification programme-linked, the pool that killed CustomCells through one customer's non-payment;[148] moderate in power tools and light e-mobility, a premium high-drain 21700 band of ₹13,334 to ₹26,667 per kWh (USD 140 to 280), two to 3.5 times the commodity floor of about ₹163 a cell (USD 1.71), on low qualification cost, where EVE's 680 million cells land; EST [44, 178] and lowest in marine and stationary, price-led, storage packs the cheapest lithium-ion category in 2025 at ₹6,667 per kWh (USD 70).[44]

VRIO, on two resources rather than one

Dry coating and in-house electrode capability are routinely treated as one asset. Scored separately they reach opposite conclusions, which is the investment case.

TEST	DRY COATING	IN-HOUSE ELECTRODE CAPABILITY, RECIPES AND FORMATION
Valuable	**Yes.** 27% lower energy per cell on two company primary documents, hitting the cost line a German plant cannot subsidise away; LG claims 17 to 30% lower electrode manufacturing cost and roughly 30% less factory land[19, 11, 169]	**Yes.** Over 1,000 tested recipes, a 1,000th customer order January 2026, in-house formation, breadth across high-nickel NMC, graphite, graphite-silicon, LFP, LCO, LMFP, LMNO, LTO and XNO[7, 11]
Rare	**Yes, narrowly.** Tesla alone has confirmed full-dry cathode at mass scale; UniverCell is credibly in the global top three on operating activity, at pilot scale on its own documents[77, 11]	**Yes, rarer.** The only named merchant dry-electrode peer is AM Batteries, pre-volume on roughly ₹657 crore (USD 69m)[176]
Inimitable	**No, three-year clock.** Fraunhofer IWS licenses DRYtraec openly at up to 240 mm and 10 m per minute; PowerCo 2026 or 2027 with Koenig und Bauer; LG 2028, and 2029 for dry LFP; Sakuu's October 2026 SPAC at a ₹5,714 crore (USD 600m) pre-money value[145, 174, 169, 177]	**Partially, for longer.** Recipes and formation windows are tacit, cumulative and customer-validated; they cannot be licensed, only hired
Organised to capture	**Unproven.** 92 people, no named CTO, CFO or VP Manufacturing, hiring twelve months behind plan, ₹320.46 crore against roughly 29% compound monthly growth for eighteen months[2, 10, 11]	**Unproven.** Already monetised; scaling untested

TEST	DRY COATING	IN-HOUSE ELECTRODE CAPABILITY, RECIPES AND FORMATION
Verdict	**Temporary advantage,** expiring around 2028 to 2029	**Potentially sustained, under-exploited:** the recipe library and the qualified electrode customer base, not the absence of solvent

Tesla proves it valuable and hard: roughly seven years and billions of dollars, its chief executive calling it a mistake and "way harder" than expected at the 2025 shareholder meeting, 4680 performance underperforming in May 2026; it sells no merchant cells, a benchmark not a rival.[77, 78] **LG Energy Solution** sets the expiry: Ochang pilot complete, mass production targeted 2028, Korean makers already 72 per cent of European production.[169, 113]

> **WHAT A BUYER IS ACTUALLY BUYING**
>
> Dry coating is a three-year option that erodes whether or not it is purchased, and it is cheaper from Fraunhofer, or from Duerr and LiCAP, who have a validated European platform and no pilot host. The recipe library, the formation window, the qualified electrode customer base and the engineers holding all three cannot be bought that way. Pay for the dry-coating story without the people and the process package and the purchase is the wasting asset, the durable one left behind.[172, 173]

The one-line strategic read

UniverCell is an electrode-process business with a cell business attached, holding a real but unpatented three-year process lead into a policy-created demand window that opens with the battery passport in February 2027 and closes as the Koreans industrialise around 2028: a licensing, offtake and process-transfer counterparty rather than an equity investment, whose merchant electrode line, not its 21700, is the asset an Indian industrial group should want.

Chapter 14

The read for India

Not demand, not capital: roughly 290 GWh of announced Indian capacity produces about 7.6 MWh a year, the import bill has grown twelvefold in seven years, and what separates them is process knowledge and the engineers who carry it, which China restricts from transfer. Europe's electrode capability is what India can buy.

Demand: one country, two forecasts, a 2.4 times spread

Three published forecasts of 2030 Indian advanced-cell demand disagree by 2.4 times and are not averaged: the spread decides whether the right plant is 1 or 10 GWh.

SOURCE	2030 FORECAST	SCOPE
IEEFA with JMK Research	**about 272 GWh by FY2030**, **36.5 per cent** compound from **28 GWh** in calendar 2025	includes grid storage; the anchor here SOFT [202]
India Cellular and Electronics Association	**115 GWh by 2030**, EV sub-segment 48 per cent	the conservative floor SOFT [204]
NITI Aayog, accelerated, 2022	**260 GWh by 2030** plus about 600 GWh cumulative storage potential	rate not stated SOFT [205]
pv magazine tally	**about 290 GWh announced, 30-plus sites**; 1.4 GWh operational nameplate	**supply, not demand**, routinely misread SOFT [208]

The gap is mostly grid storage, largely excluded by the electronics association. The 28 GWh base splits roughly 60 per cent electric vehicles to 40 per cent stationary and grid. SOFT [203] The repeated 3,100 GWh by 2030 at 16 per cent is global, not Indian. SOFT [205]

2.4x forecast spread

INDIA'S 2030 CELL DEMAND, 115 GWH AGAINST 272 GWH

Same country, same year. Size against the floor, treat the upside as option value.

The grid moves fastest: the Central Electricity Authority requirement runs **34.72 GWh in FY2026-27 to 236.22 GWh by FY2031-32**, 46.7 per cent a year on **₹3.49 lakh crore to ₹4.79 lakh crore** of investment; the August 2026 pipeline is 187 projects, **176.66 GWh**: 9.08 operational, 79.92 under construction, 24.13 awarded, 63.53 under tender, plus 69 new tenders of 102 GWh in the year, up 35 per cent on 2024. SOFT [206] EST on the rate. Each pool rests on one model:

- **Two-wheelers**: 2.36 kWh average pack, **81 GWh realistic to 154 GWh optimistic** to decarbonise the segment by 2030. SOFT [207]
- **E-buses**: 4,441 in 2025; **7.2 to 9.6 GWh cumulative** across the two largest electrification programmes; ₹6,055 crore pack market by 2030 (USD 635.73m). SOFT on GWh, EST on the value [207]
- **Telecom, data centres, rooftop**: **0.3 GWh today to 12 GWh**, period unstated, read as to 2030; Cygni Energy and Indus Towers non-binding, up to 1.5 GWh, 5 October 2026. SOFT [204]
- **Consumer electronics**: **32.2 per cent of the 2024 Indian lithium-ion market**, the largest end use; no usable forward figure, scopes differ. EST, low confidence
- **Power tools**, a named UniverCell market: **₹10,762 crore to ₹17,619 crore in 2026 (USD 1.13bn to 1.85bn)**, a 64 per cent spread across three vendors, 8.6 to 11.4 per cent a year to 2031 or 2035 on three

irreconcilable models; no Indian high-drain cylindrical source; a GWh figure **NOT FOUND**. EST, low confidence [249]

The import bill, calendar made explicit

Calendar 2025 is January to December 2025; FY2025 is April 2024 to March 2025: not comparable. Conversions use ₹95.24 to the dollar. The bill ran **₹3,663 crore in FY2018-19 (USD 384.60m)**, the source's own ₹3,571 crore; **₹26,667 crore in FY2022-23 (USD 2,800m)**; **₹27,037 crore in CY2024 (USD 2,838.87m)**; **over ₹28,572 crore in FY2024-25 (USD over 3,000m)** against the source's ₹27,900 crore, volumes not stated. SOFT [202, 82] The readings since:

PERIOD	₹ CRORE (USD M)	VOLUME, UNIT PRICE AND TAG
CY2025	**38,892** (USD 4,083.60, up 43.85 per cent)	**90.45 kt, up 75.12 per cent**; ₹43.00 lakh per tonne (USD 45,150), down 17.86 per cent; **China 93.26 per cent** SOFT [82]
LTM May 2025 to Apr 2026	**46,995 / 44,076** (USD 4,934.41 GTAIC / 4,627.93 Volza)	115,737 t, up 90.90 per cent; ₹40.59 lakh per tonne (USD 42,634); China over 93 per cent SOFT [82, 246]
FY2026, all-basis	**44,763** (USD 4,700)	China **84 per cent** across all lithium-ion SOFT [206]
China to India, LTM to mid-2026	**55,631** (USD 5,841.10, up 73.26 per cent)	mirror data, above Indian customs SOFT [80]

Three sources give three values for the same twelve months, so this report states **roughly ₹44,000 crore to ₹47,000 crore in FY2026 (USD 4.63bn to 4.93bn)**, not a point. Volza and GTAIC resell customs data: acceptable, not primary. Tradestat, the Ministry of Commerce series, sits behind a dynamic query interface: **NOT FETCHED**. SOFT China's share reconciles on wording: **84 per cent** of all lithium-ion, **93 to 94 per cent** of HS 850760 cells. EST

The growth survives all three: **twelvefold in seven years**, volume rising twice as fast as value, 75 against 44 per cent in calendar 2025, 91 against 57 per cent in the rolling year. **India is absorbing the global price collapse, not escaping the dependence.** EST on the reading, SOFT [82]

VERIFYING THE FY2030 NUMBER

IEEFA and JMK put 272 GWh of FY2030 import-met demand above USD 23bn at USD 85 per kWh. SOFT [202] It checks: at ₹8,095 per kWh (USD 85), **₹2.20 lakh crore (USD 23.12bn)**. The price input is stale: BNEF's December 2025 survey puts the pack at ₹10,286 per kWh (USD 108), the lowest observed LFP cell at ₹3,429 per kWh (USD 36). HARD [44] A point estimate pulls apart; a surface holds. At ₹4,762 and ₹6,667 per kWh (USD 50 and 70) the same 272 GWh costs **₹1.30 and ₹1.81 lakh crore**; the ICEA floor of 115 GWh costs **₹0.55, ₹0.77 and ₹0.93 lakh crore** across the three. EST [202, 44]

India is on a path to a battery import bill of ₹1.30 lakh crore to ₹2.20 lakh crore a year by FY2030 (USD 14bn to 23bn). EST

The ICCT reorders the prize: Indian demand in 2050 at **340 GWh** baseline and **573 GWh** in its momentum and ambitious cases, with a cumulative import-payment difference near **₹11.91 lakh crore (USD 125bn)**, **₹8.95 lakh crore (USD 94bn)** of it from electrification policy, only **₹2.95 lakh crore (USD 31bn)** from domestic manufacturing. SOFT [245] Localising cells is a quarter of the prize, not the principal lever.

Announced against commissioned, stated without charity

Every stage of the funnel uses a different unit. The one operating plant produced **72,418 cells, about 1.9 MWh, in Q3 FY2026, on ₹9 crore of revenue against ₹51 crore of operating expense**. SOFT [209] Annualised, **7.6 MWh a year: 0.015 per cent of the 50 GWh national PLI target, 0.0026 per cent of**

announced capacity. EST [209] Commissioned nameplate of 1.4 GWh is 2.8 per cent of target. SOFT [194]

Annex Table A23 carries all fourteen sites with status and technology source. The four largest: **JSW**, Odisha, 50 GWh by 2028 to 2030, **halted** at zero, over ₹800 crore redirected to research; **Reliance**, Jamnagar, 10 under PLI plus 40 by design, zero, in "advanced commissioning" after promising the first 40 GWh phase for 2026, **Hithium talks stalled January 2026**; **Tata Agratas**, Sanand, 20, zero, **cells expected 2027**; and **Ola**, Krishnagiri, 20 under PLI, **the only operating plant**: **1.4 GWh of nameplate against 5 "installed" per the company, near 6 per the chief executive**, in-house **dry electrode**, no Chinese partner, pilot March 2024. Sources.[208, 209, 210, 211, 212, 213, 214, 224]

Three conflicts stay unresolved. **Amara Raja**: 2 GWh reported commissioned June 2025, cells delayed to FY2027 per S&P, end-2026 in a third report; read as module or pack phases, FY2027 for cells. SOFT **Exide's** 6 GWh phase one rests on one low-trust aggregator, no exchange filing. {SOFT, low confidence} **Ola's** three values against 1.9 MWh are the finding.

Design against delivery, the whole record. The ₹18,100 crore outlay targets 50 GWh across a performance period of 1 January 2025 to 31 December 2029, on a minimum facility of 5 GWh per firm, mandatory investment of ₹225 crore for each committed GWh within two years, and domestic value addition of 25 per cent at the start rising to 60 per cent within five years. SOFT [192, 193] Of the 50 GWh, 40 was awarded: **Hyundai Global Motors won 20 GWh in March 2022 and relinquished it**, and Reliance picked up 10 GWh in September 2024. SOFT [193] Against that award, the 1.4 GWh of PLI-certified nameplate sits at one plant and the other three beneficiaries at zero, the ministry's own written reply recording 1 GWh installed there. SOFT [194, 195] **Incentive actually paid: zero**, confirmed by a Lok Sabha written reply reported on 21 July 2026. SOFT [195] Beneficiary investment stood at ₹5,180 crore and 1,277 direct jobs on 31 May 2026 against ₹9,000 crore mandated on the 40 GWh awarded, about **58 per cent of the investment contractually owed**. EST from SOFT inputs [195] The fiscal signal matches the delivery: Union Budget 2026-27 cut the ACC PLI line 44 per cent, from ₹155.76 crore to ₹86 crore, while doubling Auto PLI to ₹5,940 crore. SOFT [198] Round 3 puts the final 10 GWh, ring-fenced for grid-scale storage, to a global two-stage tender, documents 15 July 2026, bids closed 13 October 2026, technical bids opened 14 October 2026, on at least 25 per cent domestic value addition in two years and 40 per cent in five with 1 to 4 GWh installed within five years. **Result NOT FOUND**, and this is the third attempt. SOFT [196, 197]

> **COMMISSIONED IS NOT PRODUCING, AND EVERY INDIAN CAPACITY NUMBER IS ONE OR THE OTHER**
>
> Every published Indian capacity figure is an announcement, an award or a nameplate, never output. The ₹18,100 crore advanced chemistry cell programme has paid nothing, no rupee claimed on the 40 GWh awarded, one 20 GWh award handed back, and the 2026-27 Union Budget cut the allocation 44 per cent while doubling the automotive incentive. Starved by its own milestones: almost nothing commissioned, so almost no incentive payable. SOFT [192, 193, 195, 198] EST on the reading.

Why the gap exists: four constraints, none financial

One, Chinese engineers cannot enter India: roughly 200,000 visas to Chinese nationals in 2019 against about 2,000 in 2024, and Chinese firms withdrew staff from Indian plants, so the delay is physical, not financial. Amara Raja's management says partnerships could accelerate technology but have not moved, on geopolitics. SOFT [211, 212] **Two, the intellectual property is no longer transferable**: China's July 2025 licensing of cathode technology, including fourth-generation LFP and LMFP cathode preparation, and of lithium processing **was never suspended**; Announcement No. 58's controls on cells at or above 300 Wh/kg, artificial graphite anode materials with production equipment and key cell-making equipment were suspended 7 November 2025 only **to 10 November 2026**, the reported extension to 10 January 2027 resting on no numbered instrument. {SOFT, unresolved} [91, 92, 95] **Three, no material supply chain**: no LFP cathode, graphite anode, LiPF6, lithium carbonate or hydroxide at scale, though India makes copper and aluminium collectors, separators,

solvents, additives and packaging. SOFT [208] **Four, no people and no patents**: 1,000 to 1,500 process engineers per GWh, roughly **36,000** if the announcements proceed, against **seven lithium-ion patents at India's two largest battery companies and approximately 50,000 at CATL**, whose 2024 battery-only research spend was about ₹24,572 crore (USD 2.58bn), the Korean big three ₹20,000 crore (USD 2.1bn) combined, Reliance's whole group ₹4,162 crore (USD 437m) in FY2024. SOFT [209]

Value capture follows. On a 2025 storage system at about ₹11,143 per kWh (USD 117), Chinese content in cells, enclosure and power conversion is roughly ₹7,143 per kWh against ₹4,762 per kWh of local assembly and engineering: India keeps about 40 per cent of value, China 60, and integration margins compress as cell margins hold, because cells are the moat. The source's components exceed the system price, so the split is indicative. SOFT [210]

Where India is genuinely competitive

The materials bet runs ahead of the cell bet: annex Table A24 sets out the cell-adjacent layers with figures, caveats and corrections, across anode and graphite (the strongest position, Epsilon already supplying Ola), cathode, electrolyte, separator, lithium refining and recycling. SOFT [214, 215, 216, 217, 231, 232]

The arbitrage is at the pack layer: **60 GWh of pack assembly and 23.15 GWh a year of storage assembly against about 1 GWh of cells**, so pack value capture is 40 per cent and falling. The unoccupied square is the electrode: Nash Energy plans in-house lines in 2027 to 2028; beyond TDSG nobody operates. SOFT [243, 229, 214]

On domestic lithium, restraint: Reasi's **3.29 million tonnes** is laterite ore, not contained lithium, restated down from 5.9 million tonnes, two auctions have failed, no metallurgical route is proven at scale. **Reasi is optionality, not 2030 supply**; the substantive move is offshore, in Catamarca. SOFT [218, 219, 220] HARD on KABIL [233]

Specialty cells: real demand, unpublished size

Demand exists in every specialty segment, **every one on imported cells or a single government-linked licensee**, and none can be sized in GWh or rupees.

- **Space**: ISRO's Vikram Sarabhai Space Centre transferred its space-grade cell to **BHEL, March 2018**; the Bengaluru line has delivered a **100th space-grade battery** to ISRO, supplying INSAT, GSAT, IRNSS, RISAT and CMS-03 across about 16 years. VSSC **shortlisted 14 of over 100 applicants**; those 14, any second licensee in production, India's new-space suppliers: **NOT FOUND**. HARD on the transfer, SOFT on the rest [221, 222, 223]
- **Defence**: HBL Power Systems, Ministry of Defence and DRDO approved: submarine module prototypes, missile batteries, exports to Israel and the UAE, **₹100 crore** of FY2025 capex for in-house high-energy cells, defence only. DRDO built the Tapas battery and management system **but imports the cells**. Size **NOT FOUND**, none estimated: three vendor models of the 2026 global military market differ 46 per cent. SOFT [224, 209]
- **Drones**: **₹19,048 crore (USD 2bn)** of domestic procurement, 18 to 24 months, 600-plus companies, 100-plus defence-focused, on **imported cells** via small pack assemblers; Boson Cell's indigenous 18350 and 21700 high-discharge cells claim 1,000 cycles, the only Indian effort; market ₹463 crore in 2025 to ₹709 crore by 2030 (USD 48.6m to 74.4m) at 8.9 per cent. SOFT on procurement, EST on the size [225, 226, 227]
- **Railways and medical**: HBL supplies railways, size unpublished. SOFT [224] No India medical evidence, **NOT FOUND**; global portable medical device batteries ₹30,001 crore in 2026 to ₹58,287 crore by 2032 (USD 3.15bn to 6.12bn) at 11.6 per cent, the India slice unpublished. EST

The channel, and how Indian firms buy cell technology

The government-to-government architecture is thin. The Indo-German Partnership for Green and Sustainable Development, launched 2022, names battery storage, carries a storage working group and a vocational strand for the 36,000-engineer gap, tenth conversation series 19 June 2026; Germany has committed **₹1.07 lakh crore (EUR 10bn)** of climate finance by 2030, mostly soft loans, plus **₹13,246 crore (EUR 1.24bn)** under the partnership. SOFT [238, 239] The India-EU agreement concluded **27 January 2026**, talks begun 2007, signature planned December 2026, force targeted early 2027, with enhanced access to critical raw materials: nickel, copper, aluminium, steel, titanium. SOFT [234, 235, 236] India-EFTA is in force from **1 October 2025**, committing ₹9.52 lakh crore (USD 100bn) over 15 years. HARD on the agreement [237] Premier Energies works with Germany's RCT Nexor on 12 GWh of storage manufacturing, Varroc with France's Forsee Power on battery systems, around 17 to 18 October 2026. SOFT [240, 241] Four gaps, each **NOT FOUND**: the India-EU battery and cell tariff schedule, so assert nothing on HSN 8507 duties; India-EFTA's battery provisions; a KfW or DEG facility for cells or electrodes; a Germany-India battery memorandum. SOFT

Six disclosed licences, the whole record. **Exide from SVOLT Energy (China), December 2021**: irrevocable, turnkey NMC and LFP, on ₹3,600 crore of licensee equity, over ₹5,200 crore committed. **Amara Raja from Gotion-InoBat (China, via Slovakia), 2024**: LFP with setup support and supply-chain integration, plus an undated Jiangsu Highstar (China) NMC licence. **Tata Agratas from AESC Apollo (Japan, majority-owned by China's Envision), 2025**: NMC, LFP in-house. **Nash Energy from Forte Nippon (Japan), 2021**: LFP bought outright. **BHEL from ISRO and VSSC, March 2018**: space-grade cells, terms not public, a precedent, not a benchmark. SOFT [209, 210, 212, 214]

Four conclusions. **The template is an upfront fee plus a running royalty, optionally with an equity kicker**: Amara Raja's executive director confirmed it, declining numbers; AESC's 12 per cent of Agratas for about ₹66 crore, a paper valuation near ₹550 crore, is the clearest datapoint. **No licensee has disclosed a rate, so no percentage royalty should appear in any document. NOT FOUND. Every large Indian deal licensed Chinese or Chinese-controlled technology. Licensing has failed three times on geopolitics**: Reliance with CATL in 2023, Reliance with Hithium, Amara Raja with Gotion. And **a licence without absorbed know-how is a liability**: after Gotion collapsed, Amara Raja funded its own centre.

The Battery Show India, Greater Noida, 22 to 24 October 2026, has a 23-exhibitor German pavilion including **IBU-tec advanced materials and E-Lyte Innovations, both in UniverCell's dry-electrode consortium**, plus seven more equipment makers in the category of dry-coating partner IKA. **UniverCell is not on the list.** HARD [242] IKA may run an Indian subsidiary, **not verified** here.

> **THE DUTY-FREE WINDOW CLOSES 31 MARCH 2029, THE MOST ACTIONABLE FACT HERE**
>
> Customs notification 27/2026 of 2 February 2026 consolidated **a single list of 85 machine types for duty-free import**, covering coating, winding, filling, sealing, formation, welding, testing and packaging. **It runs to 31 March 2029, no longer tied to an end use.** It also adds 35 capital goods for EV battery manufacturing, 28 for mobile-phone batteries, continues the storage-cell exemption to March 2028, exempts capital goods for sodium antimonate, a sodium-ion precursor, and removes duties on lithium oxide, hydroxide and carbonate, cobalt powder, battery scrap and waste, lead and zinc. SOFT [199, 200, 201] Finished accumulators under HSN 85076000 carry **20 per cent** basic duty, a 10 per cent social welfare surcharge on the duty and 18 per cent integrated GST, a landed burden cited near 49.2 per cent, import policy free; sources conflict at 15 per cent, and 5 to 15 per cent under EV schemes, so this leans to 20 per cent, to be verified against the CBIC schedule. {SOFT, conflicting}[194, 199] **The escalation is deliberate: the machine that makes a cell lands free, the cell pays 20 per cent. A European coating line imported before 31 March 2029 captures a subsidy that then disappears, and a 24 to 36 month build puts the decision in 2026 and 2027, not 2028.** EST

The plays

Capital needs are order-of-magnitude derivations from comparables, not quotations; they and horizons are EST. No equity purchase is recommended: no UniverCell valuation was established, **NOT FOUND**.

1. **License the process, not the cell**, solvent-free electrode and dry-coating know-how: **₹240 crore to ₹570 crore** (USD 25m to 60m), 18 to 30 months to a qualified electrode, 4 years plus to cell-scale economics. Proven at scale by nobody but Tesla, which called it a mistake; **dry cathode is the hard half**.
2. **India's first merchant electrode supplier**, into stranded pack capacity and the 60 per cent value-addition cliff: **₹380 crore to ₹860 crore** (USD 40m to 90m), 24 to 36 months, **inside the 31 March 2029 duty window**. **Concentration and timing**: lines slip, Agratas to 2027, Amara Raja to FY2027, JSW halted.
3. **Buy distressed European specialty capacity, and the engineers with it**: **₹140 crore to ₹760 crore** (USD 15m to 80m), 6 to 12 months to transact, 2 to 3 years to transplant. **The asset failed once already**, usually on customer concentration; labour cost, works councils, **engineers who may not relocate**.
4. **An EU-domiciled vehicle meeting both blocs' content rules from one base**: **₹50 crore to ₹145 crore** (USD 5m to 15m) plus the asset, 6 to 18 months, against a timetable **still a proposal**. Risk: the Industrial Accelerator Act's conditionality targets thin structures.
5. **Co-develop sodium-ion for Indian grid and two-wheeler storage with a European partner**: **₹95 crore to ₹285 crore** (USD 10m to 30m) for a pilot, **over ₹950 crore** at scale, 3 to 5 years, the NTPC pilot the cheap way in. **Price, and a body count**: it must beat LFP at ₹3,429 per kWh; Log9 died here, Natron liquidated September 2025.
6. **Own the specialty and defence-qualified niche**, small volume, high price, Chinese supply excluded: **₹190 crore to ₹475 crore** (USD 20m to 50m) plus 2 to 4 years of qualification, 3 to 5 years to revenue, over 7 to scale. **Qualification time, a volume ceiling, a monopsony customer. No incentive available**: minimum facility 5 GWh, 10 to 50 times this plant.

Annex Table A25 carries each play's evidence and anchor fact, including the claimed **27 per cent lower energy consumption at cell level** HARD. **Play 2 is the highest-leverage**: a merchant electrode plant moves a licensee from 25 to 60 per cent domestic value addition without solving cathode or anode chemistry. **Play 6 is the best risk-adjusted**: the shortage does not bite on 200 people, policy excludes Chinese competition, and the March 2018 ISRO to BHEL transfer is a ready structure. Start now, or Saft and EnerSys close the window. EST

Supported. Process capability, not chemistry; the electrode step nobody occupies; the line imported before 31 March 2029. EST

Not supported. Any valuation-based equity move, any India-EU battery-duty claim, any sizing of specialty demand, any royalty rate. EST

India's constraint is not money and not demand: it is process knowledge, the engineers who carry it, and legal access to both. A distressed European specialist sector holds more of what India needs than its market value suggests. Three clocks run: the duty-free window to 31 March 2029, the lapse of China's suspended controls, Korean dry-electrode mass production in 2028 and 2029.

Chapter 15

Conclusions, unknowns and the diligence list

The evidence supports a narrow conclusion: a real electrode-process business with a cell pilot attached, a three-year process lead that is unpatented and openly licensable, and a policy window that opens in February 2027 and begins closing in 2028. It supports no valuation, no certification claim and no statement about the dry line's throughput, because those three are NOT FOUND.

What the evidence supports

Electrode foils have been made since November 2020, the 1,000th customer electrode order was logged on 26 January 2026, and FY2025 revenue is a double-digit million euro figure, ₹106.82 crore to ₹1,057.52 crore (EUR 10m to 99m).[4, 7, 23] The register is clean: active status, both founders managing directors since 2019, no litigation, no insolvency filing, no recall.[16, 17]

The capacity reconciliation rests only on the company's own numbers. Forty million cells a year at its own 21.6 Wh per cell is 0.864 GWh, below the 1.5 GWh it advertises, so 1.5 GWh is electrode coating.[10, 11] The June 2026 cell line at approximately 50,000 cells a month is 0.0130 GWh a year, 0.86 per cent of that headline, and 1.3 GWh by 2027 implies roughly 29 per cent compound monthly growth for eighteen months on one ₹320.46 crore (EUR 30m) round.[10]

The product advantage is real and bounded: cells 15 to 20 per cent lighter reproduces exactly against four named competitors, at 16.0, 15.4, 11.9 and 12.5 per cent, but the mechanism is twelve grams of aluminium housing, and at an identical 6.0 Ah the cell delivers 2.4 per cent less energy than the Molicel P60C.[11, 13] The one pending application, EP4712256A1, covers that housing and lid and nothing else.[14] The external case rests on content rules rather than climate policy: SAFE's 65 per cent EU, EEA EFTA or Ukraine floor is live now, and a United States storage project must hold 55 per cent of direct costs non-prohibited in 2026, rising to 75 per cent after 31 December 2029, or lose the entire 30 to 40 per cent investment tax credit.[124, 129] Those rules make "not Chinese" an eligibility test, the most plausible explanation for a German supplier winning a United States agreement.

What the evidence does not support

No valuation can be established: the Tracxn post-money figure is arithmetically impossible against a ₹320.46 crore round, so **NOT FOUND** stands and no equity price can be assessed.[27] No management-system or product-safety certification is publicly evidenced, so space and medical are target markets rather than qualified supply; that is an absence of evidence, not evidence of absence.[1, 2, 11] The dry line's width and throughput are **NOT FOUND**, and the only published line specification states "water or NMP" with an 80-metre twelve-zone dryer, which describes a wet line.[4] Revenue concentration is unknowable from outside and four years of accounts sit behind a fee gate.[17] Nothing supports the claim of Europe's leading gigafactory: on nameplate it is roughly five times smaller than CATL Arnstadt and 27 times smaller than PowerCo Salzgitter.

Five questions for management

#	QUESTION	WHY IT MATTERS
1	Is 1.5 GWh electrode coating or cell assembly? State which.	40 million cells a year is 0.864 GWh, so both numbers cannot describe cells. "Electrode" resolves it at the cost of a superlative
2	The **dry** line's width, speed and areal loading, in mm, m/min and mg/cm2?	The round's entire premise. An answer near Fraunhofer's 240 mm and 10 m/min makes it a pilot anyone can license[145]
3	Cell-assembly run-rate, in cells per month, for every month since June 2026?	50,000 is a starting figure against a plan needing roughly 29 per cent compound monthly growth. Three months settle it

#	QUESTION	WHY IT MATTERS
4	Which certifications are held, with numbers and issue dates: ISO 9001, ISO 13485, AS9100 or ECSS, UN 38.3, IEC 62133?	They gate the named markets, and UN 38.3 gates shipping a cell at all
5	Merchant electrode share of FY2025 revenue, the largest customer's share, and will the FY2024 Jahresabschluss be released?	CustomCells, where both founders worked, died of one customer's non-payment, and the burn cannot be modelled without the accounts

Three things that would change the conclusion

One, the dry-line specification. A verified dry-coated electrode at production width and speed with a named qualified customer converts a marketing premise into the only genuine moat here. Pilot dimensions mean the lead is licensable from Fraunhofer IWS for the price of a licence.[145]

Two, certification evidence. ISO 13485 and AS9100 or ECSS certificates move realised prices into the top band of ₹13,334 to ₹28,572 per kWh (USD 140 to 300). Continued absence caps the business at power tools, drones, light e-mobility and development programmes, where EVE Energy's single Malaysian plant already makes 680 million 21700 cells a year.

Three, a policy event. An anti-dumping initiation on HS 8507.60 rechargeable cells, absent as of 20 October 2026 despite EU imports of Chinese batteries rising 39 per cent in the first half of 2026, is the largest discrete upside for any European cell maker.[84] Reinstatement of China's Announcement No. 58 cuts both ways: bear on input and equipment cost, bull on the scarcity of a non-Chinese electrode chain.[91,92]

The dated watch-list, October 2026 to April 2028

Fifteen dated items run from the expiry of the EIC Accelerator grant on 31 October 2026 to Lyten's Heide production and LG Energy Solution's dry-electrode mass production in 2028, and Table A26 in the annex sets each of them against what it decides. Three carry the rest. **10 November 2026**, the statutory lapse of China's Decision No. 70 suspension of Announcement No. 58, is the most consequential date in this report, and the reported extension to 10 January 2027 rests on no numbered instrument.[92] **18 February 2027** makes the battery passport and QR code mandatory, the first date the Regulation rewards short, transparent chains.[96,97] Battery Booster Facility awards are due by 31 December 2026 on a call that closed 30 September 2026, outcome **NOT FOUND**, deciding whether ACC, Verkor and PowerCo draw the ₹16,023 crore (EUR 1.5bn).[105]

One date sits outside the eighteen months and belongs in the minutes anyway: India's duty-free window for cell-manufacturing capital goods closes on **31 March 2029**, and a 24 to 36 month build puts the decision in 2027.[200]

Annex

Reference tables

Twenty-six tables to consult rather than read. The chapters carry the argument; the working, the full rankings, the calendars and the row-by-row evidence sit here.

Every table is pointed at from at least one chapter, numbering runs A1 to A26 in the order the chapters first reference them, and each caption names what the table shows, the period it covers and the chapter that calls it.

Table A1. China's lithium-ion battery exports: totals, quarterly detail, rolling-year levels and destinations

China's export totals, quarterly detail, rolling-year levels and destinations, 2024 to mid-2026. Referenced from chapters 01 and 04.

METRIC, CHINA LITHIUM-BATTERY EXPORTS	2024	2025	CHANGE
Total	₹5.80 lakh crore (USD 60.9bn)	**₹7.31 lakh crore (USD 76.8bn)**	**plus 26%**
to the European Union	₹2.11 lakh crore (USD 22.2bn)	**₹2.95 lakh crore (USD 31bn)**	40% of total, share plus 4pp
to the United States	₹1.46 lakh crore (USD 15.3bn)	**₹1.13 lakh crore (USD 11.9bn)**	**minus 22% by value**
Average unit price	₹17.14 lakh/t (USD 18,000/t, Q4)	**₹14.29 lakh/t (USD 15,000/t)**	minus 18%

Quarterly detail: exports to the EU reached ₹80,002 crore (USD 8.4bn) in the fourth quarter of 2025, up 50 per cent, and ₹30,477 crore (USD 3.2bn) in December alone, up 70 per cent, while December shipments to the United States halved to ₹9,524 crore (USD 1.0bn). Source: Cui Dongshu, Secretary-General of the China Passenger Car Association, reported February 2026 SOFT. [83]

Rolling twelve months to mid-2026, mirror data: exports under HS 850760 of **₹7.80 lakh crore (USD 81,881.24m) on 4,461,413.73 tonnes**, up ₹1.12 lakh crore (USD 11,779.70m) or 14.4 per cent and up 782,524.59 tonnes, lifting China's share of world trade to 66.10 per cent by value from 61.05 and 78.18 per cent by tonnage from 75.42 SOFT. [80]

CHINA'S TOP DESTINATIONS, LTM TO MID-2026	VALUE, ₹	VALUE, USD	GROWTH	VOLUME
Germany	**₹2.70 lakh crore**	USD 28,319.15m		1,152,649 t
Australia	₹61,282 crore	USD 6,434.47m	plus 57.60%	
India	**₹55,631 crore**	USD 5,841.10m	**plus 73.26%**	
Vietnam	₹50,417 crore	USD 5,283.13m		
Netherlands	₹48,252 crore	USD 5,066.36m	plus 79.80%	

Source: mirror data, 40 reporting economies SOFT. [80]

Table A2. Western battery and battery-value-chain casualties, January 2023 to October 2026

Twelve Western casualties with dates, stated causes and capital, January 2023 to October 2026. Referenced from chapters 01 and 05.

Dates, stated causes and capital, as far as capital was disclosed. Rupee figures convert at this report's fixed rates.

COMPANY	COUNTRY	EVENT AND DATE	STATED CAUSE	CAPITAL CONSUMED OR DESTROYED
Britishvolt	UK	Administration, January 2023; the 30 GWh Blyth plan abandoned, assets to Recharge Industries	Financing failure before construction	₹49,200 crore (GBP 4bn) **plan**; actual sunk **NOT FOUND**[188]
Northvolt	Sweden	Chapter 11 on 21 November 2024; Swedish bankruptcy 12 March 2025	Ramp failure: approx 1 GWh delivered against a 16 GWh target; BMW cancelled a ₹21,364 crore (EUR 2bn) contract June 2024	approx **₹1.43 lakh crore (USD 15bn)** raised[187]
Freyr Battery / T1 Energy	Norway and United States	European LFP ambitions abandoned December 2024; Coweta County, Georgia cancelled February 2025; rebranded, relocated to Austin, pivoted to solar	Chose to leave the industry rather than compete	₹24,762 crore (USD 2.6bn) project cancelled, committed not all spent[189, 167]
Li-Cycle	Canada	Filed May 2025; Glencore acquired all assets for ₹415 crore (USD 43.6m)	Recycling economics ahead of feedstock	**₹16,191 crore (USD 1.7bn)** raised, approx 97 per cent capital destruction[186]
CustomCells	Germany	Main operating companies insolvent 30 April and 5 May 2025; rescued by family offices around ABACON with SALVIA, signed 1 July 2025; Tuebingen closed, approx 80 staff remain	Single-customer concentration: Lilium failed to pay tens of millions	**NOT FOUND**[148, 149, 150]
Morrow Batteries ASA	Norway	Bankruptcy filed 6 May 2026, weeks after commercial production began; Lyten letter of intent August 2026 secured approx 70 jobs	1 GWh plant too small for an automotive customer; no investors for expansion	over **₹4,593 crore** (NOK 5.1bn, approx EUR 430m EST), including a ₹562 crore (NOK 550m) Innovation Norway loan[158, 159, 160]
ACC	France, Germany, Italy	Kaiserslautern and Termoli paused mid-2024, **definitively cancelled February 2026**; Douvrin retained	Restart conditions unlikely to be met; shifted to LFP with CATL in Spain	Capex avoided, not destroyed[165, 166]
Cellforce Group	Germany	Porsche and CustomCells joint venture; approx 200 of 290 made redundant August 2025; **complete shutdown May 2026**, remaining approx 50 jobs lost	No sufficiently viable long-term perspective, per Porsche	**NOT FOUND**[151, 152]
Varta AG	Germany	StaRUG restructuring July 2024; second filing in self-administration at the Amtsgericht Stuttgart **24 July 2026** (one trade source gives 28 July), four entities, approx 3,260 employees affected; Noerdlingen, approx 350 jobs, to close autumn 2026	Structural 2027 financing gap, weaker demand, currency effects, loss of a major button-cell customer	**₹3,044 crore (EUR 285m)** bank write-off plus total 2024 equity wipeout; FY2024 revenue ₹8,471 crore (EUR 793m), net loss ₹689 crore (EUR 64.5m)[154, 155, 153]
Leclanche, German units	Switzerland and Germany	Liquidity constraints disclosed February 2026; insolvency applications for Leclanche GmbH and Leclanche Service GmbH at the Amtsgericht Offenburg, **23 September 2026**; German large-format cell production affected	Funding could not be secured	**NOT FOUND**[156, 157]
Natron Energy	United States	Ceased all operations 3 September 2025; Rocky Mount gigafactory cancelled; 95 staff terminated	Board failed to raise funding	₹13,334 crore (USD 1.4bn) factory cancelled[65]
Ascend Elements	United States	Battery recycler filed for bankruptcy April 2026	NOT FOUND	**NOT FOUND**[190]

Table A3. Where the 93 per cent decline went, and what Europe paid for it

Who captured the 93 per cent price decline, party by party, 2010 to 2030. Referenced from chapter 01.

PARTY OR ITEM	OUTCOME	FIGURE
Buyers	took the whole decline	a 2025 pack cost 8% of a 2010 pack in real terms [44]
Owners of the processing chain	took the thick part of the value pool	extraction and processing are 50 to 60% of value added against 15 to 30% for cell production [55]
The one scale leader	took volume and utilisation together	CATL at 96.9% utilisation in 2025 and 39.4% of global EV battery installs from January to August 2026, against a sector near 40% [54, 42, 43]
Cell makers with a captive carmaker	took survival	[163, 186]
Independent European merchant cell startups	took nothing	over ₹1.67 lakh crore (USD 17bn) of disclosed capital raised or written off
Europe as a jurisdiction	took demand rather than supply	98% of its 2025 cell production capacity was controlled by Asian manufacturers [55, 113]
First IPCEI on Batteries, approved 9 December 2019	state aid in	₹34,182 crore (EUR 3.2bn) across seven member states
EuBatIn, approved 26 January 2021	state aid in	up to ₹30,978 crore (EUR 2.9bn); ₹65,160 crore (EUR 6.1bn) across the two [119]
European Court of Auditors count	grants and loan guarantees on top	₹18,159 crore (EUR 1.7bn) [120]
EU battery imports, 2024	the demand arrived anyway	around ₹2.99 lakh crore (EUR 28bn), of which ₹2.35 lakh crore (EUR 22bn) from China [191]
Chinese battery imports, first half 2026	up 39%	₹1.75 lakh crore (EUR 16.4bn) [84]
Deloitte estimate	forgone profit for European battery companies over four years	₹1.12 lakh crore (EUR 10.5bn); lost value added possibly ₹10.68 lakh crore to ₹16.02 lakh crore (EUR 100bn to 150bn) by 2030 {EST, vendor} [55]

Table A4. BNEF December 2025 battery prices by segment, chemistry and region

The full BNEF price set by segment, chemistry and region, December 2025 survey. Referenced from chapter 03.

MEASURE, 2025	USD/KWH	₹/KWH	NOTE
Pack, all end uses	**108**	**10,286**	minus 8% YoY; record low; minus 93% vs 2010 real
Pack, battery electric vehicle	99	9,429	second consecutive year under 100
Pack, stationary storage	**70**	**6,667**	**minus 45% YoY, cheapest lithium-ion category for the first time**
Pack, two and three wheelers	133	12,667	the most expensive segment
Pack, LFP / NMC	81 / 128	7,714 / 12,191	spread USD 47, ₹4,476; LFP approx 37% cheaper
Cell, EV average	79	7,524	see the cell-series warning in chapter 03
Cell, lowest observed LFP	**36**	**3,429**	stationary storage
Pack, lowest observed LFP	50	4,762	one researcher sourced it, another could not confirm
China / North America / Europe	84 / approx 121 / approx **131**	8,000 / 11,524 / **12,476**	minus 13% / minus 4% / minus 8% YoY; **plus 44% and plus 56% over China, Europe widening**

The volume weighted industry reference.[44, 49]

Table A5. Chemistry and format positions, 2025

Chemistry and format positions with confidence tags, 2024 to 2030 figures. Referenced from chapter 03.

POSITION, 2025	VALUE
LFP share of global EV batteries deployed	**over 55%**, from nearly 50% in 2024
LFP share of Chinese EV installations	**81.2%**, 625.3 of 769.7 GWh, plus 52.9% YoY
LFP share of EU EV battery demand	more than 10%
LFP share of United States EV batteries	almost halved during 2025
LFP demand growth, all applications	**plus 48%, the fastest growing chemistry**
NMC	positive absolute growth, falling share
NCA share, any year	**NOT FOUND**. Effectively a Panasonic and Tesla cylindrical niche inside the nickel bucket
Semi-solid and all-solid-state	**8.0 GWh shipped in 2025, plus 51.9 per cent, all semi-solid. Zero all-solid cells in customer cars**
LFP 2030 share, forward estimates	37% to 75%, depending on light-vehicle or all-application scope; not averaged
46-series ramp	Samsung SDI began mass production of 46-phi cells in March 2025; LGES order backlog above 440 GWh; Panasonic has missed three 4680 deadlines, most recently end-March 2026

Sources: IEA for the global, EU and American shares, CABIA for China, Benchmark for the growth rate, EVTank for solid-state shipments.[32, 43, 36, 67]

Table A6. World trade in lithium-ion accumulators, HS 850760: exporters and importers

Exporter and importer rankings for HS 850760 with shares, 2023, the last fully reported year. Referenced from chapter 04.

RANK	EXPORTER, 2023	VALUE, ₹	VALUE, USD	SHARE
1	**China**	**₹6.19 lakh crore**	USD 65.0bn	**55.97%**
2	Poland	₹1.12 lakh crore	USD 11.8bn	10.20%
3	Hungary	₹98,097 crore	USD 10.3bn	8.90%
4	Germany	₹59,049 crore	USD 6.2bn	5.35%
5	Czech Republic	₹32,382 crore	USD 3.4bn	2.95%
	World total, 116 reporters	₹11.05 lakh crore	USD 116bn	100%

RANK	IMPORTER, LTM TO 2025	VALUE, ₹	VALUE, USD	VOLUME, T	YOY
1	**Germany**	**₹2.69 lakh crore**	USD 28,210.16m	1,098,657.59	**plus 15.73%**
2	**United States**	₹2.12 lakh crore	USD 22,220.57m	458,651.79	minus 0.09%
3	Australia	₹49,168 crore	USD 5,162.50m	97,117.00	plus 42.91%
4	Czechia	₹43,462 crore	USD 4,563.44m	183,330.03	plus 23.89%

RANK	IMPORTER, LTM TO 2025	VALUE, ₹	VALUE, USD	VOLUME, T	YOY
5	Mexico	₹40,764 crore	USD 4,280.15m	122,765.98	minus 24.42%

Sources: UN Comtrade via TrendEconomy for 2023, the last year with substantially complete filing; UN Comtrade via GTAIC, top-39 importers study, for the rolling year SOFT. [79, 81] Exporter ranks six to ten are NOT FOUND in the accessible view. Importer ranks six to ten are paywalled and NOT FOUND, and the rolling-year study publishes no share column, so import shares on that basis are NOT FOUND.

Fully reported 2023 import shares, for continuity: Germany ₹2.52 lakh crore (USD 26.5bn) at **25.13 per cent** of world imports, the United States ₹1.78 lakh crore (USD 18.7bn) at **17.76 per cent**, the Czech Republic 4.94 per cent, Belgium 3.58 per cent and Mexico 3.55 per cent, on a world import total of ₹10.00 lakh crore (USD 105bn) across 136 reporters.[79]

Table A7. Import aggregates, and India's HS 850760 imports

The 39-reporter import aggregate and India's HS 850760 line, 2024 to April 2026. Referenced from chapter 04.

AGGREGATE, 39 REPORTING IMPORTERS	2024	2025 PRELIMINARY	CHANGE
Value	₹8.91 lakh crore (USD 93.53bn)	₹8.77 lakh crore (USD 92.04bn)	minus 1.6%
Volume	2,803.29 kt	3,197.54 kt	plus 14.1%
Implied unit price	**₹31.78 lakh/t (USD 33,364/t)**	**₹27.40 lakh/t (USD 28,785/t)**	**minus 13.7%**

Source: UN Comtrade via GTAIC SOFT; unit prices computed here EST. [81]

INDIA, HS 850760	CALENDAR 2025	ROLLING YEAR TO APRIL 2026
Value	**₹38,892 crore (USD 4,083.60m)**, up 43.85%	roughly ₹44,000 crore to ₹47,000 crore (USD 4.63bn to 4.93bn) across three commercial estimates
Volume	up **75.12%**	up 90.90%
Unit price	₹43.00 lakh per tonne (USD 45,150/t), down 17.86%	down a further 17.68%
China's share	**93.26%** narrow HS 850760; **84%** all-lithium-ion	

Sources: commercial trade-data vendors SOFT. [82, 206, 246]

Table A8. UniverCell Holding GmbH: the register record

The commercial-register, purpose and officer record as filed, May 2019 to February 2026. Referenced from chapter 06.

ITEM	VALUE	NOTE
Legal name	UniverCell Holding GmbH	
Commercial register	HRB 22333 KI, Amtsgericht Kiel	prior entry HRB 14346 PI, Amtsgericht Pinneberg
First registered	20 to 22 May 2019	sources give both 20.05. and 22.05.2019; the state agency says "founded April 2019", probably the notarisation
Original seat	**Elmshorn**, Kreis Pinneberg	not Kiel, not Flintbek

ITEM	VALUE	NOTE
Seat moved to Flintbek	3 November 2020	with the capital increase to ₹2.14 crore (EUR 200,000)
Registered office today	Konrad-Zuse-Ring 1, 24220 Flintbek, Schleswig-Holstein	
Business purpose on file	production of lithium-ion accumulators; development, manufacture, distribution and trade of raw materials, cells and modules; management of own assets, notably trademarks and shareholdings; services, consulting and project planning in energy storage	wider than the marketing
Industry code	WZ and NACE C27.20, manufacture of batteries and accumulators	
Status	aktiv, economically active	no insolvency, Schutzschirm or StaRUG filing found
Trademark	"UniverCell" registered 6 February 2026	four days after the capital increase, four weeks before the Series B
Managing directors	Dr Stefan Alexander Permien, Marius Strack	each authorised to represent **jointly with another** director
Shareholders	10 known active shareholders	names sit behind a paywall: NOT FOUND

Sources.[16, 17]

Table A9. CylindricalCell 21700-HE: the published specification

The full 21700-HE cell specification from company and library data, 2026 disclosures. Referenced from chapter 06.

PARAMETER	VALUE
Designation	UC21700-HE, "UniverCell High Energy"; HE60 in the Voltt library
Nominal capacity and energy	6.0 Ah; 20.7 Wh
Nominal voltage	**3.45 V (Voltt) against 3.6 V and "+3.5 V" in the company deck. Inconsistent across the company's own documents**
Voltage window, mass, volume	2.50 to 4.20 V; approx 63 g; 24.3 cm3
Dimensions	21.0 x 70.2 mm (Voltt) against 21 x 70.5 mm (company deck)
Gravimetric and volumetric energy	**329 Wh/kg (Voltt) and "over 330 Wh/kg" (deck); 851 Wh/L**
Power density	1,971 W/kg; 5,108 W/L
Max continuous discharge and charge	36.0 A, 6.00 C; 24.0 A, 4.00 C
Cycle life	**over 750 cycles**, stated at 1C/1C and 100 per cent depth of discharge
DCIR at 50 per cent state of charge	10 mOhm or less
Cathode and anode	nickel-based NMC and other ternary materials up to 92 per cent Ni, mono and multi-crystal; graphite with silicon additive, production-proven recipes to 80 per cent SiC, up to plus 1,500 mAh/g at electrode level
Operating temperature range	**NOT FOUND**
Release year, origin	2026; Germany

PARAMETER	VALUE
Second variant	**21700-HP**, high power, samples targeted Q3 2026. No published specification for the HP variant was located

Sources.[11, 13]

Table A10. UniverCell pros and cons, with the evidence for each

The evidence behind each of the ten pros and fifteen cons, position October 2026. Referenced from chapter 06.

#	PRO	EVIDENCE
1	Electrode business real, installed, earning	foils since November 2020; 1,000th customer electrode order 26 January 2026; over 1,000 recipes tested; FY2025 revenue a double-digit million euro figure SOFT [4, 7, 22, 23]
2	Sized below the level at which European cell ventures died	1.5 GWh is roughly one per cent of Northvolt's Skellefteaa ambition; ₹320.46 crore is roughly 0.2 per cent of Northvolt's approximately ₹1.43 lakh crore (USD 15bn) EST [10]
3	End markets price-insulated by design	automotive explicitly avoided; power tools, drones, medical, marine, space and critical infrastructure are qualification-locked, low-volume, high euros per kWh [21, 28]
4	Grant dependence unusually low	approximately ₹25.50 crore (EUR 2.387m) of hard public money against roughly ₹332.39 crore (USD 34.9m) of equity EST [19, 27]
5	The weight advantage is real and independently reproduced	the 15 to 20 per cent lighter claim checks out at 16.0, 15.4, 11.9 and 12.5 per cent against the Molicel P60C, Molicel M65A, Reliance RS60 and EVE 50PL EST from HARD inputs [11, 22]
6	The one patent covers the one genuinely differentiated thing	EP4712256A1 claims the housing and lid, where effectively all the energy-density lead comes from [11, 14]
7	Adverse record clean after targeted search	no litigation, insolvency, recall, safety incident, executive departure since incorporation, or negative German-language coverage [16, 17]
8	Inside the German research establishment, with its equipment partner as investor	Permien inside a Fraunhofer FFB session April 2026; Germany's largest sodium-ion consortium; IKA both dry-coating partner and Series B co-lead [10, 12, 20]
9	Dry coating is a prerequisite, not an alternative, on most all-solid-state roadmaps	sulphide solid electrolytes are incompatible with conventional NMP and water slurry routes
10	Headcount growing, not shrinking	60 to 92, which puts it ahead of the restructured CustomCells at approximately 80 staff as Germany's largest independent custom-cell business EST [2, 10]

#	CON	EVIDENCE
1	Cell output is 0.86 per cent of the headline capacity, a 116-fold gap	series production from June 2026 at approximately 50,000 cells a month is 0.0130 GWh a year against a marketed 1.5 GWh; Chapter 07 derives it EST from HARD inputs [10, 11]
2	Two of its own headline numbers are inconsistent by 1.7 times inside one press release	40 million cells a year at its own 21.6 Wh per cell is 0.864 GWh, not 1.5 GWh EST from HARD inputs [10]
3	Dry coating, the premise of the Series B, is at pilot scale and unpatented	the installed line uses water or NMP and an 80-metre twelve-zone dryer, definitionally wet-process assets; no UniverCell dry-coating patent exists [4, 11, 14]
4	The IP position is one pending application on a mechanical packaging feature	filed September 2024, still ungranted; nothing covers dry coating, the tabless jellyroll or formation; the defence is tacit knowledge in a company of 92 [14]
5	Zero named customers, ever	the one disclosed anchor is a single United States customer, so revenue concentration is unknowable from outside; CustomCells was killed by exactly this [10, 15]
6	No certification is publicly evidenced for the markets it says it sells into	AS9100 and ECSS gate satellites, ISO 13485 critical-care medical devices, UN 38.3 every shipment; none was found [1, 2, 11]

#	CON	EVIDENCE
7	Hiring is roughly twelve months behind its own plan	approximately 100 employees promised by end-2025 in a January 2026 document, "more than 82" stated in March 2026 [10, 11]
8	The management bench is thin and the structure slow	no named CTO, CFO or VP Manufacturing among nine named staff; two quality managers; five locations on 92 people; both directors sign jointly [2, 11, 16]
9	The headline energy-density advantage does not survive a controlled comparison	against the Molicel P60C at identical 6.0 Ah it carries 2.4 per cent less energy per cell, the lead being twelve grams of housing, and 11.1 per cent less than the M65A EST from HARD inputs [11, 13]
10	The cell specification is internally inconsistent and reads as a best-case composite	over 750 cycles at 1C/1C on 0.5C pilot data; nominal voltage 3.45 V, 3.5 V and 3.6 V across its own documents; "targeted" and "typical" capacity on one slide; 329 Wh/kg with 851 Wh/L and 6C not credible together; an unresolved internal review comment on slide 18 of the deck taken to a NASA-hosted workshop [11, 13]
11	Real financials are not obtainable, so a board cannot model the burn	four years of accounts on file and none of revenue, equity or net loss is public [17, 18]
12	Excluded from the largest European instrument aimed at its own industry	the ₹16,023 crore (EUR 1.5bn) Battery Booster Facility requires at least 10 GWh of planned annual capacity and the applicant's first full commercial-scale EV cell production globally; at approximately 1.5 GWh and non-automotive UniverCell fails both [105]
13	The dry-coating lead has a visible expiry of roughly 2028 to 2029: three years, not ten	Fraunhofer licenses DRYtraec openly, including to an Indian entrant; VW PowerCo is guided for 2026 or 2027; the Duerr and LiCAP platform is hunting a pilot host; LG Energy Solution targets mass production in 2028 EST [145]
14	A well-capitalised competitor is arriving in its own federal state	Lyten, having absorbed Northvolt Sweden, the Northvolt Drei site and Morrow during 2026, targets production at Heide from 2028, roughly 60 to 70 km from Flintbek, aimed at the same defence, aerospace, drone and data-centre niches EST [162]
15	The addressable market is bounded by format as well as by capacity	the cylindrical centre of gravity is migrating from 21700 to the 46-series and cell-to-pack designs engineer small cylindrical cells out of the car, so the tabless 21700 is right for tools, drones, medical and light e-mobility and wrong for automotive volume

Table A11. Tier A European cohort: chemistry, format and segment focus by player

Chemistry, format and segment focus for twelve European players, October 2026. Referenced from chapter 08.

In chapter 09, the same twelve players are tabulated on capacity and status.

PLAYER	CHEMISTRY AND FORMAT	SEGMENT FOCUS
UniverCell, Flintbek DE	NMC class; custom tabless 21700, pouch, merchant electrode foil	power tools, marine, space, medical, e-aviation, critical infrastructure
CustomCells, Itzehoe DE	custom, pouch centric, wet process	aerospace, special applications
V4Smart, Ellwangen and Noerdlingen DE	high-power cylindrical, Group14 silicon anode	Porsche hybrids, premium automotive
Leclanche, Willstaett DE and CH	NMC large format, water-based and PFAS-free	marine, rail, stationary
Morrow, Arendal NO	LFP prismatic	automotive, storage
Northvolt into Lyten, SE, NO, DE	NMC plus lithium sulphur	storage, defence, aerospace, drones, data centres
Verkor, Dunkirk FR	NMC pouch	automotive, Alpine A390
InoBat and GIB, Surany SK	not disclosed at cell level	automotive export to the EU
ACC, FR, DE, IT	NMC	Stellantis volume
Cellforce, Kirchentellinsfurt DE	high-performance NMC	Porsche performance
Saft and EnerSys, FR, US, DE	lithium-ion plus solid state; lithium for aerospace and defence	space, defence, submarines, critical infrastructure

PLAYER	CHEMISTRY AND FORMAT	SEGMENT FOCUS
Blue Solutions, Mulhouse plan FR	lithium metal polymer, solid state	buses, stationary

Table A12. Tier A European cohort: ownership, funding and dated status changes

Ownership, funding and dated status changes across the Tier A cohort, 2025 to 2026. Referenced from chapter 08.

PLAYER	OWNERSHIP AND FUNDING	WHAT CHANGED, AND WHEN	CAPITAL RAISED OR DESTROYED	CONTESTS UNIVERCELL DIRECTLY?
UniverCell	DTCF, IKA, WIKA, EIC Fund, angels. No OEM anchor	₹320.46 crore (EUR 30m) Series B closed 4 Mar 2026; series production from Jun 2026	approx ₹332.39 crore (USD 34.9m) equity to date	n/a
CustomCells	ABACON and SALVIA family offices	insolvency 30 Apr and 5 May 2025 after customer Lilium failed to pay; purchase agreement 1 Jul 2025	NOT FOUND	**Yes, the closest structural analogue**, 50 km away, now approximately 80 staff against UniverCell's 92
V4Smart	Porsche 70% since 4 Mar 2025	renamed from V4Drive; cells in series as booster cells in the 911 Carrera GTS	not disclosed; Porsche balance sheet	**Yes, and the most under-appreciated.** German, high-power cylindrical, low volume, approximately 375 staff planned, but captive to Porsche
Leclanche	Swiss listed	liquidity warning Feb 2026; insolvency filed 23 Sep 2026 at Amtsgericht Offenburg, cases 30 IN 589/26 and 588/26	NOT FOUND	Partly. It was Europe's other champion of water-based PFAS-free coating
Morrow	Innovation Norway loan, Siemens, ABB	bankruptcy 6 May 2026, weeks after commercial production started	over ₹4,593 crore (approx EUR 430m) consumed	No on product. **Yes as the scale warning**
Northvolt into Lyten	Lyten, US owned	Northvolt Sweden acquired 27 Feb 2026 at a stated value near ₹47,620 crore (USD 5bn); Heide MOU 1 Jul 2026	Northvolt consumed approx ₹1.43 lakh crore (USD 15bn)	On talent, subsidy and narrative, yes. Heide is approximately 60 to 70 km from Flintbek
Verkor	Renault anchored	Dunkirk inaugurated Dec 2025	not disclosed	No. The proof that European cells work with an anchor OEM
InoBat and GIB	Gotion 80%, InoBat Auto 20%	construction from Oct 2025, trial production 2026	not disclosed	No on product. **Critical on policy**: the European bespoke-cell champion became the 20 per cent minority partner
ACC	Stellantis, Mercedes-Benz, TotalEnergies	Kaiserslautern and Termoli permanently cancelled Feb 2026; Stellantis went to CATL for LFP in Spain	capex avoided, not destroyed	No. It is the backdrop
Cellforce	Porsche and CustomCells	approximately 200 of 290 let go Aug 2025; complete shutdown May 2026	NOT FOUND	No longer. Its death orphaned the Duerr and LiCAP pilot
Saft and EnerSys	TotalEnergies; listed	Saft solid-state lines in Bordeaux Oct 2026; EnerSys revised ₹1,429 crore (USD 150m) DOE grant for a defence-focused lithium plant	corporate balance sheets	**Yes, and probably under-weighted.** They already hold the space, defence and submarine accounts
Blue Solutions	Bollore	Gen4 refocus and Bluebus shutdown announced 8 Sep 2026	₹23,500 crore (EUR 2.2bn) plan	No

Table A13. Tier B dry-coating competitors: model, stage and capital

Nine dry-coating competitors on model, stage and capital, 2026 to 2029. Referenced from chapter 08.

Operating-scale ranking for the same group is in chapter 08.

PLAYER	COUNTRY	MODEL	STAGE	CAPITAL	WHAT IT MEANS FOR UNIVERCELL
Tesla	US	captive cells	mass production	n/a	Benchmark, not a rival. Sells no merchant cells or electrodes[77]
VW PowerCo	Germany	captive cells	dry guided 2026 or 2027	VW balance sheet	Bigger, better funded, same country, captive to VW[174]
Sakuu	US	platform and licensing	validated, no volume	₹5,714 crore (USD 600m) pre-money SPAC, Oct 2026	Roughly 17 times UniverCell's total raise, on validation data rather than product EST[177]
AM Batteries	US	**merchant dry electrode**	pre-full-production	approx ₹657 crore (USD 69m), plus a ₹476 crore (USD 50m) DOE award	**The closest business-model competitor worldwide.** A transatlantic mirror image: EU public money against US public money[176]
LG Energy Solution	Korea	captive cells	pilot complete	corporate	Mass production 2028, LFP 2029. The clock on the lead[169,170]
Samsung SDI	Korea	captive cells	two pilots	a very large investment split between Ulsan and Cheonan, stated in won; no fixed rupee rate for the won is used in this report	No mass-production date, NOT FOUND[171]
Duerr and LiCAP	Germany and US	**equipment and process IP**	proof of concept, no host	corporate	The available entry point, and time-bound[172,173]
Fraunhofer IWS	Germany	**licensor**	240 mm, 10 m/min	₹39.52 crore (EUR 3.7m) DRYplatform	Caps the moat for everyone[145]
Anaphite	UK	process development	Series A	approx ₹215 crore (USD 22.6m)	The one genuine startup peer[27]

Table A14. Leading indicators: what tells you early which scenario is unfolding

Fourteen leading indicators with readings, triggers and observation dates, October 2026 to 2035. Referenced from chapter 10.

INDICATOR	CURRENT READING	WHAT TO WATCH FOR	READS TOWARD	OBSERVABLE FROM
China's Announcement No. 58	Suspended by Announcement No. 70 to 10 Nov 2026. An extension to 10 Jan 2027 is reported but no numbered MOFCOM instrument has been located SOFT[92,95]	Reinstatement of licensing on cells at or above 300 Wh/kg, graphite anode and cell-making equipment	Reinstatement is bear on cost and bull on European moat	10 Nov 2026, or 10 Jan 2027 if the extension is confirmed
EU trade defence on HS 8507.60	No anti-dumping or anti-subsidy case open as of 20 Oct 2026, despite EU imports of Chinese batteries at ₹1.75 lakh crore in H1 2026 (EUR 16.4bn), up 39 per cent. HARD by absence[84]	A Commission initiation notice in the Official Journal	Initiation is strongly bull for any EU producer	Any time. This is the single largest discrete upside event
Battery Booster Facility awards	Call closed 30 Sep 2026, ₹16,023 crore of interest-free loans (EUR 1.5bn), awards targeted before end-2026, outcome NOT FOUND HARD[105]	Which of ACC, Verkor and PowerCo wins, and how much	Awards made and drawn is base to bull, a failed round is bear	By 31 Dec 2026
EU BEV share of new cars	21.7 per cent Jan to Aug 2026, up from 15.8 per cent HARD[74]	Full-year 2026 outturn	Below 22 per cent supports the 40 per cent 2030 bear path, above 24 per cent supports base	ACEA full-year release, January 2027

INDICATOR	CURRENT READING	WHAT TO WATCH FOR	READS TOWARD	OBSERVABLE FROM
EU storage additions	27.1 GWh in 2025, up 45 per cent, over 50 GWh guided for 2026 HARD [53]	2026 outturn against the 50 GWh guide	Under 40 GWh is bear, over 55 GWh is bull	SolarPower Europe, late January 2027
Pack price	₹10,286 per kWh (USD 108) in 2025, minus 8 per cent HARD [44]	The December 2026 survey	Below ₹9,524 per kWh (USD 100) implies the fast learning rate	December 2026, then annually
Chinese-owned European ramp	CATL Debrecen trial production from 22 Sep 2026, Zaragoza targeted end-2026, BYD Szeged Q4 2026 SOFT [183, 184]	Whether those plants reach commercial output on schedule	On-time ramp widens the ownership gap in all scenarios	Q1 2027 output disclosures
EU 2035 CO2 file	90 per cent proposal tabled 16 Dec 2025 SOFT [75]	Adoption text, and the 2028 review mandate	Weaker than proposed is bear, local-content conditionality attached is bull	Adoption through 2026 to 2027, review 2028
Industrial Accelerator Act	Proposed 4 Mar 2026. Procurement provisions apply to procedures launched on or after 1 Jan 2029 HARD [110]	Adoption date, because the escalation to EU-origin cells follows three years after phase one	Early adoption pulls the EU-cell scarcity rent into 2028 tenders	Council and Parliament agreement
Battery Regulation carbon footprint	Delegated act establishing the methodology not adopted. The nominal 18 Feb 2025 and 18 Feb 2026 dates have not triggered SOFT [103]	Publication in the Official Journal	Publication starts a 12 to 18 month clock and is bull for low-carbon-grid producers	Any time. Realistic bite 2027 to 2029
Dry-electrode industrialisation	PowerCo guides dry start of production for 2026 or 2027, series dry use unconfirmed. LGES targets mass production in 2028 SOFT [174, 169]	A confirmed series dry-coated cell from either	Confirmation closes the specialist's lead window	Company disclosures, 2027 onward
India ACC PLI tranche 3	Technical bids for the final 10 GWh opened 14 Oct 2026, result NOT FOUND. This is the third attempt SOFT [196]	Award, or a third failure	A third failure confirms India stays an importer through 2030	Late 2026
An ALMM-style list for battery cells in India	None exists as of Oct 2026, NOT FOUND [243]	An MNRE or Ministry of Power notification	Notification is bull for anyone with Indian electrode or cell capability	Any time
EU-India FTA	Negotiations concluded 27 Jan 2026, signature planned Dec 2026, entry into force targeted early 2027. The battery tariff schedule is NOT FOUND SOFT [234]	The HS 8507 schedule in the signed text	Duty elimination on cells is bull for EU-India cell trade in both directions	December 2026 signature

Table A15. The dated technology programme board, October 2026

Twenty-one dated technology programmes with owner, guidance and status, October 2026. Referenced from chapter 10.

PROGRAMME	OWNER	WHAT IS GUIDED	GUIDED DATE	STATUS, OCT 2026
4680 with dry cathode and anode	Tesla	Full dry electrode at mass scale, Giga Texas	In production	The only confirmed full-dry cathode at mass scale, but the chief executive called the pursuit a mistake and "way harder" than expected, and 4680 performance was reported as underperforming in May 2026 SOFT [77, 78]
Dry cathode with Koenig und Bauer roller press	VW PowerCo	Both electrodes dry, about 30 per cent energy and 15 per cent floor space saved	Start of production 2026 or 2027	Cells are coming off Salzgitter from December 2025. Whether they are dry-coated is NOT FOUND SOFT [174, 175]

PROGRAMME	OWNER	WHAT IS GUIDED	GUIDED DATE	STATUS, OCT 2026
Dry electrode mass production	LG Energy Solution	17 to 30 per cent electrode cost reduction, about 30 per cent less factory land	2028, and 2029 for dry LFP specifically	Ochang pilot line complete SOFT [169,170]
S-Line and DryEV pilots	Samsung SDI	Dry process for 46-series, transfer to Ulsan	No mass-production date	NOT FOUND on timing SOFT [171]
X.Cellify DC	Duerr with LiCAP	Up to 65 per cent less space, up to 70 per cent lower energy	Pilot was to run at Cellforce in 2026	Cellforce closed in May 2026, so the host no longer exists and Duerr is publicly seeking gigawatt-scale pilot partners SOFT [172,173]
DRYtraec and DRYplatform	Fraunhofer IWS	Up to 240 mm width and up to 10 m/min, roll-to-roll double-sided	₹39.52 crore platform (EUR 3.7m) launched in Dresden 3 Nov 2025	Openly licensable, which caps everyone's moat SOFT [145]
SolidStack	Samsung SDI	All-solid-state, 900 Wh/L demonstrated	Mass production H2 2027	Pre-commercial SOFT [69]
All-solid-state	Toyota	First vehicles	2027 to 2028	Pre-commercial SOFT [69]
Eagle line with the Cobra process	QuantumScape	Separator production at scale	Inaugurated 4 Feb 2026, targeting 2027 to 2028	Pre-commercial SOFT [69]
All-solid-state pilot line	Nissan	EV launch	FY2028	Pilot running since Jan 2025 SOFT [69]
Naxtra sodium-ion	CATL	Up to 175 Wh/kg, 60 GWh of capacity by end-2026	Mass production from Dec 2025	First sodium-ion cell to pass GB 38031-2025. First mass-market car, the Changan Nevo A06, launched Feb 2026 SOFT [63,66]
European sodium-ion cathode chain	Altris with Draslovka	Industrial-scale Prussian white cathode	Production late Q3 or early Q4 2026	₹206 crore in-kind investment (EUR 19.3m), ahead of the German SIB:DE consortium industrially HARD [182]
SIB:DE	BMFTR, coordinated by EDAG	Germany's largest sodium-ion consortium, ₹154.89 crore across 25 partners (EUR 14.5m)	Mar 2026 to Feb 2029	Running. The specialist's own share is NOT FOUND SOFT [20]
Northvolt Sweden and Northvolt Labs acquisition	Lyten	16 GWh of capacity, over 160 hectares, Europe's largest battery research centre	Completed 27 Feb 2026	Near-term output is NMC for storage from H2 2026, lithium-sulphur later HARD [161]
Northvolt Drei at Heide	Lyten	About 1,000 jobs on a 110-hectare site	Production targeted 2028	Exclusive MOU signed 1 Jul 2026 HARD [162]
46-phi mass production	Samsung SDI	First Korean maker, supplying BMW, Rivian and Audi	From March 2025	In production SOFT [171]
46-series backlog	LG Energy Solution	Over 440 GWh of orders	Ramping	In production SOFT [170]
4680	Panasonic	Volume production	Missed three deadlines, most recently end-March 2026	Delayed SOFT [170]
Moses Lake silicon anode	Sila	10 GWh of material by 2026, in the electric Mercedes G-Class	Operational Apr 2025	Shipping SOFT [68]
SiMaxx	Amprius	Up to 450 Wh/kg and 1,150 Wh/L	Shipping into drones and aviation	Commercial SOFT [68]

PROGRAMME	OWNER	WHAT IS GUIDED	GUIDED DATE	STATUS, OCT 2026
P70X high-drain 21700	Molicel	340 Wh/kg at 30C	Launched Jul 2026	The direct competitive benchmark SOFT [180]

Table A16. The genuinely new 2026 demand pools

Seven new demand pools with size, durability mechanism and relevance, 2024 to 2036. Referenced from chapter 10.

POOL	SIZE, WHERE IT CAN BE STATED	MECHANISM THAT MAKES IT DURABLE	RELEVANCE TO A SPECIALIST
Data-centre backup and behind-the-meter storage	₹39,525 crore in 2026 on a backup-unit basis (USD 4.15bn, from ₹33,620 crore or USD 3.53bn in 2025, 9.2 per cent CAGR). A second vendor puts storage for data centres at ₹47,239 crore in 2026 (USD 4.96bn) growing 14.2 per cent to ₹1.79 lakh crore by 2036 (USD 18.79bn). The roughly 40 per cent spread is a scope difference, backup-only against backup plus storage, not a disagreement SOFT [71]	AI racks moved from 5 to 10 kW to 30 to 50 kW, some above 100 kW, and Open Compute Project specifications mandate dedicated lithium-ion packs at rack or row level. Lead-acid physically cannot serve this. Forced substitution, not preference	Real, and a competitor has restructured its asset base around it: Lyten is co-locating battery manufacturing with a data-centre campus at Skellefteaa HARD [161]
Drones	Ukraine received about 3 million FPV drones in 2025, roughly 2.5 times the 2024 volume, with stated plans for up to 8 million a year. At about 0.1 kWh a pack that is approximately 0.3 GWh a year EST [226]	No Chinese cell is politically acceptable in a NATO drone supply chain, and both sides are re-shoring drone cell supply, so the pool is structural rather than a spike	Small in energy, large in margin, and on the doorstep. For a 1.5 GWh plant this is arguably the most attractive single opportunity in the whole fact base
Defence	**NOT FOUND** for any credible EU-level figure in GWh or currency. Three vendors give 2026 military battery market values of ₹16,762 crore, ₹24,000 crore and ₹24,477 crore (USD 1.76bn, 2.52bn and 2.57bn), a 46 per cent spread, with CAGRs of 5.6, 4.5 and 9.3 per cent. Do not use any of them	SAFE's 65 per cent EU, EEA and Ukraine content floor is live now on a ₹16.02 lakh crore instrument (EUR 150bn), and qualification cycles of two to five years create genuine lock-in HARD [124]	The content rule is quantified, the market size is not public. Lock-in is why this suits a small producer and does not suit CATL
Space	About ₹38,001 crore in 2025 to ₹40,763 crore in 2026 (USD 3.99bn to 4.28bn) at roughly 7.3 per cent. A second source quotes ₹3,552 crore for 2026 (USD 372.9m) on a different scope. Order of magnitude only	ESA's non-dependence doctrine plus geographical return creates a procurement preference, and Saft is close to a single point of European supply	A margin and credibility play, not a volume play. Qualification runs multi-year and volumes are tiny
eVTOL	Vendor spreads run from ₹8,572 crore in 2026 to ₹2.89 lakh crore by 2035 (USD 0.9bn to 30.3bn), against ₹1,214 crore to ₹8,038 crore (USD 127.5m to 844m) on a cells-only basis. A scope artefact, whole-aircraft system against cells. Do not present either as an addressable market	Weight sensitivity makes Wh/kg the buying criterion and cost per kWh secondary	A pool that has already killed one look-alike. CustomCells, a 200-plus-staff German spin-off in the same northern cluster, filed for insolvency in April and May 2025 because its eVTOL customer Lilium failed to pay HARD [148]
Robotics	TrendForce models solid-state demand from humanoids at 8.6 GWh in 2030 and 74.2 GWh in 2035. A second vendor models the humanoid battery market at only ₹5,924 crore by 2032 (USD 622m). Two vendors differ by roughly two orders of magnitude. Report both, average neither SOFT [70]	Cell specification is small-format, high-power, high-cycle, safety-critical and high-nickel, which is exactly a specialist's product	Modest in energy, excellent in margin. Even 1 per cent of the 2030 figure is about 86 MWh, roughly 6 per cent of a 1.5 GWh plant
Medical	**NOT FOUND** for any India-specific sizing. Globally, portable medical device batteries run ₹30,001 crore in 2026 to ₹58,287 crore in 2032 (USD 3.15bn to 6.12bn) at 11.6 per cent	Extreme qualification burden, tiny volumes, very high price per kWh, near-zero price elasticity	Perfect for a 1.5 GWh plant, irrelevant to a 100 GWh one. Note that no ISO 13485 certification is publicly evidenced as of October 2026

Table A17. Battery Regulation (EU) 2023/1542: the full compliance calendar, 2023 to 2050

The full Battery Regulation compliance calendar, 2023 to 2050. Referenced from chapter 12.

Sources: the Council press release of 10 July 2023 for the collection, recycling-efficiency, recovery and recycled-content percentages, the consolidated EUR-Lex text for structural dates, a cross-checked timeline for intermediate items. HARD items are primary-sourced; the calendar overall is SOFT.

DATE	OBLIGATION	BATTERY CLASS	ARTICLE OR ANNEX	TAG
31 Dec 2023	Collection target 45 per cent	Portable	waste chapter	HARD [98]
18 Feb 2024	Regulation applies, except Art. 11, Art. 17, Ch. VI and Ch. VIII	all		HARD [96]
18 Aug 2024	Electrochemical performance and durability documentation; safety requirements for stationary storage; state of health and expected lifetime in the BMS; conformity assessment; economic-operator obligations begin; lead limit 0.01 per cent	industrial over 2 kWh, LMT, EV; lead limit portable	Ch. VI, Art. 17	SOFT [99]
18 Feb 2025	Carbon footprint **declaration**, nominal date only, not triggered	EV	Art. 7	HARD nominal [96,103]
18 Aug 2025	Separate-collection symbol; waste-management rules; Member State penalty regimes; first recycling-efficiency tier	all	Ch. VIII, Annex XII Part B(1)	SOFT [99]
31 Dec 2025	Recycling efficiency achieved: lithium-based 65 per cent, lead-acid 75 per cent, nickel-cadmium 80 per cent, other 50 per cent, by average weight	by chemistry	Annex XII Part B	HARD [98]
18 Feb 2026	Carbon footprint **declaration**, nominal date only, not triggered	industrial over 2 kWh excluding exclusively-external-storage, and LMT	Art. 7	HARD nominal [96,103]
26 Jul 2026	Commission due-diligence **guidelines** due, moved from 18 Feb 2025	all in scope	Art. 48(5)	HARD [100]
18 Aug 2026	**Printed labelling** begins: general information label; capacity label; minimum average duration. Also carbon-footprint **performance class**, nominal	label: all; capacity label: rechargeable portable, LMT, SLI; class: EV	Annex VI Part A, Art. 7	SOFT [99]
18 Feb 2027	**QR code on every battery. Battery passport mandatory.** Removability and replaceability. Carbon-footprint declaration	QR: all. Passport: LMT, industrial over 2 kWh, EV. Removability: portable and LMT. Declaration: LMT	Annex VI Part C, Art. 77 and Annex XIII, Art. 11	HARD on QR and passport [96,99]
18 Aug 2027	**Due diligence applies**, above the turnover threshold. Also carbon-footprint performance class; minimum performance values	due diligence: all in scope. Class and minimum values: industrial over 2 kWh	Art. 48, Annex X	HARD on due diligence [100,101]
31 Dec 2027	Collection target 63 per cent. Material recovery: Co 90, Ni 90, Cu 90, Pb 90, Li 50 per cent	collection: portable. Recovery: by material	Annex XII Part C(1)	HARD [98]
18 Aug 2028	Carbon-footprint declaration; **recycled-content documentation** for Co, Li, Ni, Pb must accompany the battery; minimum performance; collection target 51 per cent; zinc-air button-cell lead exemption ends	declaration and minimum performance: LMT; documentation: in-scope classes; collection: LMT	Art. 8	SOFT [99]
18 Feb 2029	**Maximum life-cycle carbon-footprint threshold** applies	EV and rechargeable industrial over 2 kWh, excluding external storage	Art. 7	SOFT [99]
18 Feb 2030	Carbon-footprint performance class	LMT	Art. 7	SOFT [99]

DATE	OBLIGATION	BATTERY CLASS	ARTICLE OR ANNEX	TAG
18 Aug 2030	Carbon-footprint declaration	industrial with exclusively external storage	Art. 7	SOFT [99]
31 Dec 2030	Collection target 73 per cent; recycling-efficiency step-up	portable; lead-acid and lithium-based	Annex XII Part B(2)	HARD [98]
18 Aug 2031	**Recycled-content minimums, phase 1.** Also maximum carbon-footprint threshold; collection target 61 per cent	recycled content: industrial over 2 kWh, EV, SLI. Threshold and collection: LMT	Art. 8	HARD on the percentages [98]
31 Dec 2031	Material recovery steps up: Co 95, Ni 95, Cu 95, Pb 95, Li 80 per cent	by material	Annex XII Part C	HARD [98]
18 Feb 2032	Carbon-footprint performance class	industrial with exclusively external storage	Art. 7	SOFT [99]
18 Aug 2033	Maximum carbon-footprint threshold; recycled-content documentation	threshold: industrial with external storage; documentation: LMT	Art. 7, Art. 8	SOFT [98,99]
18 Aug 2036	**Recycled-content minimums, phase 2**	industrial over 2 kWh, EV, SLI	Art. 8	HARD on the percentages [98]
by 2050	Commission duty to set thresholds consistent with climate neutrality	all in scope	Art. 7	SOFT [98,99]

Table A18. EU benchmarks, state aid and the content instruments

EU benchmarks, state aid and the content instruments, 2019 to 2040. Referenced from chapter 12.

INSTRUMENT	STATUS	THE NUMBERS THAT MATTER	TAG
Critical Raw Materials Act, Reg. (EU) 2024/1252	Adopted 11 Apr 2024, in force 23 May 2024	2030 benchmarks: 10 per cent of EU annual consumption from domestic extraction, 40 per cent from domestic processing, 25 per cent from recycling, and no single third country above 65 per cent of EU consumption of any strategic raw material (Art. 5)	HARD on the benchmarks [115]
RESourceEU Action Plan	Adopted 3 Dec 2025, COM(2025) 945 final	European Critical Raw Materials Centre to be set up in 2026 for market intelligence, strategic stockpiling and joint purchasing; ₹32,046 crore (EUR 3bn) referenced for 2026; 46 new strategic raw-material projects	HARD / SOFT [116]
Net-Zero Industry Act	In force 29 Jun 2024	40 per cent of EU annual deployment needs manufactured domestically by 2030; 15 per cent of global market value by 2040; battery-specific indicative objective of 90 per cent of EU annual demand produced domestically in 2030, equal to 550 GWh of capacity	SOFT / HARD via EPRS [117,118]
Clean Industrial Deal	Published 26 Feb 2025	An Industrial Decarbonisation Bank of ₹10.68 lakh crore (EUR 100bn) planned for Q2 2026. Its state-aid arm, CISAF, is the legal basis for Germany's new industrial power price	SOFT / HARD [104]
IPCEI Batteries 1	Approved 9 Dec 2019	₹34,182 crore (EUR 3.2bn) of state aid, 17 direct participants, 7 Member States, expected to unlock about EUR 5bn of private investment	HARD [119]
IPCEI European Battery Innovation	Approved 26 Jan 2021	Up to ₹30,978 crore (EUR 2.9bn), 42 direct participants, 12 Member States	HARD [119]

INSTRUMENT	STATUS	THE NUMBERS THAT MATTER	TAG
Combined IPCEI position		**59 companies, 12 Member States, up to ₹65,160 crore (EUR 6.1bn) of state aid**, with expected private leverage of ₹1.50 lakh crore (EUR 14bn); Germany intended up to ₹32,046 crore (EUR 3bn) of it. The European Court of Auditors separately found ₹18,159 crore (EUR 1.7bn) of EU grants and loan guarantees to the battery industry. **UniverCell does not appear in any participant list: NOT FOUND**	SOFT / HARD via EPRS [118, 119, 120]
Industrial Accelerator Act	Proposed 4 Mar 2026, directly applicable, no national transposition	Covers battery technologies and storage alongside solar, heat pumps, wind, electrolysers, nuclear fission. Union-origin phasing for storage: assembly in the EU with an EU battery management system and other main specific components, escalating after three years to include **the cells**; reported trigger above 1 MWh. Non-price criteria in auctions rise from at least 30 per cent to at least **40 per cent** of auctioned volume per Member State per year. **Public procurement provisions apply to procedures launched on or after 1 January 2029**	HARD, SOFT on the 1 MWh threshold [110, 111, 112]
IAA foreign-investment conditionality	same instrument	Investments above ₹1,068 crore (EUR 100m) by investors from a country controlling over 40 per cent of global manufacturing capacity, in practice China, require approval on **at least four of six conditions**: ownership below 49 per cent; a joint venture with Union entities; licensing intellectual property to the Union target; 1 per cent of gross annual revenue into EU research; a 50 per cent Union workforce; 30 per cent of inputs sourced in the Union	HARD [110]

Table A19. Germany: battery research funding, pilot capacity and the industrial power price

German research funding, pilot capacity and the industrial power price, 2023 to 2028. Referenced from chapter 12.

ITEM	POSITION	TAG
Federal battery research, the crash	Cut to ₹1,656 crore (EUR 155m) in 2024; in September 2024 the ministry confirmed no new battery research projects would be funded from 2025, following the November 2023 Constitutional Court ruling on the Climate and Transformation Fund; about ₹1,260 crore (EUR 118m) in 2025, down 23.9 per cent, with no new commitments	SOFT [141, 142]
Federal battery research, the 2026 restart	₹2,425 crore (EUR 227m) in the 2026 budget under the Hightech Agenda Deutschland, which names battery technology one of six key technologies. New competence clusters on materials, on production including equipment manufacturing, and on solid state, first projects from end-2026; ₹534 crore (EUR 50m) for a fast-battery-customisation cluster; call published 2 June 2026	SOFT / HARD via BMFTR [139, 140]
Public pilot capacity	Fraunhofer FFB Fab, Muenster, completing end-2027, electrode machine capacity reported at about 6.8 GWh per year. Funding reported both as ₹5,341 crore (EUR 500m) of federal project finance plus ₹3,418 crore (EUR 320m) from North Rhine-Westphalia, and as up to ₹8,012 crore (EUR 750m) in total. This report carries both and leans to the later, ministry-sourced figure	SOFT, conflicting [143, 144]
Industrial power subsidy, from 1 Jan 2026	Target ₹5.34 (5 EUR cents) per kWh on 50 per cent of annual consumption, paid as compensation equal to half the wholesale reference price with the target as a floor. 2026 reference price 8.744 cents, differential ₹4.00 (3.744 cents). Programme 2026 to 2028, total about ₹33,114 crore (EUR 3.1bn), administered by BAFA, applications 31 March to 30 September 2027. At least half the aid must be reinvested in decarbonisation within 48 months	HARD unless noted [132, 133]
Eligibility for it	The 91 sectors of Annex I List 1 of the KUEBLL guidelines as carried into CISAF, verified by NACE code. **NACE 27.20, manufacture of batteries and accumulators, is reported to be on the eligible list**	SOFT, needs direct confirmation against the CISAF annex [134]
Schleswig-Holstein grid, 2025	29.9 million MWh fed into the grid, of which 27.3 million MWh or **91.2 per cent renewable**, against only 10.0 million MWh supplied to end-users in the state: roughly three times as much generated as consumed	HARD [136, 137]
Schleswig-Holstein battery policy	**NOT FOUND.** No state battery strategy, cluster programme or dedicated funding line. The state's levers are power, permitting and land	NOT FOUND

Table A20. Porter's five forces: the published basis for each score

The published basis behind each of the ten Porter scores, evidence of 2010 to 2027. Referenced from chapter 13.

In chapter 13, the scores are given for both arenas.

FORCE	SPECIALTY NICHE: PUBLISHED BASIS	COMMODITY EV CELL: PUBLISHED BASIS
Competitive rivalry	The peer set was removed by failure, not competition: CustomCells filed on 30 April and 5 May 2025, restructured to roughly 80 staff with Tuebingen closed; Cellforce closed in May 2026; Morrow filed on 6 May 2026; Leclanche's two German subsidiaries filed at the Amtsgericht Offenburg on 23 September 2026; Varta filed in self-administration on 24 July 2026.[148, 151, 158, 157, 154] What survives is Taiwanese: Molicel, approximately 1.5 GWh at Tainan, a 1.8 GWh Kaohsiung project funded at ₹3,333 crore (NT$10bn, about USD 350m) and a 3.3 GWh target, same high-drain 21700, already holding Archer Aviation, Vertical Aerospace and McMurtry Automotive; a Kaohsiung fire forced a production shift to Tainan[180]	Arithmetic. Volume-weighted pack prices fell eight per cent in 2025 to ₹10,286 per kWh (USD 108), a record low and 93 per cent below 2010 in real terms; Chinese planned capacity of roughly 4,800 GWh sat against roughly 1,200 GWh of demand.[44, 54] CATL will not run a 50,000-cell-a-month line: at 39.9 per cent of global installs and 96.9 per cent utilisation, its scarce resource is line time[41, 54]
Buyer power	Qualification cycles of two to five years and pack-level design-in create genuine switching costs EST. Against that, buyers are few and large and exposure here is extreme: no customer named anywhere, only "first multi-million-euro commercial contracts" and a single United States customer.[10] The precedent is documentary and close: CustomCells Itzehoe, holder of four patents that name both UniverCell founders as inventors, was pushed into insolvency when its largest customer, Lilium, failed to pay tens of millions[15, 148]	Demonstrated, not theorised. BMW cancelled an approximately ₹21,364 crore (EUR 2bn) Northvolt contract in June 2024 after the ramp failed; Stellantis, Mercedes-Benz and TotalEnergies definitively cancelled ACC's Kaiserslautern and Termoli plants in February 2026, Stellantis replacing them with a 50:50 CATL joint venture at Zaragoza.[187, 165, 184] A buyer reading an LFP pack at ₹7,714 per kWh (USD 81) against an NMC pack at ₹12,191 per kWh (USD 128) needs no more leverage[44]
Supplier power, which is the finding	Scale buys supplier leverage and the niche has no scale: a 92-person buyer taking tonne-scale quantities has no purchasing power against a cathode producer whose other customer is CATL. The hedge is structural, not commercial: all anodes and cathodes developed and produced in-house, more than six validated silicon suppliers, multiple validated suppliers for all raw materials, 100 per cent EU and North American sourcing stated as viable[1, 11]	A single-country monopoly in all but name, and it sits above both arenas. China holds approximately 85 per cent of cathode active material capacity, over 90 per cent of anode active material, 94 per cent of battery-grade graphite refining and 98 per cent of graphitization; cathode active material is roughly 50 per cent of cell manufacturing cost, anode roughly 10 per cent.[32, 118] Already weaponised: MOFCOM Announcement No. 58 placed cells at or above 300 Wh/kg, artificial graphite anode, advanced cathode materials and key cell-making equipment under dual-use licensing, suspended and not repealed, statutory expiry 10 November 2026, while the July 2025 licensing of fourth-generation LFP and LMFP cathode technology was never suspended. HARD on the dates, SOFT on the numeric thresholds[91, 92] The DRC has capped cobalt exports at 96,600 tonnes a year for 2026 and 2027 against roughly 204,000 tonnes of 2024 output, with cobalt up about 350 per cent in fourteen months[61]
Threat of entry	Gated by qualification and tacit process knowledge, not capital: AS9100 or ECSS for satellites, ISO 13485 for critical-care medical devices, UN 38.3 to ship a cell by any mode at all, plus formation control and a recipe library. This company stands in front of that gate, not behind it: none of those certifications is publicly evidenced as of October 2026.[1, 2, 11] The imitation route is open and priced: Fraunhofer IWS licenses DRYtraec to any comer at up to 240 mm width and 10 m per minute across four prototype systems and launched the ₹39.52 crore (EUR 3.7m) DRYplatform in Dresden on 3 November 2025; Duerr and LiCAP hold a validated European platform and lost their pilot host when Cellforce closed in May 2026[145, 172, 173]	Blocked for the Western startup, wide open for the capitalised Chinese incumbent. CATL began trial production at Debrecen on 22 September 2026 with 34 GWh in phase one and 100 GWh planned; Stellantis and CATL target end-2026 at Zaragoza on 50 GWh of LFP; BYD targets Szeged in the fourth quarter of 2026; Gotion-InoBat is building 20 GWh in phase one at Surany[183, 184, 185, 164]

FORCE	SPECIALTY NICHE: PUBLISHED BASIS	COMMODITY EV CELL: PUBLISHED BASIS
Threat of substitutes	Format substitution is the live threat to a 21700 franchise, and one the company can only position around, not engineer away.[171, 170] Product substitution is already visible: Molicel's P70X was cited in July 2026 at 340 Wh/kg and 30C, above UniverCell's 329 to 330 Wh/kg; Amprius SiMaxx is quoted up to 450 Wh/kg and 1,150 Wh/L.[180, 68] Lithium-sulphur is the long-dated threat and now has a well-capitalised owner in Lyten, aimed at defence, aerospace, drones and data centres[161]	The main event. LFP passed 55 per cent of global EV batteries deployed and 81.2 per cent of Chinese installations, growing 48 per cent in 2025; CATL's Naxtra sodium-ion entered mass production from December 2025 at up to 175 Wh/kg, with 60 GWh of capacity targeted by end-2026 and a mass-market car launched in February 2026[32, 43, 36, 63, 66]

Table A21. PESTEL: all twenty-four entries, each dated or quantified

All twenty-four PESTEL entries, each dated or quantified, 2019 to 2036. Referenced from chapter 13.

Chapter 13 carries six of these entries in full. Tags read from the perspective of a European specialty cell maker. Ec is economic, En environmental.

	ITEM	DATE OR QUANTITY	TAG
P	US prohibited-foreign-entity rules under OBBBA, with the material assistance cost ratio	Enacted 4 July 2025; 55 per cent non-prohibited direct costs in 2026 to 75 per cent after 2029; guidance in Notice 2026-15, February 2026 SOFT [129]	**Tailwind**
P	SAFE and the wider ReArm Europe plan	Up to ₹16.02 lakh crore (EUR 150bn) inside a ₹85.46 lakh crore (EUR 800bn) plan; 65 per cent EU, EEA or Ukraine content floor; live now[124, 125]	**Tailwind**
P	China's export-control regime	July 2025 cathode technology licensing in force and never suspended; Announcement No. 58 suspended only to 10 November 2026, extension to 10 January 2027 reported but not confirmed by a numbered instrument[91, 92]	**Ambiguous**
P	EU trade defence on rechargeable lithium-ion cells	No anti-dumping or anti-subsidy case open on HS 8507.60 as of 20 October 2026, with Chinese battery imports into the EU up 39 per cent in H1 2026 to ₹1.75 lakh crore (EUR 16.4bn)[84]	**Ambiguous**
Ec	The commodity price floor	Pack ₹10,286 per kWh (USD 108) in 2025, minus eight per cent; 2030 pack floor of roughly ₹6,667 to ₹8,095 per kWh (USD 70 to 85) on two independent methods EST [44]	**Headwind**
Ec	The European cost premium	European packs about 56 per cent above Chinese, and the premium widened relative to North America's 44 per cent[44]	**Headwind**
Ec	German industrial power	22.64 ct/kWh in Eurostat band IC, H2 2025, roughly 23 per cent above the EU average; the 2026 to 2028 industrial price scheme pays 3.744 ct/kWh on half of consumption[131, 132]	**Headwind**
Ec	Europe's demand mix	Storage at eight per cent of European lithium-ion demand in 2025, down from sixteen; EU storage additions 27.1 GWh, up 45 per cent, with over 50 GWh forecast for 2026[36, 53]	**Ambiguous**
S	The engineer constraint, which is the social fact that decides outcomes	1,000 to 1,500 trained process engineers per GWh; India would need roughly 36,000 if announced gigafactories proceeded; India issued roughly 2,000 visas to Chinese nationals in 2024 against about 200,000 in 2019[209, 211]	**Tailwind** for a European holder of engineers
S	Competition for the same German engineers	Lyten hiring more than 600 additional staff within twelve months in Sweden and targeting roughly 1,000 jobs at Heide from 2028[161, 162]	**Headwind**
S	A displaced German battery workforce	Varta's July 2026 filing affected roughly 3,260 employees with Noerdlingen's 350 jobs closing in autumn 2026; Cellforce shed roughly 200 of 290 in August 2025 and the remaining 50 in May 2026[154, 151]	**Ambiguous**
S	Employer footprint, as a data gap rather than a finding	No Kununu and no Glassdoor profile for a 92-person German employer; a self-administered widget shows seven reviews, which this report does not treat as evidence. NOT FOUND[31]	**Ambiguous**

	ITEM	DATE OR QUANTITY	TAG
T	Dry electrode industrialisation by others	PowerCo guided 2026 or 2027; LG Energy Solution 2028, and 2029 for dry LFP; Samsung SDI two named pilots with no date, NOT FOUND [174, 169, 171]	**Headwind**
T	Dry processing as a prerequisite for solid-state	Sulphide solid electrolytes are incompatible with conventional NMP and water slurry routes, so dry coating is a precondition on most all-solid-state roadmaps, which reach volume between 2030 and 2035	**Tailwind**
T	Silicon anode commercialisation	Already in drones and eVTOL, mass EV between 2028 and 2032; Amprius SiMaxx up to 450 Wh/kg and 1,150 Wh/L; high-silicon anodes are hard to wet-coat SOFT [68]	**Tailwind**
T	Format migration	Samsung SDI in 46-phi volume since March 2025; LG backlog above 440 GWh SOFT [171, 170]	**Headwind**
En	Battery Regulation carbon footprint	Nominal dates of 18 February 2025 and 18 February 2026 have not triggered because the delegated act does not exist; the maximum threshold is set for 18 February 2029 {SOFT, strong} [103, 99]	**Tailwind, deferred**
En	Recycled-content minimums	18 August 2031: 16 per cent cobalt, 6 per cent lithium, 6 per cent nickel, 85 per cent lead; 18 August 2036: 26, 12, 15 and 85 per cent [98]	**Tailwind**
En	Grid intensity at the site	Schleswig-Holstein at 91.2 per cent renewable feed-in in 2025, generating roughly three times what the state consumes EST on the multiple [136]	**Tailwind**
En	Solvent removal as an environmental and permitting item	The dry route eliminates NMP handling and recovery, a regulated emission source, and an 80 metre twelve-zone dryer; the company's own figure is 27 per cent lower energy per cell [4, 19, 11]	**Tailwind**
L	Battery passport and QR code	Mandatory 18 February 2027 for LMT, industrial above 2 kWh and EV batteries; printed labelling from 18 August 2026 [96, 99]	**Tailwind**
L	Due diligence under Article 48	Applies 18 August 2027 above ₹427 crore (EUR 40m) of net turnover, so it is a customer-side burden this company supports with data rather than bears HARD [100, 101]	**Tailwind**
L	CBAM	Definitive regime from 1 January 2026 covering cement, iron and steel, aluminium, fertilisers, electricity and hydrogen. Batteries are not in scope. Whether CN 8507 appears in the 2028 downstream list is NOT FOUND [121, 122]	**Neutral, and widely misstated**
L	EU CO2 standards for 2035	Commission proposed on 16 December 2025 to soften the target from 100 per cent zero-emission to a 90 per cent CO2 reduction, with the 2030 van target cut from 50 to 40 per cent SOFT [75]	**Headwind**

Count: eleven tailwinds, seven headwinds, five ambiguous, one neutral.

Table A22. The battery value chain: who controls each stage

Who controls each of the eight value-chain stages, 2020 to 2027. Referenced from chapter 13.

Chapter 13 carries the value-added bands and UniverCell's participation.

STAGE	WHO CONTROLS IT, WITH THE SHARE
1. Mining and extraction	Geographically diverse except cobalt: DRC quota 96,600 t/yr for 2026 and 2027 against roughly 204,000 t of 2024 output [61]
2. Refining and processing	China: lithium 74%, cobalt 76%, battery-grade graphite 94%, graphitization 98%, nickel 31%, copper 47%; overall concentration rose from 82% in 2020 to 86% in 2024 [32]
3. Active materials, cathode and anode	China approximately 85% of cathode capacity, up to 90% by chemistry, approximately 95% of LFP cathode, over 90% of anode. Cathode is roughly 50% of cell manufacturing cost, anode roughly 10% [32, 113, 118]

STAGE	WHO CONTROLS IT, WITH THE SHARE
4. Electrode manufacture	**No controlling party and almost no merchant market. The named merchant players worldwide are UniverCell and AM Batteries; Fraunhofer FFB is a public pilot platform at roughly 6.8 GWh of electrode machine capacity, not a supplier**[176, 143]
5. Cell assembly and formation	Over 80% of production located in China; in Europe over 85% of active capacity is non-European-owned and South Korean makers are 72 per cent of production[32, 113]
6. Module, pack and BMS	Fragmented. India holds roughly 60 GWh of pack assembly and captures roughly 40 per cent of storage system value against China's 60 per cent, with margins compressing[214, 210]
7. System integration and application	OEMs, defence primes, tool brands, satellite integrators
8. Recycling and second life	Emerging and capital-hostile: Li-Cycle's assets sold to Glencore for ₹415 crore (USD 43.6m) against ₹16,191 crore (USD 1.7bn) raised, roughly 97 per cent capital destruction. EU recycling efficiency of 65 per cent for lithium-based batteries applied from 31 December 2025[186, 98]

Table A23. India: announced against commissioned cell capacity, every site, October 2026

Fourteen Indian sites, announced against commissioned, October 2026. Referenced from chapter 14.

Chapter 14 carries the four largest announcements in full.

OWNER AND SITE	ANNOUNCED GWH	COMMISSIONED CELL GWH	STATUS, OCTOBER 2026	TECHNOLOGY SOURCE
Ola Cell Technologies, Krishnagiri	20 under PLI, 100 long-run	**1.4 nameplate (PLI-certified); 5 "installed" per company; "near 6 built-in" per the chief executive**	**India's only operating cell plant.** 72,418 cells and about 1.9 MWh in Q3 FY2026. Pilot from March 2024	In-house. **Dry electrode**, NMC 4680 and 46100 LFP, 275 Wh/kg published. Cathode from Umicore (Belgium), anode from Epsilon (India). **No Chinese technology partner**
Reliance New Energy Battery, Jamnagar	10 under PLI plus 40 design, modular to 100 to 120	**0**	"Advanced commissioning", ramp guided to the second half of 2026; the June 2026 annual meeting promised the first 40 GWh phase "this year"	Faradion (sodium-ion, about ₹1,000 crore, 2021), Lithium Werks (219 LFP patents, about ₹464 crore, 2022). **Hithium licensing talks stalled as of January 2026**
Reliance New Energy Battery Storage, Dharwad	5 under PLI	**0**	No commissioning reported	as above
Exide Energy Solutions, Bengaluru	12, of which 6 in phase one	**Phase one 6 GWh reported "commissioned 23 September 2026"; commercial output not evidenced**	₹3,600 crore of equity invested, over ₹5,200 crore committed. Also runs 1.5 GWh of module and pack capacity	**Licensed from SVOLT Energy (China)**, December 2021: irrevocable licence to NMC and LFP, turnkey including plant, equipment and quality systems
Amara Raja, Divitipally	16 by FY2031, ₹9,500 crore, 262 acres	**conflicted**	100 MWh qualification line commissioning. S&P reports cell production delayed to FY2027	Gotion-InoBat LFP licence plus Jiangsu Highstar NMC licence. **Gotion deal derailed on geopolitical grounds**; now about ₹952 crore (USD 100m) into an in-house research and engineering centre
Tata Agratas, Sanand	20	**0**	Topped out April 2026, production equipment arrived September 2026, **cells now expected 2027**	NMC licensed from AESC Apollo (Japanese entity majority-owned by China's Envision), which **took 12 per cent of Agratas for about ₹66 crore**. LFP in-house. Over ₹3,810 crore (USD 400m) Bengaluru research centre

OWNER AND SITE	ANNOUNCED GWH	COMMISSIONED CELL GWH	STATUS, OCTOBER 2026	TECHNOLOGY SOURCE
JSW, Odisha	50 by 2028 to 2030, 10 in phase one	**0**	**Cell gigafactory halted.** Over ₹800 crore redirected into a battery research centre with a Pune university; still seeking a technology partner	**None secured**
TDSG, Hansalpur	1.2 (LTO, hybrids)	**operating**	**The only plant in India ever to achieve electrode-level localisation in lithium-ion, August 2025, reaching over 80 per cent domestic value creation**	Toshiba SCiB
Nash Energy, Dobbaspet	roadmap to 10	sub-GWh, **but shipping**	**2.5 million LFP cells shipped in 15 months**, 40-plus pack-builder customers. **In-house electrodes planned 2027 to 2028**	LFP chemistry acquired outright from Forte Nippon (Japan), 2021
GODI India, Hyderabad	12.5	**0 at scale**	NMC 21700 and LFP BIS-certified; about 108 employees, about ₹114 crore raised (USD 12m); Graphite India took 46 per cent in 2025	In-house
International Battery Company, Bengaluru	10 prismatic NMC-622, ₹9,524 crore stated (USD 1bn)	**0**	₹390 crore facility announced January 2025, not evidenced as operating. Joint venture with Mahanagar Gas	US research with Korean manufacturing, **explicitly non-China and US content-rule compliant**
HBL Power Systems, Hyderabad	not applicable	pilot	**About ₹100 crore of capex in FY2025 to make high-energy cells in-house, for defence use only.** Ministry of Defence and DRDO approved	In-house
Log9 Materials, Bengaluru	LTO	**0, ceased**	**Insolvency.** About ₹150 crore invested in manufacturing, admitted to the tribunal after defaulting on ₹6.7 crore, having raised ₹571 crore (USD 60m). **The LTO bet lost to cheap Chinese LFP**	In-house
Adani	20 appears on announcement lists	**0 cells**	**No announced Indian cell project.** The real position is deployment: 6.63 GWh of operational storage, over half of India's total, including a 3.37 GWh single-location system	Discussed CATL; nothing announced

Sources.[208,209,210,211,212,213,214,224]

Table A24. India: the cell-adjacent layers where India is genuinely competitive

The cell-adjacent layers where India is genuinely competitive, 2025 to 2031. Referenced from chapter 14.

LAYER	WHAT EXISTS, WITH NUMBERS	READ
Anode and graphite	**Epsilon Advanced Materials:** 2,500 tonnes a year of meso coke today; a **₹4,000 crore** synthetic graphite anode plant cleared in August 2026 for 30,000 tonnes by 2028 and 100,000 in phase two; targets of 100,000 tonnes synthetic plus 50,000 natural by 2030; a group commitment of **₹15,350 crore** in Karnataka. **Already supplies anode to Ola. Himadri Speciality Chemical:** about 20,000 tonnes near-term, 100,000 by 2028. **HEG** exploring	**The strongest cell-adjacent position India holds.** An Indian anode supplier already qualified into an operating cell plant is a real asset SOFT [215,214]

LAYER	WHAT EXISTS, WITH NUMBERS	READ
Cathode	**Altmin**, Telangana: **₹750 crore**, 20 acres, 20,000 tonnes a year of LFP, roughly 8 GWh equivalent, operations in the fourth quarter of 2026, about 10 per cent of Indian LFP demand, plan 100,000 tonnes by 2030. **Allox**, Hyderabad: patented LFP route at 50 kg a day today, target 10 to 15 tonnes a day, **₹210 crore** memorandum for 3 GWh a year. **Epsilon** cathode: 10,000 tonnes by 2026 to 100,000 by 2031	Early but real, and against roughly 95 per cent Chinese control of LFP cathode globally, strategically disproportionate SOFT [216, 214]
Electrolyte and separator	Neogen Chemicals and Gujarat Fluorochemicals in electrolyte. Daramic (Asahi Kasei) makes lead-acid separators today with a greenfield lithium-ion separator project planned	Partial. **LiPF6 and battery-grade lithium salts remain absent at scale** SOFT [214]
Lithium refining	**Lohum** announced India's first battery-grade lithium refinery in March 2025, designed to take both black mass and spodumene feed	A genuine midstream entry. Note the corrections: **Coromandel International is an agrochemicals company and is not in lithium refining, and Vedanta Ltd has no lithium mining plans**; the recycling joint venture with Belgium's Silox belongs to NAN GreenMet, a separate vehicle SOFT [217, 214]
Recycling, India's strongest card	Formal installed capacity about **35,000 to 45,000 tonnes a year in 2026 at only 30 to 50 per cent utilisation** because of feedstock shortage. Attero: 16,000 tonnes adding 10,000, plan 60,000 within three years, plus **₹150 crore** for 100,000 tonnes of e-waste. Lohum targeting 50,000 tonnes by 2026 on **₹1,210 crore** raised (USD 127m). BatX at 4,000 tonnes. Only about half of throughput converts to usable black mass	One source claims capacity passed 100,000 tonnes by 2025, a 2.5 times disagreement. **This report trusts the lower, utilisation-aware figure**, consistent with the stated feedstock problem. The National Critical Mineral Mission carries **₹34,300 crore over seven years (₹16,300 crore budgetary plus ₹18,000 crore from public enterprises)** including a **₹1,500 crore recycling incentive** to FY2031 targeting 270,000 tonnes a year SOFT [214, 231, 232]
Domestic lithium and offshore supply	The Salal-Haimana block in Reasi covers **4.18 square kilometres** and completed G-2 exploration in the 2025-26 field season. Two auctions failed, the first for want of qualified bidders, the second annulled for non-receipt of bids; a third is queued for the eighth tranche in **FY2027**, and no proven metallurgical route exists for lithium from aluminous laterite at scale. Offshore, KABIL has acquired about **15,703 hectares** in Catamarca, Argentina for lithium exploration and mining	SOFT [218, 219, 220], HARD on KABIL [233]
Packs, BMS and storage assembly	About **60 GWh of pack assembly against about 1 GWh of cells**. Storage assembly commissioned in 2026: GoodEnough Energy 7 GWh, Waaree 5.15 GWh, JSW 5 GWh, about **23.15 GWh a year** across four firms. Indigenous energy-management software is already mandated under the storage funding guidelines from 4 August 2025	**Commoditised and oversupplied, which is exactly why pack value capture is 40 per cent and falling.** The arbitrage any cell-technology partner sells into is 23 GWh of assembly sitting on 1 GWh of cells SOFT [214, 243, 229]

Table A25. The six India plays: supporting evidence and provenance

Supporting evidence and provenance for each of the six plays, 2018 to 2029. Referenced from chapter 14.

Chapter 14 carries each play with its capital need, main risk, horizon and anchor fact.

PLAY	SUPPORTING EVIDENCE AND PROVENANCE
1. License the process, not the cell	A solvent-free electrode process licence is manufacturing capability, not chemistry intellectual property: cathode-agnostic, outside the Chinese control regime, aimed at the constraint India cannot buy around. The holder's own documents claim **27 per cent lower energy consumption at cell level** HARD and up to 60 per cent smaller factory footprint, against an industry range of 17 to 30 per cent on electrode manufacturing cost SOFT. **Ola already runs a dry electrode process for NMC 4680**, so the precedent exists in India and nobody owns it. EST **The fact it rests on:** China restricted LFP and LMFP cathode technology in July 2025 and **never suspended it**; equipment and anode controls lapse back into force 10 November 2026. [91, 92]
2. Merchant electrode supplier	Moves a licensee **from 25 to 60 per cent domestic value addition without requiring them to solve cathode or anode chemistry first**, and every incentive beneficiary is bound to those thresholds at two and five years with zero rupees claimed. The model is proven in Europe and absent in India: custom coated electrode foil sold standalone, over 1,000 tested recipes, a thousandth customer electrode order in January 2026. Capital is scaled from Altmin's ₹750 crore for 20,000 tonnes of cathode and Allox's ₹210 crore for 3 GWh. EST **The fact it rests on: exactly one plant in India has ever localised at electrode level** and reached over 80 per cent domestic value by doing so. [214]

PLAY	SUPPORTING EVIDENCE AND PROVENANCE
3. Distressed European specialty capacity	Six European cell makers failed or restructured between January 2023 and September 2026, over 700 GWh of announced capacity is delayed or cancelled with about 240 GWh in Germany alone, and Fraunhofer's risk model puts realisation at 54 to 75 per cent of announcements. Relocating a plant fails. **Keeping the European site as the qualification, research and EU-market entity and moving only the process package and 10 to 20 engineers to India** preserves EU domicile for Play 4 and solves the talent constraint. EST **The fact it rests on:** 1,000 to 1,500 process engineers per GWh and about **36,000 needed**, against roughly 2,000 Chinese visas in 2024.[209,211]
4. EU-domiciled vehicle	The European Union's Industrial Accelerator Act, proposed 4 March 2026, requires storage systems assembled in the Union with a Union battery management system, escalating after three years to include cells, with procurement provisions applying to procedures launched on or after **1 January 2029** and conditionality on investors from countries holding more than 40 per cent of global manufacturing capacity. India mandates a **minimum 20 per cent local content on every funded storage project** under a ₹5,400 crore, 30 GWh scheme at ₹18 lakh per MWh; the third incentive tranche required 25 per cent domestic value addition in two years and 40 per cent in five, and **closed on 13 October 2026**, so this targets a fourth tranche, state schemes or the funding route. One technology base can bid Indian content in India and Union content in Europe. EST [110,228,230,196] **The fact it rests on:** both blocs now pay for local content and **neither has enough**: EU cells required after three years with procurement from 1 January 2029, India at 20 per cent on funded storage.[110,228]
5. Sodium-ion co-development	A federally funded European sodium-ion consortium of 25 partners runs from March 2026 to February 2029 on ₹154.89 crore (EUR 14.5m). In India, NTPC invited bids in September 2026 for a sodium-ion storage pilot under Indian grid and climate conditions, KPIT with Trentar is building 3 GWh for two-wheelers, Tata Chemicals has an indigenous pilot with customer test units targeted by end-FY2027, and Reliance is fast-tracking Faradion's chemistry at Jamnagar. SOFT [244] China's structural advantage is weakest here: sodium-ion needs neither lithium nor graphite anode, and graphite anode sits inside the suspended Chinese control list. But the cost case is fragile, and CATL shipped sodium-ion into a mass-market car in February 2026 while building 60 GWh. EST **The fact it rests on:** India's storage requirement runs **34.72 GWh in FY2027 to 236.22 GWh in FY2032**, the largest and least density-sensitive pool in the country.[206,244]
6. Specialty and defence-qualified niche	The only one of the six where India's engineer shortage does not bite, because a 200-person specialty plant is staffable; where Chinese price competition is excluded by policy rather than tariff; and where a documented unmet import dependency can be quoted to a government customer. The 2018 ISRO to BHEL transfer is the existing government-sanctioned structure for moving specialty cell technology into an Indian industrial. The incumbents are Saft and EnerSys, holding the space, defence, submarine and aerospace accounts with decades of qualification history and flight heritage, **and a satellite programme does not change cell supplier because the new one has better coating.** Against them, India's indigenous high-discharge effort is one early-stage start-up. EST **The fact it rests on: DRDO built the Tapas battery and its management system indigenously but imports the cells**; HBL's entire in-house commitment is about ₹100 crore.[224]

Table A26. The dated watch-list, October 2026 to April 2028

Fifteen dated items with what each decides, October 2026 to April 2028. Referenced from chapter 15.

DATE	ITEM	WHAT IT DECIDES
31 Oct 2026	EIC Accelerator grant GA 101161461 expires	Any "ongoing EIC project" claim is past tense from November[19]
10 Nov 2026	Statutory lapse of China's Decision No. 70 suspension of Announcement No. 58	The most consequential date in the report. The reported extension to 10 January 2027 rests on no numbered instrument[92]
Dec 2026	BloombergNEF's next price survey	A pack below ₹9,524 per kWh (USD 100) implies the fast learning rate[44]
Dec 2026	EU-India FTA signature planned; the HS 8507 schedule is NOT FOUND	Whether cells and electrode foil move duty-free either way[234]
by 31 Dec 2026	Battery Booster Facility awards, call closed 30 September 2026, outcome NOT FOUND	Whether ACC, Verkor and PowerCo draw the ₹16,023 crore (EUR 1.5bn)[105]
Q4 2026	Zaragoza start-up, BYD Szeged mass production, Altris and Draslovka cathode production	Whether Chinese-owned European capacity arrives on schedule
Late 2026	India ACC PLI third tranche, technical bids opened 14 October 2026	A third failure confirms India stays an importer through 2030[196]
10 Jan 2027	Reported new expiry of the Chinese suspension	Verify against a numbered MOFCOM instrument before relying on it

DATE	ITEM	WHAT IT DECIDES
Jan 2027	ACEA full-year 2026 registrations; SolarPower Europe's EU storage outturn	Below 22 per cent BEV share, or storage under 40 GWh against a 50 GWh guide, is the bear path[74, 53]
Q1 2027	CATL Debrecen output disclosures	Whether 34 GWh of phase one produces or merely stands
18 Feb 2027	Battery passport and QR code mandatory; removability applies	The first date the Battery Regulation rewards short, transparent chains[96, 97]
2027	EU 2035 CO2 file adopted; carbon-footprint delegated act unpublished	A weaker text is the largest downside in European cell demand; the act starts a 12 to 18 month clock[103]
2027	The 1.3 GWh milestone; the 21700-HP variant, samples targeted Q3 2026	The ramp test, and whether a second product exists
18 Aug 2027	Due diligence applies above ₹427 crore (EUR 40m) turnover	A customer-side burden to support with data rather than bear
2028	Lyten production at Heide; LG Energy Solution dry-electrode mass production	The lead window closes, 60 to 70 km away and one generation out[162, 169]

Appendix

Source register

249 sources, one line each, numbered in the order first cited. Eighteen entries carried in the research file but cited nowhere in the report have been dropped, so every chapter citation resolves here and the numbering runs without gaps.

Trust ratings are this report's, not the publisher's. PRIMARY is the originating body's own publication and may be stated as fact. HIGH is an established institution or news organisation reporting a primary fact it names. MEDIUM is trade press, law-firm or consultancy secondary reporting: attribute it and cross-check the number. LOW is a vendor market model, data aggregator or self-administered rating: direction only, never a point estimate.

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